How Hidden Local Government Debt Shapes Shadow Banking Competition

Evidence from Wealth Management in China

Kevin Zheyuan Cui[1] Tao Shen[2] Tianqi Zhi[2]

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Abstract

Using granular holdings data on China's RMB 33 trillion market for wealth-management products operated by banks or their subsidiaries, we study how government support for borrowers shapes competition in shadow banking. Smaller banks sought an edge over larger rivals by reaching for yield in home-province bonds issued by local government financing vehicles (LGFVs). The 2023 debt resolution initiative strengthened expected support for LGFVs and compressed their bond spreads. The estimated product-return premium associated with home LGFV bonds fell by 69%. Larger local banks expanded into higher-yielding, LGFV-linked private credit, while smaller banks substituted little and lost ground in relative product returns and wealth-management balances. Our model explains this unequal adjustment through fixed contracting costs that make investment scale important. Government support for borrowers can thus erode smaller banks' competitive edge and shift shadow banking toward larger intermediaries and more opaque credit.

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[1] Department of Economics, Princeton University

[2] School of Economics and Management, Tsinghua University

1 Introduction

Borrowing by local government financing vehicles (LGFVs) fueled the expansion of China's shadow banking system. After the 2009 fiscal stimulus, LGFVs refinanced maturing bank loans through bonds and shadow-banking channels (Chen et al. 2020). Wealth-management products (WMPs), which pool savers' money outside banks' own balance sheets, became central to this system (Acharya et al. 2025). At the end of 2025, WMPs held RMB 33 trillion, nearly a quarter of China's GDP, for about 143 million investors, almost all of them individuals.[1] At the end of 2024, WMPs held an estimated 28.1% of outstanding LGFV bonds (Figure A5). The exposure was especially important for local banks: LGFV bonds accounted for 38.5% of identified standard-bond holdings in their WMPs, compared with 13.2% at national banks.[2] Much of the literature examines how shadow banking finances LGFVs. Using granular WMP portfolio data, we study the other side of this relationship: how LGFV bonds shape competition for household savings within shadow banking.

Because WMPs cannot guarantee principal or returns, the performance their products deliver is an important way for banks to compete for savers. Smaller banks, with less extensive brands and branch networks, reached for yield in LGFV bonds from their home provinces. China's 2023 local-government debt resolution undercut this strategy. By allowing provinces to issue special refinancing bonds to repay selected LGFV obligations, the program strengthened expected support for LGFVs and compressed their bond spreads. The estimated product-return premium associated with home LGFV bonds fell by 69%. Larger local banks expanded into higher-yielding, LGFV-linked private credit. Smaller banks substituted little, and their product returns and WMP balances declined relative to those of larger rivals.

[1] Industry statistics are from the Banking Wealth Management Registration and Depository Center's annual market reports (2025; 2026), referred to below as the annual WMP market reports. China's 2025 GDP of RMB 140.2 trillion is the National Bureau of Statistics' preliminary estimate.

[2] These are shares by holding value over 2022Q1–2023Q3. The denominator includes all positions identified as standard bonds in the holdings data, including government and policy-bank bonds, but excludes private credit. Internet Appendix D.8 gives the calculation and distinguishes these bond-only shares from the debt-allocation shares used below.

In the WMP market, large state-owned and nationwide joint-stock banks, which we call national banks, compete with local banks serving provincial, city, and rural markets. Higher-yielding investments accessible to WMPs are scarce. Products can hold government and policy-bank bonds, deposits, negotiable certificates of deposit, and corporate debt, but safer instruments offer limited spreads and China's public high-yield bond market is narrow (Schipke et al. 2019). LGFV bonds help fill this gap. All eligible investors can buy them, and local banks can draw on familiarity with the borrowers. Private credit (PC), classified in China as non-standard debt, can finance the same borrowers through less transparent, nontraded claims such as trust loans and financing through asset-management plans. Unlike buying a marketable bond, arranging these investments entails sourcing, evaluation, and contracting costs. This distinction motivates the scale mechanism we develop below.

We link quarterly holdings of 60,396 WMPs from 171 banks to product returns, investor flows, issuer credit events, and bank-level WMP balances. Local banks concentrated their LGFV bond holdings in their headquarters provinces; we refer to these as home LGFV bonds. Across all banks, smaller institutions held larger home-LGFV shares. These holdings mattered commercially: before debt resolution, home LGFV exposure carried the largest estimated product-return premium among the asset categories we examine, including LGFV bonds from other provinces. Within risk and withdrawal categories, top-return-quintile products attracted subsequent net inflows exceeding those of bottom-quintile products by 12.2% of beginning assets. The portfolio and demand evidence thus connects home bonds to local banks' competition for savings.

Why do local banks seek yield at home rather than elsewhere? We examine whether local information and support for local borrowers help explain this choice. Among otherwise similar LGFV bonds, greater ownership at issue-quarter end by banks elsewhere in the issuer's province is associated with lower issue prices relative to public secondary-market valuations. These banks appear to obtain better investment terms, consistent with informed

purchasing.[3] For banks in the issuer's own city, the pricing pattern reverses: greater ownership accompanies higher issue prices and lower yields, reducing the borrower's financing cost. Greater same-city ownership is also associated with fewer subsequent issuer credit events, conditional on observed initial risk. Together, these patterns are consistent with informed selection and borrower support alongside reaching for yield.

Debt resolution eroded this source of yield. Average LGFV issuance spreads over Treasury yields fell from about 200 to 97 basis points, with greater compression in more intensively refinanced provinces. For a 20-percentage-point allocation, the annual product-return premium from home LGFV bonds fell from about 21 to 7 basis points, the largest decline among the asset categories. Banks with greater pre-program exposure to these bonds also experienced larger product-return declines, even after accounting for initial bank size. As refinancing progressed, institutions initially more tilted toward home investments reduced their home LGFV bond shares more.

Private credit retained considerably more yield than LGFV bonds: from 2023Q2 to 2024Q4, its average contractual yield fell from 6.6% to 5.9%, compared with 4.6% to 2.7% for bonds, widening the gap from 2.0 to 3.2 percentage points.[4] Home private credit also retained a positive product-return premium. Before resolution, national banks held 82.7% of the identified private credit in our sample, mostly outside their headquarters provinces. Afterward, larger local banks expanded their use at home. The largest third increased home private credit from 7.4% to 18.3% of disclosed bonds and private credit, more than offsetting their reduction in home bonds, while the smallest third kept this share below 1%. The difference persists after accounting for initial portfolios and common provincial conditions. Most of this private credit expansion remained within the LGFV sector. Smaller banks continued launching products but rarely included this substitute.

[3] Issue-quarter-end ownership proxies for participation at issuance; primary-market orders are not observed.

[4] The financing-amount-weighted series are in Figure 2. Comparing the same legal borrower and financing month, with maturity controls, also shows a widening gap: from 1.9 percentage points in 2023Q2 to 2.9 in 2024Q4 (Figure 3).

The unequal portfolio adjustment was accompanied by a widening gap in the returns offered to savers. After implementation, annual product returns at a local bank fell by about 10 basis points relative to those at a bank with twice its initial assets. Smaller banks also lost ground in the scale of their WMP business: balances grew about 11% more at the larger bank. Greater initial home-LGFV exposure predicts lower total balances in newly launched products, even after accounting for initial bank size; greater private-credit expansion accompanies stronger new business. Our model connects the incentive to offer high returns with the ability to replace lost yield. Reputation attracts cautious savers to larger banks, leaving smaller banks to compete more heavily on yield. Establishing private credit requires a fixed cost that a larger investor base can cover through additional fee income. Banks with this capacity can expand private credit when bond yields fall; banks for which entry remains uneconomical lose more of the return margin they use to compete.

These findings reveal a less-explored side of the two-way relationship between LGFV borrowing and shadow banking. Shadow banks finance local public investment, but the yield available on LGFV debt also shapes how they compete for savings. Stabilizing those borrowers can weaken smaller intermediaries' return advantage while encouraging larger institutions to finance the same sector through more opaque claims.

Our paper contributes to three strands of the literature. First, we connect shadow banks' role in financing local public investment to their competition for savings. Existing work traces off-budget fiscal expansion through LGFVs (Bai et al. 2016), the refinancing of stimulus loans through bonds and shadow banking (Chen et al. 2020), and the crowding out of private investment (Huang et al. 2020). Related studies show how deposit competition (Acharya et al. 2025), liquidity rules (Hachem and Song 2021), and monetary tightening (Chen et al. 2018) encouraged banks' shadow-banking activities. Implicit guarantees also shaped the pricing and funding of trust products and WMPs (Allen et al. 2023; Huang et al. 2023).

More broadly, research explains shadow banking through demand for claims perceived as safe (Gennaioli et al. 2013), regulation and technology (Buchak et al. 2018), and competition

with insured deposits (Xiao 2020); low interest rates can also change funds' product choices and industry structure (Di Maggio and Kacperczyk 2017). We identify an investment channel of competition: LGFV bond yield helped smaller banks offset weaker franchises, while the scale required to replace that yield reshaped their position relative to larger intermediaries.

Second, we connect local specialization to competition on returns. Distance and organizational form shape lenders' use of borrower information (Petersen and Rajan 2002; Degryse and Ongena 2005; Hauswald and Marquez 2006; Agarwal and Hauswald 2010; Stein 2002; Berger et al. 2005). Geographic and industry concentration are associated with investment performance (Coval and Moskowitz 2001; Kacperczyk et al. 2005; Loutskina and Strahan 2011), while lending specialization makes credit relationships difficult to substitute (Paravisini et al. 2023). Performance-sensitive flows and redemption risk shape asset managers' incentives (Sirri and Tufano 1998; Chen et al. 2010; Goldstein et al. 2017); competition (La Spada 2018; Choi and Kronlund 2018) and regulatory constraints (Becker and Ivashina 2015) can encourage reaching for yield. We show how familiar local bonds supplied a return-based competitive strategy for smaller banks, and why replacing that yield required greater scale.

Third, we study how borrower support affects financial intermediaries. Governments influence credit allocation (Sapienza 2004; Khwaja and Mian 2005) and, under fiscal stress, banks' holdings of government debt (Becker and Ivashina 2018; Ongena et al. 2019). In China, creditor political power affects selective defaults (Gao et al. 2021), and expected government support shapes the pricing gap between state-owned and private borrowers (Geng and Pan 2024). Geng et al. (2025) trace the government-led LGFV credit cycle and show that debt resolution narrows local banks' subordinated-bond spreads while reducing profitability and market shares at banks in riskier cities. Our contribution is to identify the investments through which this borrower support changes competition in banks' off-balance-sheet savings businesses. Granular WMP holdings reveal both the source of the initial return margin—home LGFV bonds—and the higher-yielding substitute increasingly used by larger local banks. Initial bond exposure predicts subsequent return losses, while the portfolios show financing

shifting toward opaque private credit within the same borrower sector. The distinctive implication is that stabilizing borrowers can change not only intermediaries' performance, but also the instruments through which they compete and supply credit.

Section 2 introduces the market, policy, and data. Section 3 establishes the local investment strategy, its contribution to returns, investor demand, and evidence on information and borrower support. Section 4 studies yield compression and bond retreat. Section 5 examines unequal substitution, product returns, and WMP balances. Section 6 explains these findings with a model in which savers differ in risk aversion and private credit requires a fixed cost. Section 7 concludes.

2 Setting and Data

This section describes the WMP market and its place in household savings and shadow banking, the investments available to WMPs, the debt-resolution program, and the data we use.

2.1 Banks, products, and investment opportunities

A wealth-management product pools savers' money in a separately accounted investment portfolio run by a bank or its wealth-management subsidiary. For Chinese households, WMPs are a major alternative to bank deposits, and they form a central part of the country's shadow banking system (Acharya et al. 2025; Chen et al. 2020). Since the 2018 asset-management reform, products must report net asset values (NAVs) and cannot guarantee principal or returns.[5] Banks sell WMPs through branches and online platforms. National banks comprise large state-owned and nationwide joint-stock commercial banks, with extensive branch networks and nationwide name recognition. Local banks include provincial, city, and rural

[5]See the asset-management reform guidelines (2018).

institutions. They compete for savings with national banks and with local banks of different sizes.

WMPs invest primarily in fixed-income and similar assets. Permitted investments primarily include government and policy-bank bonds, deposits, negotiable certificates of deposit, and corporate debt.[6] Most readily accessible investments offer limited credit spreads, while the public high-yield corporate bond market is narrow (Schipke et al. 2019). Figure A1 shows the industry's asset mix, and Figure A2 traces its growth.

LGFV bonds provide one source of higher yield. LGFVs are state-owned companies that finance local infrastructure and public services. They borrow in their own names, separately from formal government debt. Repayment depends on project assets, local fiscal resources, and expected government support (Hendy et al. 2024; Yang and Jiang 2021). Favorable issuance review helped qualifying vehicles obtain bond financing.[7] In our pre-program sample, LGFV issuers were smaller and their bonds offered higher issuance yields than those of other issuers. Median FY2021 assets were RMB 28.4 billion, compared with RMB 35.3 billion for other nonfinancial issuers and RMB 52.9 billion for financial issuers (Table A3, Panel E). Average fixed-rate issuance yields were 4.26%, 3.03%, and 2.77%, respectively. The LGFV yield differences persist after controlling for issue characteristics and ratings (Table A15). All eligible investors can buy these bonds, but local banks can draw on lending, deposit, and payment relationships to assess familiar but opaque borrowers.

WMPs can also finance the same borrowers through private credit, classified by Chinese regulators as non-standard debt. These investments include identified trust loans, leasing receivables, and claims held through trust or asset-management plans. Unlike buying marketable bonds, investing in private credit requires arranging, documenting, and monitoring

[6] Article 35 of the commercial-bank WMP rules lists permitted investments (2018).

[7] The enterprise-bond review notice (2013) provided accelerated and simplified review for a qualifying urban investment company's first application when its debt-to-assets ratio was below 50%. Exchange-market corporate bonds and interbank corporate debt instruments follow separate review and registration frameworks (2021; 2015).

a contract without a continuously observable trading price. These activities entail costs even for a small investment. The same borrower can issue both bonds and private credit.

The WMP–LGFV bond link is economically important for both markets. At year-end 2024, we estimate that WMPs held about RMB 3.1 trillion in LGFV bonds, representing 10.2% of total WMP net assets and 28.1% of outstanding LGFV bonds (Figure A5). These industry totals are extrapolated from disclosed holdings; Internet Appendix D.8 explains the calculation and its denominators.

2.2 Debt resolution and the linked data

The policy announcement of July 24, 2023 called for comprehensive local-debt resolution. Provinces began issuing special refinancing bonds in October, raising about RMB 1.4 trillion in 2023Q4 (Xinhua News Agency 2023; Wang 2023; Wong 2024). The proceeds repaid selected obligations as part of a broader program that also included loan restructuring and restrictions on new borrowing. These refinancing bonds are provincial-government obligations, distinct from the LGFV bonds held by WMPs. An expansion announced in November 2024 added RMB 6 trillion of debt-swap capacity for 2024–2026 (Ministry of Finance 2024). The program strengthened expected government support without formally guaranteeing every LGFV claim. The resulting spread compression can raise the prices of existing bonds while reducing the yields available when banks reinvest. We date implementation to 2023Q4, or October in monthly tests.

We start from Wind's product-level WMP holdings and link them to NAVs, bond issuance, and borrower-side private credit disclosures from a commercial data vendor. The holdings data identify each product's largest disclosed positions at the individual-asset level.[8] Quarterly disclosures over 2022Q1–2025Q3 contain 2,279,252 holding-level records for 60,396 products, covering 253,484 product-quarters. The sample includes 171 banks headquartered in 30 provinces: 18 national banks (six state-owned and twelve joint-stock) and 153 local banks

[8]The headline allocation and return-premium findings survive disclosure controls and alternative exposure denominators (Appendix E.2).

(100 provincial or city commercial banks and 53 rural banks). Tables A1–A3 define the variables and report sample coverage and descriptive statistics.

3 Local Credit and Competition for Savings

Local banks compete for savings against national banks with stronger reputations and broader distribution networks. A return advantage can offset that disadvantage. We begin by asking which investments provide this margin, whether it attracts savers, and why local banks seek it through familiar borrowers.

Our investment and performance outcomes concern banks' WMP businesses. Throughout, a bank's holdings and product returns refer to its WMPs, whether operated directly or through a wealth-management subsidiary. We assign each WMP its underlying bank's national or local category. An investment is home when its borrower is in that bank's headquarters province; either national or local banks' WMPs can hold home and non-home assets.

Throughout the product regressions, i denotes a product, a its issuing bank or subsidiary, b the underlying bank, q a quarter, m a month, and p_b the bank's headquarters province. Monthly returns distinguish separately priced NAV series. Variable names describe what is measured; units and timing are specified with each regression. Log bank size$_b$ always refers to fixed FY2021 bank size, whereas log bank assets in the control rows varies over time. Controls collects the covariates listed for that specification, so its contents can differ across tests. Fixed-effect subscripts identify the units held fixed; $\mathbb{1}\{\cdot\}$ indicates that a condition holds.

3.1 Overall asset allocation

We first compare LGFV bonds with private credit, the two higher-yielding WMP investments central to the analysis. To place them on a common basis, we express each holding as a share of the combined value of identified bonds and private credit with known borrower

locations, excluding central-government, central-bank, and policy-bank obligations. These shares describe this part of the WMP portfolio, not total WMP assets.[9]

The initial WMP portfolios differ sharply by bank type. Within these bond and private credit holdings, LGFV bonds accounted for 43.1% at local banks, compared with 12.6% at national banks; private credit accounted for 9.3% and 24.1%, respectively (Table D1). Local banks' WMPs thus relied more on marketable LGFV bonds, whereas national banks' WMPs already made substantial use of private credit.

The bank-type comparison leaves open how these investments vary with size. We therefore estimate the initial size gradient across national and local banks:

$$\text{Asset share}_{iq} = \beta_S \,\text{Log bank size}_b + \gamma' \,\text{Controls}_{iq} + \alpha_q + u_{iq}, \tag{1}$$

Here the outcome is the total LGFV bond share or total private credit share, in percentage points of the WMP bond and private credit holdings defined above. Log bank size$_b$ is log FY2021 underlying-bank interest-bearing assets in RMB trillion; bank and subsidiary products share that measure. A doubling increases Log bank size$_b$ by $\ln 2$. The ten variables in Controls$_{iq}$ cover product size, risk, and redemption terms; bank nonperforming loans, cost–income ratio, and net interest margin; and home-city GDP, growth, LGFV debt/GDP, and urban income.[10] The coefficient β_S compares allocations across bank sizes, holding these characteristics and quarter conditions fixed.

Across national and local banks, smaller banks' WMPs allocate more to LGFV bonds and less to private credit (Table 1). For a bank twice as large, the estimated LGFV bond

[9]The tables refer to this denominator as included debt holdings. It also excludes deposits, funds, and other assets. The introduction's 38.5% and 13.2% instead divide LGFV bond value by identified standard-bond value in the broader disclosed sample; those bond-only shares exclude private credit and retain government and policy-bank bonds. Neither comparison uses total WMP net assets, the denominator of the 10.2% industry estimate in Section 2. Appendix D.8 reconciles the samples and denominators; Table A11 reports the broader disclosed portfolio.

[10]The sample is 2022Q1–2023Q3. Category zeros remain; shares are not benchmark-adjusted. Regressions weight by included debt value and cluster by underlying bank. Fixed size replaces contemporaneous bank assets; quarter effects enter without bank effects or a local-bank indicator. Appendix D.9 gives the sample and weighting definitions.

share is about 4.9 percentage points lower and the private credit share 4.7 points higher. The underlying coefficients are −7.13 and 6.83 points per log point of bank assets. Private credit is also scarce at the smallest local banks: it accounts for only 1.1% of their WMP bond and private credit holdings, compared with 11.4% at the largest third (Table D1, Panel B). Smaller banks thus enter the program with portfolios more dependent on marketable LGFV bonds and little use of the privately arranged alternative.

3.2 Home-province allocation

The asset mix is only part of the specialization: local banks also concentrate investment in their home provinces. Home LGFV bonds account for more than one-fifth of the WMP bond and private credit holdings in every local-bank size group (Table D1, Panel B). Applying equation (1) to the home LGFV share, rather than the total LGFV share, gives a size coefficient of −5.46 percentage points per log point of bank assets (Table 1). Across all banks, a bank twice as large therefore has a home LGFV share about 3.8 points lower: RMB 3.8 less in home LGFV bonds per RMB 100 of analyzed WMP bonds and private credit, holding the controls fixed. The greater LGFV allocation at smaller banks thus has a substantial home-province component.

Large home holdings could reflect a large local bond market rather than a preference for nearby borrowers. To distinguish the two, we first measure geographic concentration within each asset class. For LGFV bonds, we divide the product's home LGFV holdings by all its LGFV holdings, then subtract the fraction of nationwide disclosed LGFV holdings issued by borrowers in that province. We call this difference within-category home bias. Its denominator is LGFV bonds alone, not the product's entire debt portfolio. The same calculation applies separately to other bonds and private credit. Over 2022Q1–2025Q3, we

estimate

$$\begin{aligned}\text{Within-category home bias}_{iq}^{n} = \beta_n \, \mathbb{1}\{\text{Local bank}\}_b \\ + \gamma_n' \, \text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}.\end{aligned} \tag{2}$$

Here n identifies the asset class. Zero within-category home bias means that, within that class, the product invests in its home province at the nationwide portfolio weight; a positive value means it invests more at home. The local-bank indicator identifies products issued by local banks or their subsidiaries, so β_n measures their additional within-category home bias relative to national-bank products. Only products holding the asset class enter, since the within-class ratio is otherwise undefined.[11]

Local banks' WMPs invest disproportionately at home (Table 2). The LGFV coefficient of 0.3205 means that their within-category home bias is 32.0 percentage points greater than that of national-bank products. For every RMB 100 invested in LGFV bonds, this represents about RMB 32 more invested at home, relative to each province's nationwide portfolio weight and holding the controls fixed. The analogous differences are 16.1 points within other bonds and 48.9 points within private credit. The private-credit estimate measures geographic concentration among products holding that asset class. Local banks' specialization is therefore geographic as well as a choice of asset class.

Product design provides another way to see how local banks compete. Periodic-open and closed-end products account for 80.2% of their product assets, compared with 47.3% at national banks (Table A6). These products restrict withdrawals, giving banks more room to hold investments that are difficult to sell quickly. Daily-open products instead compete more directly on access to cash.

To connect withdrawal terms to investment, we need to know how much of the portfolio goes to home borrowers, not just where a category's holdings are located. We therefore

[11] Controls are product size, risk, withdrawal terms, disclosure count and coverage; the bank's average product term and risk; bank nonperforming loans, cost–income ratio, net interest margin; and home-city GDP growth. Errors cluster by headquarters city–quarter. Appendix D.9 defines both denominators.

change the denominator. For LGFV bonds, we now divide home LGFV holdings by all the product's analyzed bonds and private credit, rather than by LGFV bonds alone. The benchmark changes accordingly: nationwide holdings of LGFV bonds from that province are divided by nationwide holdings of all analyzed bonds and private credit. Portfolio home overweight is the product's allocation minus this benchmark.[12] This portfolio-level measure captures both the amount invested in the asset class and its geographic concentration. We estimate

$$\begin{aligned}\text{Portfolio home overweight}^{n}_{iq} = \beta_1 \, \mathbb{1}\{\text{Local bank}\}_b \\ + \beta_2 \, \mathbb{1}\{\text{Restricted redemption}\}_i \\ + \beta_3 \, \mathbb{1}\{\text{Local bank}\}_b \times \mathbb{1}\{\text{Restricted redemption}\}_i \\ + \gamma' \, \text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}.\end{aligned} \tag{3}$$

$\mathbb{1}\{\text{Restricted redemption}\}_i$ identifies products without daily withdrawals. Unlike within-category home bias, Portfolio home overweight$^{n}_{iq}$ is defined even when the product holds none of class n: its raw allocation is zero before subtracting the benchmark, so these products remain in the sample. The coefficient β_1 measures the local–national allocation gap among daily-open products; β_3 measures how much wider that gap is among restricted-redemption products.[13]

The estimated interaction is 20.0 percentage points when home investments are pooled across asset classes (Table A7). Per RMB 100 of analyzed bonds and private credit, the local–national gap in home investment above the nationwide benchmark is therefore RMB 20 larger among restricted-redemption products than among daily-open products, holding the controls fixed. Of this additional gap, RMB 7.7 is in LGFV bonds and RMB 7.8 in

[12] For example, consider a WMP with RMB 100 in analyzed bonds and private credit, including RMB 20 in LGFV bonds, of which RMB 10 is from home. Its within-LGFV home share is 50%, but its home LGFV allocation across the analyzed debt portfolio is 10%. Each measure subtracts the matching nationwide benchmark. Neither denominator is total WMP assets.

[13] The controls are those in equation (1), excluding redemption, which enters separately, and adding time-varying log bank assets. The sample is 2022Q1–2025Q3. Quarter effects enter; errors cluster by bank city–quarter. Appendix D.3 gives the construction.

private credit. The comparison is across redemption types, conditional on quarter effects and the other controls. Local banks' greater use of withdrawal restrictions thus accompanies greater investment in home borrowers: their product terms and portfolios support the same return-based strategy.

The geographic and product-design evidence shows how local banks use these investments to differentiate their WMPs: they concentrate on home borrowers and offer withdrawal terms that accommodate less liquid holdings. For the smallest banks, substantial home LGFV positions coexist with little private credit and few accessible high-yield alternatives. We next ask whether their reliance on home bonds delivers a return margin to savers.

3.3 Local credit and product returns

Did these portfolios help local banks offer higher returns before debt resolution? We compare product returns across otherwise similar portfolios with different asset allocations, distinguishing home LGFV bonds from bonds elsewhere and from private credit. To compare initial return premiums with their subsequent changes on the same basis, we estimate one model that allows the allocation–return relationship to differ before and after implementation. This subsection focuses on the initial premiums. We relate monthly returns to holdings at the preceding quarter end:

$$\begin{aligned}\text{Monthly return}_{im} = {} & \beta' \,\text{Asset shares}_{i,h(m)} \\ & + \delta' \,\text{Asset shares}_{i,h(m)} \times \mathbb{1}\{\text{Post}\}_m \\ & + \gamma' \,\text{Controls}_{i,h(m)} + \lambda\, \mathbb{1}\{\text{Local bank}\}_b + \alpha_m + \varepsilon_{im}.\end{aligned} \tag{4}$$

The outcome Monthly return$_{im}$ is the change in log distribution-adjusted NAV, in basis points. In this specification i is a separately priced NAV series, assigned its product's holdings. The five fractions Asset shares$_{i,h(m)}$ are home LGFV bonds, home private credit, home other bonds, non-home LGFV bonds, and non-home private credit, relative to included

debt holdings. Non-home other bonds are omitted. Holdings and controls are measured at the preceding quarter end $h(m)$; June holdings explain July–September returns. Post starts in October 2023. Column 1 of Table 3 includes product controls; column 2 adds bank and home-city controls on the same sample and is our baseline.[14]

Before debt resolution implementation, Post equals zero, so β measures each asset's return premium relative to non-home other bonds. These are the coefficients we interpret here. The interactions δ allow the premiums to change afterward; Section 4.3 examines those changes and the remaining premiums $\beta + \delta$. For a 20-percentage-point allocation replacing non-home other bonds, the annualized comparison is 12×0.20 times the coefficient (Table 3).

Home LGFV bonds provide the largest estimated pre-program return premium (Table 3). Reallocating 20 percentage points from non-home other bonds to home LGFV bonds is associated with about 21 annual basis points more product return, compared with about 7 points for LGFV bonds from other provinces. The raw monthly coefficients are 8.852 and 3.041 per unit share. The home premium is almost three times the non-home premium, and their difference is statistically significant.[15]

Private credit also has a larger return premium at home: the same 20-point allocation corresponds to about 16 annual basis points for home claims and 7 for claims elsewhere, from raw coefficients of 6.765 and 2.776. Home other bonds contribute little, with a coefficient of 0.506. This pattern ties the return margin to LGFV bonds and private credit rather than to home investment generally. Home LGFV bonds also have enough weight to matter: their initial mean allocation is 26% and the 90th percentile 72% (Table A3).

We also ask whether higher reported returns came with more frequent visible losses. Using the full-period specification in equation (D.10), we relate reported volatility and downside

[14]The eleven controls add time-varying log bank assets to the ten controls in equation (1). The regression includes a local-bank indicator and month effects, with underlying-bank-clustered errors. It uses April 2022–December 2025 returns, excluding observations outside the 1st–99th percentile ranges of returns or trailing volatility. September 2025 holdings support October–December returns. The estimates measure conditional premiums in reported returns. Internet Appendix E examines measurement sensitivity.

[15]The pre-program home-minus-non-home LGFV contrast is 5.811 monthly basis points per unit share (bank-clustered SE 1.756; $p = 0.0012$), calculated from the same regression and its covariance matrix. At a 20-point allocation difference, the contrast is about 14 annual basis points.

outcomes to benchmark-adjusted home LGFV exposure, retaining the eleven controls, local-bank indicator, and month effects (Table A8). A 20-point exposure difference is associated with a 0.8-percentage-point lower probability of a negative month, against a 9.9% mean. Savers therefore observed relatively smooth performance alongside the return margin.[16]

3.4 Performance and investor demand

A return margin gives banks a commercial advantage when savers direct more money to better-performing products. We test that link by ranking products' prior-quarter returns within the same risk rating and withdrawal category, including products from both local and national banks. Following the performance–flow approach in Sirri and Tufano (1998) and La Spada (2018), we estimate

$$\begin{aligned}\text{Net flow}_{iq} = &\sum_{k=2}^{5} \beta_k \mathbb{1}\{\text{Return quintile}_{i,q-1} = k\} \\ &+ \gamma' \,\text{Controls}_{i,q-1} + \alpha_{g(i,q-1),q} + \varepsilon_{iq}.\end{aligned} \tag{5}$$

The outcome Net flow$_{iq}$ is asset growth net of investment gains, as a percentage of beginning assets. Return quintile$_{i,q-1}$ is prior return quintile, with the bottom fifth omitted; $g(i, q-1)$ identifies prior risk and withdrawal categories. Category–quarter effects hold common demand conditions fixed, and Controls$_{i,q-1}$ contains the lagged covariates listed below. The coefficient β_5 is the top–bottom flow difference.[17]

Flows increase across the performance ranks (Table A10). Top-quintile products receive subsequent net inflows 12.2% of beginning assets greater than bottom-quintile products—RMB 12.2 million for a product initially holding RMB 100 million—and are 8.1 percentage points more likely to receive positive flows. Comparing products of banks in the same

[16] Exposures lag one quarter; errors cluster by bank city–month. These outcomes measure fluctuations in reported NAVs. Appendix D.6 defines the outcomes and specification.

[17] The eleven controls and local-bank indicator enter; risk and withdrawal levels are absorbed. Product assets are dated at the flow quarter's start and other controls one quarter earlier. The sample is 2022Q3–2025Q3; errors cluster by underlying bank. Top and bottom groups differ by about one point in prior quarterly return. Appendix D.5 explains inferred assets and peer groups.

home province and quarter gives a similar 12.8% flow difference. This evidence explains the incentive behind the investment strategy: higher reported performance helps attract savings, and home LGFV bonds supplied local banks with a return margin.

3.5 Local information and borrower support

Higher returns help WMPs attract savers, but reaching for yield alone does not explain why local banks concentrate on home bonds: they can also buy LGFV bonds elsewhere. Local knowledge may help them obtain better investment terms or select borrowers more likely to repay. Close ties may also encourage them to support local borrowers by accepting lower yields. To examine these motives, we compare WMP investment by banks in the issuer's own city with investment by banks elsewhere in its province.

The pricing test asks whose interests are reflected in the terms at issuance. We measure each group's ownership at the end of the issuance quarter as a proxy for its participation at issuance, and compare the issue yield with CSI's public secondary-market valuation of the same bond. A lower issue price relative to that benchmark gives the buyer a higher yield for the same promised cash flows; a higher relative price lowers the borrower's financing cost. Our outcome, the yield concession, is the valuation spread minus the issue spread in basis points. A positive concession therefore means borrower-favorable pricing; a negative concession means investor-favorable pricing.[18]

For otherwise similar bonds j issued in the same city and quarter q, we estimate

$$\begin{aligned} \text{Yield concession}_{jq} = {} & \beta_C \, \text{Same-city ownership}_{jq} \\ & + \beta_P \, \text{Other-city provincial ownership}_{jq} \\ & + \gamma' \, \text{Controls}_j + \alpha_{c(j),q} + \varepsilon_{jq}. \end{aligned} \tag{6}$$

[18] Quarter-end ownership proxies for issuance participation and can include subsequent trading. The comparison uses CSI valuations rather than transaction prices.

The ownership shares are percentages of the bond's issued amount and enter together. Issuer-city–quarter effects $\alpha_{c(j),q}$ compare bonds issued in the same local environment; $c(j)$ denotes bond j's issuer city. Controls account for bond terms, issuer administrative level, and overall WMP participation.[19]

The two groups display opposite pricing patterns (Table A12). Greater ownership by banks elsewhere in the issuer's province is associated with lower relative issue prices—better terms for buyers at issuance, consistent with informed purchasing. Greater same-city ownership instead accompanies higher relative prices and lower yields. Replacing one percentage point of other-city provincial ownership with same-city ownership is associated with a 1.01-basis-point larger yield concession, holding their combined ownership fixed. Where proximity should make information especially useful, pricing favors the borrower instead. This contrast points to local borrower support, rather than better investment terms alone.

The distress test examines borrower selection alongside these pricing concessions. We compare subsequent credit outcomes among issuers with similar observed initial risk. We replace the pricing outcome in equation (6) with Issuer distress$_{jq}$, equal to 100 for an issuer's first credit event within eight quarters and zero otherwise. We retain both ownership shares, the controls, and city–quarter effects, and add the public valuation spread to account for initial risk.[20]

Greater same-city ownership is associated with less subsequent issuer distress, including relative to other-city provincial ownership (Table A13).[21] The two tests concern different decisions. Pricing indicates that same-city borrowers receive favorable financing terms; lower

[19]Controls are log issue amount, maturity, guarantee and call indicators, coupon, county- and prefecture-level issuer indicators, WMP holder count, and total WMP ownership. The sample is 2022Q1–2025Q3. Table errors are heteroskedasticity-robust; issuer-clustered inference is also reported for the ownership contrast.

[20]The sample comprises previously undistressed issuers of 2022Q1–2023Q4 bonds, with complete outcomes through 2025Q4. Distress concerns the issuer, not necessarily default on the held bond. Coefficients measure percentage-point differences in distress probability per ownership point.

[21]A one-percentage-point shift from other-city provincial to same-city ownership corresponds to about 0.10 percentage points less subsequent distress, against a 2.7% mean. The issuer-clustered p-value is 0.132 with issuer-city-by-quarter effects. Alternative event windows are in Table A4.

subsequent distress is consistent with informed borrower selection. A WMP can support a borrower on price while selecting one it expects to repay.

Does support also mean absorbing borrowers' refinancing needs as debts mature? We measure repayment pressure by scheduled LGFV principal repayments relative to the issuer city's 2020 budget revenue. We compare home- and outside-province WMP buyers of the same city's bonds in the same quarter. When more debt comes due, home buyers reduce net purchases and favor guaranteed bonds among newly added holdings relative to outside buyers (Table A9).[22] Local WMPs thus limit exposure and favor protection as refinancing needs rise.

The evidence points to informed investment alongside selective borrower support: local familiarity helps explain investment terms and borrower selection, while repayment pressure brings greater caution. Accepting lower yields on selected same-city issues can coexist with a product-return advantage from allocating more to higher-yielding LGFV bonds.

4 Debt Resolution and the Loss of Bond Yield

The debt-resolution program changed the yield available from this strategy by strengthening expected government support for LGFVs. Following the July 2023 announcement, provinces began issuing special refinancing bonds in October. Their proceeds repaid selected LGFV obligations, alongside measures to restructure loans and restrict new borrowing. These refinancing bonds were obligations of provincial governments, distinct from the LGFV bonds held by WMPs. The program reduced refinancing pressure and reinforced expectations of support.

For WMPs, stronger support had two different implications: falling spreads could raise the value of bonds already held, but reduced the yield available when funds were reinvested. The latter threatened the return margin that local banks used to compete for savers. We first

[22] Equations (D.7) and (D.8) in Appendix D.4 also include buyer–quarter effects; the net-purchase regression further absorbs persistent buyer–city ties. Newly added holdings need not be newly issued bonds.

document spread compression and compare bond yields with those on private credit, then examine the loss of the WMP bond-return premium and the response of WMP portfolios.

4.1 LGFV bond-spread compression

LGFV bonds lost much of the extra yield they offered over government bonds. Among newly issued fixed-rate bonds, the average spread over the Treasury yield fell from 200 basis points in 2022Q1–2023Q3 to 97 basis points in 2023Q4–2025Q4, roughly halving. Average LGFV issuance yields fell from 4.18% to 2.70%, while the corresponding Treasury yields fell from 2.18% to 1.72%. The decline thus went well beyond the general fall in government borrowing rates.[23]

The regional gap also narrowed. Provinces in the highest and lowest quintiles of initial spreads differed by 564 basis points in 2023Q1 but only 64 by 2025Q4 (Figure A4). Keeping group membership fixed, almost nine-tenths of the gap disappeared: formerly high-spread regions supplied much less additional yield for reinvestment.

A quarterly event study links this compression to the program (Figure 1). Comparing changes since 2023Q2, and accounting for province and quarter effects, spreads fall more where special refinancing is larger relative to pre-program GDP. The difference emerges with implementation and deepens through 2024; regressions using refinancing completed by each date show the same negative association (Table A5; Appendix D.2). The central implication for WMPs is that home LGFV bonds now offer substantially less additional yield to compete for savers.

4.2 Contractual yields on alternative claims

Replacing lost bond income is possible only if another accessible instrument retains yield. We therefore compare new LGFV bonds with new private credit contracts (Figure 2). From

[23] These are descriptive, equal-issue-weighted averages, without conditioning on WMP ownership. Appendix D.2 gives sample construction and the supporting regressions.

2023Q2 to 2024Q4, average private credit yields fall from 6.6% to 5.9%, while bond yields fall from 4.6% to 2.7%, widening the gap from about 2.0 to 3.2 percentage points. To see whether this difference reflects the financing instrument rather than different borrowers or maturities, we next hold the legal borrower and financing date fixed and control for tenor.

We use an instrument-by-quarter event study to compare the yield gap before and after the announcement. For contract or bond issue j of borrower f in month m, we estimate

$$\begin{aligned}\text{Yield}_{jfm} = \alpha_{fm} + \sum_q \beta_q \mathbb{1}\{\text{Private credit}\}_j \mathbb{1}\{q(m) = q\} \\ + \sum_q \gamma_q \,\text{Maturity}_j \mathbb{1}\{q(m) = q\} + e_{jfm}.\end{aligned} \tag{7}$$

The annual outcome Yield_{jfm} is in percentage points, $\mathbb{1}\{\text{Private credit}\}_j$ identifies private credit, and Maturity_j is maturity in years. Borrower–month effects compare the same legal borrower's two financing channels, excluding affiliates; quarter-specific maturity slopes adjust tenor. Thus β_q measures the quarterly PC-minus-bond yield gap, not a change in borrower composition.[24]

To present the event study relative to 2023Q2, Figure 3 plots

$$\delta_q = \beta_q - \beta_{q_0}, \qquad q_0 = 2023\text{Q2}, \qquad \delta_{q_0} = 0. \tag{8}$$

The same-borrower comparison confirms that private credit retains more yield. Zero on the plot means an unchanged gap from the 1.90 percentage points in 2023Q2, not equal yields. The gap widens from the second half of 2023, reaching 2.86 points in 2024Q4—0.96 points above the reference—and remaining 1.06 points above it in 2025Q3. Confidence intervals use the covariance of both gap estimates. This comparison measures the changing yield available on alternative claims against the same borrower. Its role is to identify the potential replacement investment, complementing the province-level refinancing evidence above.

[24]There are 17,303 observations for 589 borrowers over 2021Q1–2025Q3. Errors cluster by borrower province. Mixed-channel borrower-months identify the direct comparison; other months inform maturity controls. Appendix D.2 gives the matching rules.

Alternative maturity controls, financing-cell weights, a fixed borrower cohort, and inference methods retain the later widening (Tables D4–D5). The fixed-borrower rate series explain the pre-announcement trough in 2023Q1: bond yields temporarily rise while private credit yields remain stable (Figure D1; Appendix D.2). The plot therefore describes relative financing terms rather than a parallel-trends estimate of the policy's effect.

A broader reporting-issuer comparison, which includes financing by affiliates, provides a second check on the financing-channel difference. Within issuer and origination quarter, the private credit-minus-bond gap widens by 1.30 percentage points between 2022 and 2024, controlling for maturity and persistent instrument differences. On common issuer-years, it widens by an additional 1.04 points in the twelve priority provinces relative to elsewhere (Table 4; equations (D.3) and (D.5)). Both comparisons show that private credit retained yield within the local borrower sector. Section 5 examines which banks used this alternative.

4.3 Compression of the bond-return premium

Contractual yields measure the income available on new financing. We now ask whether the bond-yield decline also reduced the return margin that WMPs offered savers, and how that loss compares across investment categories. Section 3.3 established the initial premiums using β in equation (4); we now turn to the Post interactions δ in that same regression. Each interaction measures the change in an asset's premium, and $\beta + \delta$ gives its post-program premium. We retain the same annualized comparison for a 20-point allocation replacing non-home other bonds (Table 3).

Home LGFV bonds lose the most of the initial return margin. Their annual premium falls from 21.2 to 6.5 basis points for the common 20-point allocation: a loss of 14.7 basis points, or 69%. With product controls alone, the corresponding loss is 16.1 basis points, or 61%. The corresponding non-home LGFV premium falls from 7.3 to 1.9 basis points, a loss of 5.4. Home bonds therefore lose about 9.3 basis points more than LGFV bonds elsewhere.

The asset supplying the strongest initial home return advantage also experiences the largest erosion.[25]

Private credit retained more of its contribution to product returns. Its home premium fell from 16.2 to 10.5 annual basis points, a loss of 5.7; its non-home premium rose from 6.7 to 10.0. Home other bonds had little premium in either period, at 1.2 and 1.4 basis points. The decline in the home LGFV premium is statistically larger than the change in each of these categories.[26] In particular, home LGFV bonds lost about 9.0 annual basis points more than home private credit. This difference matters for substitution: private credit still offered a positive return premium from local borrowers after much of the bond premium had disappeared.

Does lower income from the bonds help explain the lost WMP return premium? We reestimate the return regression with and without a proxy for the portfolio's bond income: the holdings-weighted average yield at which its bonds were originally issued. We also interact this yield with Post, allowing its relationship with WMP returns to change after the program. Both regressions use exactly the same product-month observations. Once bond yields are taken into account, the estimated annual home-LGFV premium loss falls from about 15 to 6 basis points for the same 20-point allocation difference (Table E1). The smaller remaining decline is consistent with lower bond income contributing to the loss of the return advantage.

Could the result instead reflect when gains and losses appear in reported NAVs? A loss recorded in one month may be offset by a gain in another. We therefore replace one-month returns with average monthly performance over complete six- and twelve-month windows, using portfolio shares observed before each window. The home-LGFV premium still falls: the annualized loss is about 16 and 13 basis points, respectively, for the same 20-point allocation

[25]The raw monthly coefficients for home LGFV exposure are 8.852 before implementation and -6.133 for its Post interaction. Their sum gives the remaining premium; multiplying each comparison by 0.20×12 gives the annualized 20-point allocation comparison.

[26]Bank-clustered tests using the regression's full covariance matrix give $p = 0.014$ for the difference in changes between home and non-home LGFV bonds, $p = 0.015$ between home LGFV bonds and home private credit, $p = 0.002$ against home other bonds, and $p < 0.001$ against non-home private credit. The 20-point comparisons multiply the raw monthly coefficients by 0.20×12; they do not standardize the regressors.

difference. Home private credit retains positive post-program premiums of about 13 and 12 basis points (Table E2).[27] The contrast therefore persists over longer measurement windows, rather than appearing only in month-to-month reported returns.

Together, the financing and product-return comparisons establish the change in investment opportunities: home LGFV bonds lost much of their yield contribution, while private credit retained a return premium. We next ask how banks adjusted the bond positions that had supported their strategy.

4.4 Retreat from home LGFV bonds

As the return advantage narrowed, did banks reduce the home-bond positions that had supported it? We first ask whether WMPs initially more oriented toward home investments reduced home LGFV bonds more as refinancing proceeded. Comparing local-bank products in the same home province and quarter, we estimate

$$\begin{aligned}\text{Home LGFV bond share}_{iq} = \beta\, \text{Initial home overweight}_{a} \times \text{Refinancing}^{debt}_{p_b,q} \\ + \gamma' \,\text{Controls}_{iq} + \alpha_a + \alpha_{p_b,q} + \varepsilon_{iq}.\end{aligned} \tag{9}$$

The outcome is the product's home LGFV bond share in percentage points. Initial home overweight is the issuing institution's product-size-weighted 2022Q1–2023Q3 average home investment above provincial market weights, summed across bonds and private credit as defined in Section 3.2. Refinancing is cumulative special-refinancing bond issuance divided by 2020 LGFV debt in the underlying bank's home province. Institution and home-province–quarter effects hold persistent portfolio differences and common regional conditions fixed. Thus β asks whether initially more home-oriented institutions reduce home bonds more as refinancing rises.[28]

[27] These comparisons use the winsorized-return specifications in columns 2 and 4. Internet Appendix E gives the window dates, starting exposures, controls, and bank-clustered inference.

[28] Both interaction variables are fractions. The eleven time-varying controls enter; errors and wild-bootstrap draws cluster by home province. The refinancing denominator is LGFV debt, not GDP as in the spread tests.

The answer is yes (Table 5). For two institutions differing by ten percentage points in initial home overweight, a ten-point rise in refinancing relative to initial LGFV debt is associated with 0.46 percentage points more home-bond retreat at the more home-oriented institution (wild-bootstrap $p = 0.010$). Using strictly pre-announcement reliance or allowing it separate quarterly paths gives similar estimates. The reduction is therefore greater where WMP portfolios were initially more concentrated at home.

Could this retreat simply reflect fewer local bonds being issued? We compare home and outside WMP buyers of the same issuer city's bonds in the same quarter:

$$\begin{aligned}\text{LGFV bond share}_{acq} = \beta\, \mathbb{1}\{\text{Home buyer}\}_{ac} \times \mathbb{1}\{\text{Post}\}_q \\ + \mu_{ac} + \mu_{aq} + \mu_{cq} + \varepsilon_{acq}.\end{aligned} \tag{10}$$

Here a is the WMP-issuing institution, c the issuer city, and q the quarter. The outcome is that city's LGFV bonds as a percentage of the institution's total WMP bond holdings. Home identifies a local bank investing in its headquarters province. Buyer–city effects absorb persistent ties, buyer–quarter effects absorb buyer-wide changes, and city–quarter effects hold local bond-supply conditions fixed. The Home–Post coefficient therefore measures home buyers' additional retreat relative to outside buyers facing the same local supply.[29]

Home buyers reduce their allocation by an additional 0.45 percentage points per issuer city, about 12% of their initial mean allocation (Table 6). The decline is greatest in their own city; net buying and new purchases also fall, with larger purchase reductions in formerly higher-spread cities. The change thus extends beyond a common contraction in local bond supply and is concentrated in familiar, previously higher-yielding investments.

Figure 4 shows how this retreat develops over time. We aggregate destinations to provinces and replace Home–Post with quarterly home-buyer interactions, retaining buyer–destination, buyer–quarter, and destination–quarter effects. Relative to the 2023Q2 home–outside gap, the

[29] Home and Post levels and bank/city controls are absorbed. Errors cluster by bank city. Appendix D.4 defines the additional outcomes and event-study specification.

home allocation falls by 2.3 percentage points in 2024Q1 and 6.1 by 2025Q3; pre-announcement confidence intervals include zero. These are province-wide changes, not the per-city estimates above. The gradual adjustment can include letting bonds mature without replacing them as well as sales. Together, the tests show WMPs pulling back from the home bonds whose return advantage had diminished. The next question is which banks could replace that lost yield.

5 Bank Size, Private Credit, and Competition

Debt resolution changed the relative yields on two ways of financing familiar borrowers: home LGFV bonds lost much of their premium, while private credit retained one. The substitute offered a potential way to preserve the returns with which local banks competed for savings. We follow the adjustment from portfolio choices to product returns and WMP business. New products are informative at two stages: their holdings reveal whether banks used private credit when choosing fresh portfolios, and their total balances show how new business subsequently differed with initial bond reliance and private-credit expansion.

5.1 Which local banks shifted into private credit

We first locate the shift across bank types and borrower geography (Table 7). Before implementation, national banks held 82.7% of the identified private credit value in our sample, mostly outside their headquarters provinces. Local banks' share subsequently rose from 17.3% to 31.3%. Within local banks' portfolios, home private credit increased from 6.1% to 13.9% of included debt holdings, more than the increase outside their provinces. The higher-yielding instrument previously concentrated at national banks thus became a more important way for local banks to finance home borrowers.[30]

[30]These are descriptive, holding-value-weighted shares over 2022Q1–2023Q3 and 2023Q4–2025Q3. The national–local percentages describe each group's share of all private credit in the sample; the portfolio percentages divide home private credit by each group's included bonds and private credit.

Splitting local banks by their pre-program size reveals who made greater use of the substitute. Among banks observed in both periods, the largest third increased home private credit from 7.45% to 18.26% of included debt holdings. This increase more than offset their decline in home LGFV bonds: together, the two investments rose from 28.1% to 33.2%. At the smallest third, home private credit remained negligible, falling from 0.72% to 0.31%, while the combined share fell from 25.8% to 21.1%. The middle group substituted modestly. The large-bank adjustment thus involved expanding an instrument already present in their portfolios, while the smallest banks reduced home investments without building a comparable substitute. The size groups use fixed FY2021 assets, so these differences do not arise from reclassifying banks after the program.

To establish what larger banks were substituting into, we match the private credit borrowers to LGFV bond issuers and financing-platform names. These matches account for 85% of the largest local banks' post-program home private credit value (Internet Appendix D.1). The shift largely replaced bonds with less transparent financing within the familiar LGFV sector.

5.2 Bank size and the shift into private credit

The portfolio averages show where the substitute was used. We next compare changes across the three fixed size groups, accounting for product, bank, and local economic characteristics. For local-bank product i, issued by bank b, in quarter q, we estimate

$$\begin{aligned}\text{Private credit share}_{iq} = {} & \beta_M \, \mathbb{1}\{\text{Middle tercile}\}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_L \, \mathbb{1}\{\text{Largest tercile}\}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \gamma' \, \text{Controls}_{iq} + \alpha_b + \alpha_q + u_{iq}.\end{aligned} \tag{11}$$

The outcome is the product's home or total private credit as a percentage of its disclosed bonds and private credit. The smallest tercile is the comparison group; bank and quarter effects

absorb the fixed group indicators and the common Post indicator. We weight each product-quarter by its disclosed bond and private-credit value and cluster errors by underlying bank. Thus β_L measures the change in the largest group's investment-value-weighted private-credit share relative to the smallest group's change.

The largest group's home private-credit share rises by 8.21 percentage points more than the smallest group's share (Table 7a). The corresponding difference for total private credit is 10.46 points. A complementary comparison of the largest tenth with the remaining banks gives a 9.49-point difference in home private credit when each bank-quarter receives equal weight, while products within a bank-quarter remain weighted by their disclosed investment value (Table D3). This comparison retains bank and quarter effects and the same controls.

Figure 5 places this shift beside national banks' portfolios. National banks already invested heavily in private credit, predominantly outside their headquarters provinces. By contrast, the largest local banks' expansion was concentrated at home, while the smallest local banks remained largely absent from the instrument. The home and total-allocation panels thus distinguish local banks' replacement of familiar borrowers' bonds from national banks' geographically broader use of private credit.

The fixed groups make the large-bank shift visible. A continuous-size specification describes how the relationship between scale and allocation changes across the observed local banks. For local-bank product-quarters, we estimate

$$\begin{aligned}\text{Private credit share}_{iq} = \beta_S \,\text{Log bank size}_b + \delta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_q \\ + \gamma' \,\text{Controls}_{iq} + \alpha_q + u_{iq}.\end{aligned} \tag{12}$$

where Private credit share$_{iq}$ is product i's total or home private credit share in quarter q, in percentage points of included holdings. Log bank size$_b$ is its underlying bank's log FY2021 assets in RMB trillion, without standardization. The ten controls Controls$_{iq}$ are those in equation (1); quarter effects α_q remove common movements. The coefficient β_S measures how investment differed with bank size before implementation. The coefficient of interest, δ_S,

measures whether larger banks increased investment more afterward. Their sum gives the size difference remaining after implementation (Table D2).[31]

The size gradient widens after implementation. The home private credit slope rises from 4.72 to 12.18 percentage points per log point of assets, a change of 7.47 points. A local bank twice as large therefore increases its home private credit share by about 5.2 points more. The widening is significant under bank wild-bootstrap inference and remains positive with bank fixed effects. Larger banks expand private credit's portfolio weight disproportionately after bond yield compresses.

Does size still predict substitution after accounting for initial portfolios and provincial conditions? We compare local banks in the same province and quarter and allow both size and initial home investment to predict their response to refinancing:

$$\begin{aligned}\text{Private credit share}_{iq} = {} & \beta_S \,\text{Log bank size}_b \times \text{Refinancing}^{debt}_{p_b,q} \\ & + \beta_B \,\text{Initial home overweight}_a \times \text{Refinancing}^{debt}_{p_b,q} \\ & + \gamma' \,\text{Controls}_{iq} + \alpha_a + \alpha_{p_b,q} + \varepsilon_{iq}.\end{aligned} \tag{13}$$

The outcome is the product's home or non-home private credit share, in the same percentage-point units as above. Initial home overweight measures pre-program investment at home above provincial market weights across both bonds and private credit (Section 4.4). Refinancing is cumulative special-refinancing issuance divided by 2020 LGFV debt in the bank's headquarters province p_b, not the borrower's province. The second interaction allows initially more home-oriented institutions to adjust differently as refinancing proceeds. The coefficient of interest, β_S, then asks whether larger banks expand private credit more, holding that initial orientation constant.[32]

[31] The sample is 2022Q1–2025Q3. Observations are weighted by included debt value, and errors cluster by underlying bank. Quarter effects absorb the Post level. Bank-fixed-effect versions absorb fixed size and retain contemporaneous bank assets as a control. No standard-deviation normalization is used.

[32] Institution effects absorb persistent portfolio differences and the fixed size and home-overweight levels. Home-province–quarter effects absorb common regional conditions, including refinancing intensity. The eleven time-varying controls enter; observations are weighted by included debt value and errors cluster at the bank-city-by-quarter level.

Larger banks still expand more. As refinancing rises by ten percentage points of initial provincial LGFV debt, a bank twice as large increases its home private credit share by about 1.7 percentage points more (Table 8, Panel A). Replacing refinancing with Post in both interactions gives 4.1 points more expansion after implementation (Panel B). The size gradient thus remains within provincial markets after accounting for initial home investment.[33]

Does the instrument switch simply offset bond retreat, or leave larger banks with a greater combined allocation to home borrowers? We replace the outcome with the combined share of home LGFV bonds and home private credit, retaining size–Post and reliance–Post interactions, institution and regional time effects, and the controls. The size–Post coefficient is 4.76 (Table A14): a bank twice as large increases its combined home allocation by about 3.3 percentage points more. Larger banks therefore retain a greater portfolio share in the home borrower sector as they change instruments, while smaller banks make less use of the replacement.

5.3 Portfolio choice in new products

Existing portfolios may adjust slowly because of their maturity structure or withdrawal terms. New products let banks choose a fresh portfolio. We therefore compare the first eligible holdings of WMPs launched in 2023Q4–2025Q3 across the same fixed size groups, asking both whether private credit appears at all and how much of the new portfolio it occupies. This separates limited use of the instrument from an inability to launch products.[34]

The smallest banks did launch new products, but almost none used the substitute. Only two of their 253 new products held home private credit, compared with about 21% at the largest banks (1,118 of 5,248 products; Table 9). Home private credit represented just 0.78% of the smallest banks' new-product debt holdings, versus 27.5% at the largest

[33]The raw size–intensity coefficient is 24.21 per log point of assets and unit of the debt-based refinancing fraction, giving $24.21 \times 0.10 \times \ln 2 = 1.68$ percentage points. The raw size–Post coefficient is 5.85, giving $5.85 \times \ln 2 = 4.06$ points. Both versions retain the same controls and fixed effects.

[34]Launch is identified by the income-start date rather than first disclosure. Seven products disclosed before their recorded start date are excluded. Banks retain their FY2021 size groups. Portfolio shares are holding-value weighted; product counts are unweighted.

banks. Home LGFV bonds still accounted for 29.4% at the smallest banks. Thus, their new portfolios continued to use the accessible bond investment rather than taking up private credit. Requiring disclosure in the launch quarter or the next quarter gives the same pattern.

The substitution gap therefore extends to freshly chosen portfolios, rather than arising only from slow adjustment of old products. Launching another WMP did not by itself give small banks a replacement for lost bond yield. We next examine how initial bond reliance and subsequent substitution relate to product returns. Section 5.5 then uses total balances across newly launched products to examine whether initial reliance and substitution also predict differences in new business.

5.4 Relative product returns

To connect portfolio adjustment to the returns offered to savers, we distinguish the yield at risk from the replacement banks used. Initial home-LGFV exposure measures reliance on the investment whose premium contracted; the subsequent increase in private credit measures adjustment toward the substitute. We first ask whether banks more exposed before the announcement lost more return, holding initial size constant. Freezing exposure before the announcement keeps subsequent portfolio choices out of the reliance measure.

For underlying bank b, Initial home LGFV share$_b$ is the value of home-province LGFV standard bonds divided by total analyzed bond and private-credit holdings, aggregated over 2022Q1–2023Q2. Unlike Initial home overweight$_a$, it measures the actual bond allocation rather than a deviation from market weights. The outcome Monthly return$_{bm}$ is the bank's average WMP log return in month m, in basis points. Equations, tables, and plots retain monthly units; we multiply by 12 when discussing annualized magnitudes. We estimate

$$\begin{aligned}\text{Monthly return}_{bm} = {} & \beta_L \,\text{Initial home LGFV share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_m + \gamma' \,\text{Controls}_{b,h(m)} \\ & + \alpha_b + \alpha_{g_b,m} + u_{bm}.\end{aligned} \tag{14}$$

The coefficient β_L measures the post-program return change associated with initial home-LGFV reliance; a negative value means more exposed banks lost more return. The size interaction allows different return changes by FY2021 log bank size. Bank effects remove persistent performance differences; g_b denotes pre-program subsidiary status, whose month effects allow the two groups separate time paths. Where included, $\text{Controls}_{b,h(m)}$ contains eight prior-quarter bank and home-city controls.[35]

Initial reliance predicts where the return loss is greatest (Table E5). A bank with a ten-percentage-point higher initial home-LGFV bond share experiences about 13 basis points more annualized return decline, after accounting for initial size and the eight lagged controls (wild $p = 0.003$). The relationship also survives province-by-month effects and separate size slopes in every quarter. Thus the decline is concentrated at banks whose WMPs entered the program relying more heavily on the investment whose yield premium contracted.

A bank offering many products could disproportionately influence a product-level comparison, while requiring a holdings match could select particular products. Table E6 checks these two possibilities separately. On the same bank-months, with initial size, the eight controls, and both province-by-month and subsidiary-status-by-month effects held fixed, we give products or banks equal weight. We then broaden NAV coverage without changing initial exposure. Greater initial home-LGFV reliance predicts return losses under both weights and in the broader sample. These checks support the exposure result beyond a particular weighting scheme or set of holdings-matched products; Appendix E.3 reports the magnitudes and inference.

Initial reliance identifies where the loss of bond yield mattered most. A separate comparison asks whether returns also diverged along the size dimension on which private-credit substitution differed. Following the allocation design in equation (13), this product-level

[35]We average monthly log returns across NAV series within product-month, then across products within bank-month. The sample is July 2022–September 2025, excluding July–September 2023; Post begins in October 2023. Initial exposure is holdings-value weighted; returns give each bank-month equal weight. Errors and wild-bootstrap draws cluster by bank. Appendix E.3 gives the measurement checks.

comparison retains broad initial home overweight rather than the specific home-LGFV share:

$$\begin{aligned}\text{Monthly return}_{im} = {} & \beta_S \,\text{Log bank size}_b \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \beta_B \,\text{Initial home overweight}_a \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \gamma' \,\text{Controls}_{i,h(m)} + \alpha_a + \alpha_{p_b,m} + \varepsilon_{im}.\end{aligned} \tag{15}$$

The size measure Log bank size$_b$, prior reliance Initial home overweight$_a$, and home-province intensity retain their earlier definitions. The outcome Monthly return$_{im}$ is in basis points, and the eleven controls are measured at the preceding quarter end $h(m)$. The quarterly intensity series is assigned to the months of its quarter. Issuing-institution effects remove persistent performance differences; home-province-by-month effects compare banks facing the same regional conditions. A second version replaces intensity with $\mathbb{1}\{\text{Post}\}_m$ in both interactions. In each version, β_S measures the widening of the size-related return gap, conditional on prior reliance.[36]

The relative return gap widens alongside unequal substitution. After implementation, the annualized product-return difference between a bank and one twice as large widens by about 10.3 basis points (Table 8, Panel B). The intensity version gives the same direction: when refinancing rises by ten percentage points of initial provincial LGFV debt, the annualized return difference associated with a doubling of bank size widens by about 11.6 basis points (Panel A).[37] For perspective, applying the lost home-LGFV premium to local banks' average initial holdings implies about 19 annualized basis points of lost return. The return gap is therefore economically meaningful relative to the bond-income advantage that disappeared.

When did smaller banks begin losing ground in returns? We extend equation (15) to a size-by-quarter event study, replacing both refinancing interactions with log bank size

[36] These regressions are unweighted, and errors cluster at the bank-city-by-month level. Issuing-institution effects absorb fixed size and reliance; province-by-month effects absorb intensity or Post levels. A positive size interaction denotes a widening return gap in favor of larger banks.

[37] The raw size–Post coefficient is 1.2345 monthly basis points per log point of assets, giving $1.2345 \times \ln 2 \times 12 = 10.3$ annualized basis points. The size–intensity coefficient is 13.93, giving $13.93 \times 0.10 \times \ln 2 \times 12 = 11.6$ annualized basis points.

interacted with quarter indicators and omitting 2023Q3. The monthly return outcome, eleven controls, and fixed effects are unchanged; prior-reliance interactions are not included. Figure 6 plots the change in the size–return relationship in each quarter relative to 2023Q3, in monthly basis points per log point of assets.[38]

The event study shows the return gap developing gradually rather than jumping at the announcement. There is a short-lived positive estimate in 2022Q2, followed by pre-program fluctuations. The sustained widening begins in 2024Q3 and remains positive through 2025Q3. For a bank twice as large, the plotted changes imply about 33 annualized basis points more relative return in 2024Q3 and 45 in 2025Q3.[39] Initial exposure identifies where bond-income losses mattered most; the event study dates the subsequent widening of the size-related return gap.

The size comparisons follow the dimension along which substitution differed. We now measure substitution directly, relating the pre/post change in bank-average WMP returns to the increase in the bank's home-private-credit share, holding initial home-LGFV exposure, initial private credit, and FY2021 bank size constant. This DiD includes bank and month effects and the eight lagged bank and city controls; Appendix E.4 gives the specification and measurement. A ten-percentage-point larger private-credit increase is associated with about 11 basis points less annualized return decline (bank-clustered $p = 0.039$; Table E7, column 2). In the same regression, ten points more initial home-LGFV exposure predicts about 16 basis points more decline ($p < 0.001$). Initial bond reliance and subsequent substitution therefore enter the return comparison separately: one identifies the income exposed to compression, while the other captures banks' use of the higher-yielding replacement.

[38]The sample covers 2022Q2–2025Q3. Observations are unweighted. Errors cluster by underlying bank rather than bank city–month as in the average comparisons. Fixed size is absorbed by institution effects, and quarter indicators by province–month effects.

[39]The plotted coefficients remain in monthly units: 3.98 and 5.46 basis points per log point of assets. Multiplying by $\ln 2 \times 12$ gives the annualized doubling comparisons. These are quarter-specific differences relative to 2023Q3, not the average post-program difference reported above.

5.5 New business and total WMP balances

The return evidence identifies losses associated with initial bond reliance and return preservation associated with private-credit expansion. Do these differences also appear in the business banks build after the program? We first examine total balances in newly launched products, combining the number of products a bank launches with the balances they accumulate. This is our main measure of new WMP business. We then examine total outstanding WMP balances, which also include older products.

Section 5.3 showed that smaller banks continued launching WMPs but rarely used private credit in their new portfolios. To examine the business associated with these choices, we sum the balances of all observed products launched by bank b in quarter q, measuring each balance at the end of quarter $q+1$. Products are then roughly three to six months old. This cohort total captures new business across product designs, rather than the size of a typical product. For local banks, we estimate

$$\begin{aligned}\text{Log new-product total}_{bq} = {} & \beta_L \,\text{Initial home LGFV share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_H \,\text{Initial home PC share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \gamma' \,\text{Controls}_{b,q-1} + \alpha_b + \alpha_q + u_{bq}.\end{aligned} \tag{16}$$

Initial shares measure the bank's home-LGFV bonds and home private credit as fractions of its disclosed bonds and private credit over 2022Q1–2023Q2. Size is log FY2021 bank interest-bearing assets. The eight lagged bank and home-city controls match those in the return comparison. Bank effects α_b absorb persistent differences, and launch-quarter effects α_q absorb common changes; Post starts with launches in 2023Q4. Each bank–launch quarter receives equal weight, and errors cluster by bank. The sample covers launches in 2022Q2–2025Q2, excluding 2023Q3; unobserved cohorts are not assigned zero balances. Appendix E.5 gives the construction and sample requirements.

Column 1 of Table 11 uses 647 bank–launch quarters at 72 banks. A ten-percentage-point greater initial home-LGFV share predicts 14.4% lower new-product totals after the program, relative to the pre-program difference ($p = 0.004$). A bank twice as large has 16.8% greater relative growth in these totals ($p = 0.009$). Initial bond reliance therefore predicts weaker new business as well as larger return losses, even after accounting for bank size and initial private-credit holdings.

Column 2 adds the increase in the bank's home-private-credit share interacted with Post, using the same expansion measure as the return comparison. A ten-point larger increase is associated with 14.6% greater relative growth in new-product totals ($p = 0.034$), while ten points more initial home-LGFV reliance continues to predict a 15.3% decline ($p = 0.002$). The additional size coefficient falls from 0.224 to 0.138 once expansion is included. This comparison separates initial exposure to the disappearing bond yield from banks' subsequent use of the substitute: more exposure predicts weaker new business, whereas greater substitution accompanies stronger new business.[40]

We next examine the overall WMP business, combining an underlying bank's own products and its subsidiary's products. Unlike the new-product totals, these outstanding balances include existing products; unlike the earlier flow measure, they also reflect product entry and maturity. We first compare banks by initial size, then relate balance growth directly to private-credit expansion.[41]

For local bank b at half-year end s, we estimate

$$\begin{aligned}\text{Log WMP balances}_{bs} = \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_s \\ + \gamma' \,\text{Controls}_{bs} + \alpha_b + \alpha_s + \varepsilon_{bs}.\end{aligned} \tag{17}$$

[40] Column 2 uses 641 bank–launch quarters at 71 banks with sufficient pre- and post-program holdings to measure expansion. Re-estimating column 1 on these same observations gives nearly unchanged reliance and size comparisons: −14.1% for ten points more reliance and +16.2% for a doubling of size. Tables report unstandardized coefficients; percentage effects in the text use $100[\exp(\beta\Delta x) - 1]$.

[41] Appendix D.5 distinguishes the flow and balance measures and translates the linear flow coefficient into the units of the size-related return gap.

where Log WMP balances$_{bs}$ is log total WMP balances for local bank b at half-year end s. Size is log FY2021 bank assets in RMB trillion, and Post begins with the December 2023 report. Bank effects α_b remove persistent differences in WMP balances. Half-year effects α_s remove common growth. The coefficient β_S asks whether the proportional balance change after implementation is greater at initially larger banks. The baseline omits Controls$_{bs}$; subsequent columns add initial reliance, bank and home-city controls, or province-by-half-year effects.[42]

Smaller banks' weaker relative performance is accompanied by weaker WMP-business growth (Table 10). A bank twice as large experiences about 11% greater proportional WMP-balance growth after implementation, relative to the initial difference. The raw size–Post coefficient is 0.152 log balance points per log point of assets. Comparing banks within the same province and half-year gives 0.144, implying about 10.5% greater growth for a doubling of size. The result also remains with controls for initial reliance or bank and home-city characteristics. Across the fixed size groups, the largest third grows about 40% more than the smallest. [43]

A fixed-bank comparison makes the contraction at the smallest institutions concrete. Among the same 19 banks reporting in June 2023 and December 2024, aggregate WMP balances fall by 22.0%, with declines at all 19 banks.

To trace when smaller banks lost ground in WMP business, we replace the size–Post interaction in equation (17) with size interacted with each half-year reporting date, omitting June 2023. This event study retains the baseline's bank and half-year effects, bank-clustered errors, and absence of additional controls. Figure 7 plots the change in the size–balance relationship relative to June 2023, per log point of FY2021 bank assets. Zero denotes an

[42]The panel covers 2022H1–2025H2. Fixed size is absorbed by bank effects and Post by time effects. The baseline has no additional controls because the bank-level panel is sparse; errors cluster by bank. For an asset ratio a, the implied relative growth difference is $100[\exp(\beta_S \log a) - 1]\%$. Missing reports are not zeros. Balances measure WMP business scale, combining investment returns, product entry and maturity, and investor transactions. Internet Appendix E.1 gives coverage and measurement details.

[43]For a 0.981-log-point standard deviation of size, the baseline and province-time estimates imply about 16% and 15% greater relative growth. These are calculated from the raw coefficients; size is not standardized.

unchanged size–balance relationship relative to June 2023. The sample covers 2022H1–2025H2; the quarterly axis shows actual reports at Q2 and Q4, without interpolating Q1 or Q3 balances.

The two 2022 estimates are close to zero, and little of the later separation is present by December 2023. The size-related growth gap then increases during 2024 and is larger in 2025. For a bank twice as large, the path implies roughly 11% greater relative growth by December 2024 and 26% by December 2025, compared with June 2023.[44] The late-period difference exceeds the average 11% Post comparison because that average also includes the earlier reports when the gap was much smaller. The event study thus shows a growing business difference, broadly alongside the later separation in product returns.

We check whether this business divergence depends on who reports or on organizational form. Table E3 retains the same 29 banks at every date, separately excludes the final period, and allows banks with different organizational forms separate time paths. The positive size-related growth difference remains, as it does separately below and at or above RMB50 billion of June 2023 WMP balances. The result therefore extends beyond the comparison across this cutoff or between subsidiary and other banks. Appendix E.1 gives the sample definitions and results.

Finally, we apply the initial-characteristics and private-credit-expansion specifications to total WMP balances, with half-year reporting-date effects in place of launch-quarter effects (Table E7, columns 3–4). Both use the same eight controls and retain all eligible observations for this outcome. A ten-percentage-point larger increase in home private credit is associated with 7.2% greater proportional balance growth after the program ($p = 0.024$, column 4). Thus, the positive association between substitution and business growth extends from new-product cohorts to the overall WMP business.

The business comparisons reinforce the distinction between initial reliance and subsequent adjustment. Greater initial home-LGFV exposure predicts return losses and lower new-

[44] The raw log-balance coefficients are 0.146 and 0.334 per log point of assets. The doubling comparisons are $100[\exp(0.146 \ln 2) - 1] \approx 11\%$ and $100[\exp(0.334 \ln 2) - 1] \approx 26\%$. The joint test for the two 2022 coefficients gives $p = 0.094$.

product totals; greater private-credit expansion accompanies stronger returns, new business, and total WMP-balance growth. Together with the portfolio evidence, these patterns connect the erosion of bond yield to unequal adjustment and commercial outcomes. The model explains how a bank can have a strong incentive to replace lost yield yet lack the investor base needed to make private credit's fixed setup cost worthwhile.

6 A Model of Unequal Adjustment

The model explains why relying on high yields and being able to replace them are different advantages. A weaker reputation makes yield more important for winning savers, but a smaller investor base limits the fee income available to cover the fixed cost of establishing private credit. Banks that have paid this cost can expand their positions when bond yields fall; other banks must first find entry worthwhile. We take the documented home orientation of investment as given and derive how saver choice and these portfolio decisions interact.[45] Internet Appendix B derives the conditions and computes an illustrative full equilibrium; Internet Appendix C gives proofs.

6.1 Savers and banks

A bank in the model represents its WMP business, including any wealth-management subsidiary; the investments below belong to its products, not its own balance sheet. A unit mass of savers each invests one unit with a bank or an outside saving option. Savers differ in risk aversion γ_h, drawn from a nondegenerate distribution G on a bounded positive interval. Saver h obtains utility

$$U_{hb} = \gamma_h a_b + \beta\left(R_b - \frac{\gamma_h}{2} V_b\right) + \varepsilon_{hb} \tag{18}$$

[45] The two layers of demand follow Darmouni et al. (2024): savers choose institutions, which choose assets. Heterogeneous saver demand draws on Xiao (2020) and random-coefficient demand models (Berry et al. 1995); fee-based portfolio incentives follow La Spada (2018).

from bank b, and utility ε_{h0} from the outside option. Each saver chooses the alternative with the highest utility. Here R_b and V_b are expected excess return and return variance, so $R_b - \frac{\gamma_h}{2} V_b$ is the risk-adjusted return to saver h, and $\beta > 0$ measures how strongly choices respond to it. The positive return–flow relationship in Section 3 motivates this demand channel. The term a_b captures reputation and distribution. We assume cautious savers place greater weight on reputation, setting that weight equal to γ_h. The taste term ε_{hb} captures preferences such as a nearby branch or an existing account.[46] Among savers with risk aversion γ, the share choosing bank b is

$$s_b(\gamma) = \frac{\exp\left(\gamma a_b + \beta(R_b - \gamma V_b/2)\right)}{1 + \sum_j \exp\left(\gamma a_j + \beta(R_j - \gamma V_j/2)\right)}, \tag{19}$$

and the bank's market share is $s_b = \int s_b(\gamma)\, dG(\gamma)$. The sum in the denominator runs over the three banks; the term 1 represents the outside option. The banks differ in reputation: a national bank N, a large local bank L, and a small local bank S, with $a_N > a_L > a_S$. Holding portfolios fixed, a stronger reputation attracts more savers. Market share measures scale in the model; the empirical comparisons use the bank's log FY2021 assets, Log bank size$_b$, as predetermined scale.

Each bank invests its products' funds in one portfolio. It places a share x_b in LGFV bonds, a share n_b in private credit linked to the same borrower sector, and the rest in accessible low-yielding assets with riskless return r_f, with $x_b, n_b \geq 0$ and $x_b + n_b \leq 1$. Bonds and private credit earn positive expected excess returns μ_B and μ_{PC} over r_f, net of expected credit losses. Their variances are σ_B^2 and σ_{PC}^2. We assume a positive covariance $\sigma_{B,PC} > 0$ to capture their common exposure to the LGFV sector. The portfolio's expected excess return and variance are

$$R_b = \mu_B x_b + \mu_{PC} n_b, \qquad V_b = \sigma_B^2 x_b^2 + 2\sigma_{B,PC} x_b n_b + \sigma_{PC}^2 n_b^2. \tag{20}$$

[46] The taste shocks, including the outside-option shock, are independent of risk aversion and independently and identically distributed type I extreme value across alternatives and savers. This logit assumption yields equation (19); see Berry (1994) and, for bank deposits, Xiao (2020).

We assume $\sigma_{PC}^2 > \sigma_B^2 > \sigma_{B,PC}$: private credit has greater return variance than bonds, and the two are imperfectly correlated. All banks face the same asset returns and risks. Their portfolio differences arise from the savers they attract and whether entering private credit is worthwhile.

A bank earns a fee f per yuan of savings. Initiating private credit requires a one-time fixed cost $F > 0$ for sourcing, evaluation, and contracting. The bank pays it when it first arranges private credit, rather than for each product or contract, and the cost is sunk afterward. Taking rivals' portfolios as given, the bank chooses its portfolio to maximize

$$f\, s_b - F\, \mathbb{1}\{n_b > 0\}\, e_b, \tag{21}$$

where $e_b = 1$ if the bank has not arranged private credit before and $e_b = 0$ otherwise. Conditional on whether it enters, a bank maximizes the savings it attracts. Entry must also generate enough additional fee income to pay the fixed cost.[47]

6.2 Portfolio choice

A bank chooses its portfolio for the savers whose decisions it can influence. Holding rivals' offers fixed, a unit increase in the deterministic utility offered to type γ raises its share of that type at rate $s_b(\gamma)\big(1 - s_b(\gamma)\big)$. This response is small when the bank attracts almost none or almost all of that type. Weighting savers by this response gives their mean risk aversion at the margin:

$$\bar{\gamma}_b = \frac{\int \gamma\, s_b(\gamma)\big(1 - s_b(\gamma)\big)\, dG(\gamma)}{\int s_b(\gamma)\big(1 - s_b(\gamma)\big)\, dG(\gamma)}. \tag{22}$$

We write $\mathrm{Var}_b(\gamma)$ for risk-aversion variance under the same weights. At an optimum with positive holdings and $x_b + n_b < 1$, the marginal effect of each asset held on market share

[47] We compare two one-period portfolio-and-entry games, before and after an unanticipated reform. Banks maximize fee income in each state; prior entry carries over through $e_b = 0$. Thus, the model does not assign an additional continuation value to entering before the reform. A Nash equilibrium consists of portfolios and entry decisions that are best responses to the other banks' choices in that state.

must be zero. The resulting first-order conditions are

$$\mu_B = \bar{\gamma}_b\Big(\sigma_B^2 x_b + \sigma_{B,PC} n_b\Big), \qquad \mu_{PC} = \bar{\gamma}_b\Big(\sigma_{B,PC} x_b + \sigma_{PC}^2 n_b\Big). \tag{23}$$

Each condition equates an asset's expected excess return to its marginal risk, valued by the bank's marginal savers. The first condition applies when the bank holds bonds; the second applies when it also holds private credit. To make the portfolio choice explicit, we hold marginal-saver composition, and hence $\bar{\gamma}_b$, fixed. This benchmark maximizes their risk-adjusted return; the full demand problem also accounts for changes in who is marginal. Appendix B.6 solves that problem numerically. For the interior portfolios considered here, the bond-only regime sets $n_b = 0$ and gives

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b\, \sigma_B^2}. \tag{24}$$

If the bank arranges private credit, the bond condition gives $x_b = x_b^B - (\sigma_{B,PC}/\sigma_B^2) n_b$: private credit adds exposure to the same borrower sector, so the bank reduces its bond share by $\sigma_{B,PC}/\sigma_B^2$ for each unit of private credit. Substituting this bond share into the private credit condition gives

$$\mu_{PC} - \frac{\sigma_{B,PC}}{\sigma_B^2}\,\mu_B = \bar{\gamma}_b\left(\sigma_{PC}^2 - \frac{\sigma_{B,PC}^2}{\sigma_B^2}\right) n_b. \tag{25}$$

The left side is the return that private credit earns beyond the bonds it displaces, and the right side is the risk it adds beyond those bonds. Solving,

$$n_b^{PC} = \frac{\mu_{PC} - (\sigma_{B,PC}/\sigma_B^2)\,\mu_B}{\bar{\gamma}_b\Big(\sigma_{PC}^2 - \sigma_{B,PC}^2/\sigma_B^2\Big)}, \qquad x_b^{PC} = x_b^B - \frac{\sigma_{B,PC}}{\sigma_B^2}\, n_b^{PC}, \tag{26}$$

and private credit raises the risk-adjusted return to the bank's marginal savers by

$$g_b = \frac{\Big(\mu_{PC} - (\sigma_{B,PC}/\sigma_B^2)\,\mu_B\Big)^2}{2\bar{\gamma}_b\Big(\sigma_{PC}^2 - \sigma_{B,PC}^2/\sigma_B^2\Big)}, \tag{27}$$

We impose $\mu_{PC} > (\sigma_{B,PC}/\sigma_B^2)\mu_B$, so private credit earns more than the bonds it displaces and n_b^{PC} and g_b are positive. We also restrict attention to parameters for which both portfolios are interior.[48] With marginal savers held fixed, the bank chooses (x_b^{PC}, n_b^{PC}) if it arranges private credit and $(x_b^B, 0)$ otherwise.

6.3 Who serves cautious savers

Reputation determines not only how many savers a bank attracts but also which ones. For any two banks, equation (19) gives

$$\log \frac{s_b(\gamma)}{s_{b'}(\gamma)} = \beta\Big(R_b - R_{b'}\Big) + \gamma\Big[\Big(a_b - a_{b'}\Big) - \frac{\beta}{2}\Big(V_b - V_{b'}\Big)\Big]. \tag{28}$$

The bracket combines the reputation gap with the difference in portfolio risk. When it is positive, more risk-averse savers are relatively more likely to choose the more reputable bank. Portfolio choice depends on marginal savers. To obtain transparent comparative statics, we use the small-share approximation: when $s_b(\gamma)$ is small, the marginal-saver weight $s_b(\gamma)(1 - s_b(\gamma))$ is approximately $s_b(\gamma)$. The two saver distributions coincide in this limit. Holding portfolios fixed, reputation then raises marginal risk aversion and market share at the rates

$$\frac{\partial \bar{\gamma}_b}{\partial a_b} = \mathrm{Var}_b(\gamma) > 0, \qquad \frac{\partial \log s_b}{\partial a_b} = \bar{\gamma}_b \tag{29}$$

(Proposition B.1). Under the scale condition below, allowing portfolios to adjust preserves these signs. Higher-reputation banks attract more cautious savers; lower-reputation banks offer more bond exposure to savers willing to bear risk. The higher bond share in equation (24) therefore helps compensate for a weaker reputation in attracting savings.

[48] In each state, $0 < x_b^B < 1$, $x_b^{PC} > 0$, $n_b^{PC} > 0$, and $x_b^{PC} + n_b^{PC} < 1$. Appendix B.2 gives the constrained conditions when these inequalities do not hold.

6.4 Why private credit requires scale

A bank that has not yet arranged private credit enters only if the fees from the additional savers it attracts cover the fixed cost. Let s_b^B and s_b^{PC} be its maximized market shares without and with access to private credit, holding rivals' portfolios fixed. These exact choices allow marginal-saver composition to adjust. Taking entry at indifference, the bank enters if and only if

$$f\left(s_b^{PC} - s_b^B\right) \geq F. \tag{30}$$

This exact comparison is the entry rule. For intuition, in the small-share case the share gain is approximately $\beta s_b g_b$, so to first order the bank enters when

$$f\,\beta\, s_b\, g_b \geq F \tag{31}$$

(Proposition B.2). Here βg_b is the approximate proportional increase in demand from offering the substitute. Multiplying it by the investor base s_b gives the additional savings attracted, and multiplying again by f gives the fee income available to cover F. For a given proportional demand response, a larger bank attracts more money and earns more fees, while paying the same fixed cost.

Less cautious savers accept more risk in exchange for yield, making the substitute more valuable to the small bank's marginal savers. But the larger bank can offer it to a broader investor base. Under the scale condition, that broader base dominates, and its additional fee income is greater.[49] For an intermediate fixed cost, only the large local bank enters. The small bank also wants to offer higher returns, but doing so through private credit would cost more than the additional fees it earns. Launching a new product does not remove this bank-level cost. Figure 8 illustrates the resulting entry threshold.

[49] For the small-share, first-order gain, Proposition B.2 gives the sufficient condition $\mathrm{Var}_b(\gamma)(1+\beta R_b) < \bar{\gamma}_b^2$ along the reputation comparison. It allows the bond portfolio to adjust and ensures that the investor-base effect outweighs the change in marginal savers' preferences. The numerical example checks the exact fee gains directly.

6.5 Debt resolution and the three predictions

We represent the persistent loss of bond income by a fall in expected excess return from μ_B to $(1-\tau)\mu_B$, with $0 < \tau < 1$. To isolate this yield channel, the benchmark holds private credit returns, asset risks, and marginal-saver composition fixed. Expected returns are net of credit losses; the contractual-yield evidence motivates the direction of the shock, not a calibration of its magnitude. The predictions use the sorting, scale, and interior-portfolio conditions above; Internet Appendix B also allows bond risk to fall and solves a numerical equilibrium with changing saver composition. Throughout, Δ denotes the post-resolution change.

Prediction 1: smaller banks rely more on higher-yielding local bonds. Before entry into private credit, the optimal bond share and its response to reputation are

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b \sigma_B^2}, \qquad \frac{dx_b^B}{da_b} = -\frac{\mu_B}{\bar{\gamma}_b^2 \sigma_B^2} \frac{d\bar{\gamma}_b}{da_b} < 0. \tag{32}$$

Under the sorting condition, higher reputation raises marginal risk aversion, $d\bar{\gamma}_b/da_b > 0$, and lowers the optimal bond share. Smaller banks instead use more bond yield to attract their less cautious savers. Prior entry into private credit further reduces bond holdings through equation (26). Table 1 examines the predicted all-bank size gradient; Table 2 establishes that local banks pursue this strategy mainly through home borrowers.

Prediction 2: scale determines which banks substitute into private credit. For a first-time entrant, the small-share, first-order entry benchmark gives

$$\begin{aligned} n_b^* &= \mathbb{1}\left\{ s_b/\bar{\gamma}_b \geq C(\tau) \right\} n_b^{PC}, \\ C(\tau) &= \frac{2F\left(\sigma_{PC}^2 - \sigma_{B,PC}^2/\sigma_B^2\right)}{f\beta\left[\mu_{PC} - (\sigma_{B,PC}/\sigma_B^2)(1-\tau)\mu_B\right]^2}. \end{aligned} \tag{33}$$

Here s_b is the bond-only market share at the comparison date, and n_b^{PC} is the conditional position in equation (26) at the lower bond return. As τ rises, private credit becomes more valuable relative to bonds and the entry cutoff $C(\tau)$ falls. Under the scale condition, the

large local bank can cross this cutoff while the small bank remains below it. Earlier users do not pay again: they expand n_b^{PC} as bonds become less attractive, with the positive response derived in equation (B.17). The empirical allocation tests measure expansion across size groups, combining existing users and new participants (Tables 7a and 8). The new-product comparison shows that the difference also appears in freshly chosen portfolios, not only in legacy holdings (Table 9).

Prediction 3: smaller local banks lose more product return and savings business. Comparing the two initially bond-only local banks when only the large bank enters,

$$\begin{aligned}\Delta R_L - \Delta R_S = \tau(2-\tau)\frac{\mu_B^2}{\sigma_B^2}\left(\frac{1}{\bar{\gamma}_S} - \frac{1}{\bar{\gamma}_L}\right) \\ + \left(\mu_{PC} - \frac{\sigma_{B,PC}}{\sigma_B^2}(1-\tau)\mu_B\right)n_L^{PC} > 0.\end{aligned} \tag{34}$$

The smaller bank loses more bond income because it relied more on bonds before resolution; this is the first term. Only the larger bank obtains replacement income from private credit; this is the second. The small bank therefore loses part of the return advantage it used to compete against the larger bank's reputation. In the small-share benchmark, holding the national bank's offer fixed, this deterioration in its relative offer also reduces its relative share of savings (Proposition B.5). The empirical tests examine return losses by initial bond exposure in Table E5, relative returns by size in Table 8, and WMP-business growth in Table 10. Table 11 relates initial bond exposure and subsequent private-credit expansion to total balances in newly launched products. Table E7 directly relates actual private-credit expansion to changes in returns and balances, conditional on initial portfolios and size. Figure 6 and Figure 7 show the timing of the size-related differences.

7 Conclusion

In China's WMP market, LGFV bonds did more than finance local public projects: they gave smaller banks a source of yield with which to compete against larger rivals. Following the 2023

debt resolution, LGFV bond spreads compressed and the estimated product-return premium associated with home LGFV holdings fell by 69%. Banks initially more exposed to these bonds experienced larger return declines and lower total balances in newly launched products, even after controlling for initial size. Greater private-credit expansion was associated with stronger returns and new business.

Private credit retained higher contractual yields and a positive reported product-return premium. Larger local banks expanded these opaque claims, largely within the same LGFV borrower sector, while smaller banks substituted little and lost ground in relative product returns. WMP balances also grew more at larger banks. The model explains why reliance on bond yield can coexist with limited scope to replace it. Reputation attracts cautious savers to larger banks, leaving smaller banks to compete more heavily on yield. A larger investor base makes the fixed cost of establishing private credit worthwhile; once that capacity exists, banks can expand the substitute when bonds become less attractive. Banks for which entry remains uneconomical lose more of the return margin that helped them compete.

The findings reveal how borrower financing feeds back into competition among the intermediaries providing it. Government support changes not only borrower risk but also the investment opportunities through which banks compete for savings. When replacing lost yield requires scale, stabilization can erode smaller intermediaries' competitive edge and redirect shadow banking toward larger institutions and more opaque credit.

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Table 1
Initial credit specialization by bank size

Dependent variable	Total LGFV bond share (pp) (1)	Total private credit share (pp) (2)	Home LGFV bond share (pp) (3)	Home private credit share (pp) (4)
Log bank size	−7.130***	6.831***	−5.458***	0.622
	(1.778)	(1.803)	(1.123)	(0.572)
Log product assets	−3.028***	−3.480***	−0.677	−0.820***
	(0.828)	(0.914)	(0.431)	(0.282)
𝟙{Restricted redemption}	2.405	16.943***	−0.264	1.123
	(2.138)	(4.113)	(1.323)	(0.947)
Product risk rating	−3.065**	4.162	0.392	0.467
	(1.316)	(2.956)	(0.867)	(1.117)
Bank nonperforming-loan ratio	−7.968	10.855**	−3.180	−0.675
	(6.171)	(5.471)	(4.021)	(1.338)
Bank cost–income ratio	−0.245*	−0.582***	−0.085	−0.110**
	(0.131)	(0.184)	(0.092)	(0.049)
Bank net interest margin	−5.013	9.229	2.795	3.955
	(7.203)	(8.397)	(4.126)	(2.737)
Home-city GDP (RMB trillion)	0.064	−1.051	0.810	−1.932**
	(3.585)	(1.945)	(2.118)	(0.864)
Home-city GDP growth (%)	−0.381	−0.182	0.527	−0.012
	(1.211)	(1.046)	(0.768)	(0.508)
Home-city LGFV debt/GDP	27.483**	3.752	22.502***	5.983**
	(11.977)	(4.867)	(6.930)	(2.381)
Home-city income (RMB 10,000)	1.325	3.696**	−1.132	1.656**
	(3.213)	(1.595)	(2.049)	(0.767)
Wild p: log bank size	0.002	0.009	< 0.001	0.364
Observations	68,331	68,331	68,331	68,331
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column regresses the stated allocation share (pp) on unstandardized log FY2021 parent-bank interest-bearing assets in RMB trillion. A one-unit change is one log point, not one standard deviation. Zero holdings are retained. All ten other controls are displayed; fixed size replaces current bank assets. Quarter effects enter, but no national-bank indicator or bank effects. Weights use included debt value. Parentheses contain parent-bank clustered errors, grouping subsidiaries and aliases; stars indicate 10%, 5%, and 1% significance. Wild tests use 4,999 Rademacher draws. Appendix D.9 defines the comparison. Estimation sample: 2022Q1–2023Q3.

Table 2
Within-category home bias by bank type and asset class

Asset class	All credit	LGFV	PC	Other standard
Dependent variable	Within-category home bias (fraction) (1)	Within-category home bias (fraction) (2)	Within-category home bias (fraction) (3)	Within-category home bias (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	0.2713***	0.3205***	0.4885***	0.1611***
	(0.0092)	(0.0144)	(0.0192)	(0.0086)
Log product assets	−0.0004	−0.0019	0.0016	−0.0083***
	(0.0018)	(0.0031)	(0.0055)	(0.0021)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.0182***	−0.0189**	−0.0285	−0.0107*
	(0.0068)	(0.0090)	(0.0260)	(0.0058)
Product risk rating	0.0059*	0.0035	0.0067	−0.0021
	(0.0035)	(0.0070)	(0.0095)	(0.0043)
Log(1 + disclosed positions)	−0.0965***	0.0077	0.0001	−0.0629***
	(0.0145)	(0.0285)	(0.0128)	(0.0182)
Total disclosed weight (%)	0.0007***	0.0016***	0.0008***	0.0003*
	(0.0002)	(0.0003)	(0.0003)	(0.0002)
Log average product maturity	0.0203***	0.0337***	0.0178***	0.0217***
	(0.0027)	(0.0048)	(0.0058)	(0.0027)
Average product risk rating	−0.0513**	−0.1744***	0.1068***	−0.1189***
	(0.0231)	(0.0538)	(0.0369)	(0.0199)
Bank nonperforming-loan ratio (%)	−0.0371***	0.0296	0.0490*	−0.0142
	(0.0121)	(0.0216)	(0.0255)	(0.0116)
Bank cost–income ratio (%)	−0.0008*	−0.0003	0.0019**	−0.0001
	(0.0005)	(0.0007)	(0.0009)	(0.0005)
Bank net interest margin (%)	0.0939***	0.2127***	0.2460***	0.0412**
	(0.0203)	(0.0241)	(0.0330)	(0.0161)
Home-city GDP growth (%)	0.0188***	0.0310***	0.0158*	0.0088**
	(0.0046)	(0.0083)	(0.0086)	(0.0040)
Mean of dependent variable	0.1851	0.2562	0.1944	0.1305
Observations	186,731	108,313	43,803	158,812
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Bank city × quarter	Bank city × quarter	Bank city × quarter	Bank city × quarter

Notes: PC denotes private credit. This table tests whether local banks overweight home credit beyond what provincial market size predicts. Estimates use equation (2). The unit of observation is a product-quarter. The dependent variable is the product's home-province share within the asset class minus the home province's share of nationwide disclosed holdings in the class, Within-category home bias$^{n}_{iq}$, as a fraction. Each column uses products with positive holdings in the class. Local bank equals one for products of banks other than state-owned and joint-stock commercial banks. Product size and the number of disclosed positions enter as $\log(1+x)$. Menu term and menu risk are the average contractual term and numerical risk rating of the bank's products in the quarter, weighted by holding records. Standard errors clustered at the bank-city-by-quarter level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q3.

Table 3
Home credit exposure and product returns before and after the program

Dependent variable	Monthly NAV return (bp)	
	(1)	(2)
Home LGFV bond share	11.009***	8.852***
	(1.937)	(1.856)
Home LGFV bond share × 𝟙{Post}	−6.711***	−6.133***
	(1.603)	(1.501)
Home private credit share	7.517***	6.765***
	(1.454)	(1.291)
Home private credit share × 𝟙{Post}	−2.800*	−2.395
	(1.642)	(1.472)
Home other-bond share	1.271	0.506
	(1.611)	(1.552)
Home other-bond share × 𝟙{Post}	−0.317	0.077
	(1.617)	(1.538)
Non-home LGFV bond share	3.512**	3.041**
	(1.527)	(1.348)
Non-home LGFV bond share × 𝟙{Post}	−2.621*	−2.261
	(1.489)	(1.398)
Non-home private credit share	2.586**	2.776***
	(1.092)	(0.957)
Non-home private credit share × 𝟙{Post}	1.499*	1.382*
	(0.853)	(0.796)
𝟙{Local bank}	1.690*	0.193
	(0.914)	(0.677)
Log product assets	−0.649***	−0.567***
	(0.153)	(0.142)
𝟙{Restricted redemption}	3.875***	3.711***
	(0.676)	(0.625)
Product risk rating	−2.690***	−2.326***
	(0.729)	(0.607)
Log bank assets	—	−0.346
		(0.452)
Bank nonperforming-loan ratio	—	0.794
		(1.227)
Bank cost–income ratio	—	0.052*
		(0.027)
Bank net interest margin	—	0.106
		(1.045)
Home-city GDP (RMB trillion)	—	0.203
		(0.655)
Home-city GDP growth (%)	—	0.043
		(0.276)
Home-city LGFV debt/GDP	—	3.588**
		(1.690)
Home-city income (RMB 10,000)	—	−0.438
		(0.684)
Observations	554,555	554,555
Month fixed effects	Yes	Yes
Standard errors	Parent bank	Parent bank

Notes: Each column is one regression of monthly log adjusted-NAV returns (bp) on lagged exposure shares, Post interactions, controls, and month effects. Column 1 includes product controls; column 2 adds bank and home-city controls. Both use the column 2 estimation sample. Non-home other bonds are the reference. Shares are fractions; multiply coefficients by 0.2 for a 20-point allocation difference. Post starts in October 2023. In column 2, the implied post-program premiums are 2.719 (SE 1.613) for home LGFV bonds and 4.370 (0.996) for home private credit; the difference in their premium changes is −3.738 (1.522). These linear combinations use the full covariance matrix. Centrally administered issuers are never home. Parentheses contain parent-bank clustered errors; stars indicate 10%, 5%, and 1% significance. Internet Appendix E gives robustness checks. Estimation sample: Apr. 2022–Dec. 2025.

Table 4
Contractual-rate movements: private credit versus LGFV bonds

Panel A. Nationwide annual rate changes (pp)

Dependent variable	Borrowing yield (1)	PC yield (2)	Bond yield (3)
$\mathbb{1}\{\text{Private credit}\} \times \mathbb{1}\{\text{Year} = 2023\}$	-0.146^{***} (0.047)	—	—
$\mathbb{1}\{\text{Private credit}\} \times \mathbb{1}\{\text{Year} = 2024\}$	1.299^{***} (0.072)	—	—
$\mathbb{1}\{\text{Year} = 2023\}$	—	-0.089^{**} (0.037)	0.163^{***} (0.033)
$\mathbb{1}\{\text{Year} = 2024\}$	—	-0.245^{***} (0.053)	-1.436^{***} (0.042)
Observations	102,310	36,251	27,769
Issuer × quarter fixed effects	Yes	No	No
Issuer × debt type fixed effects	Yes	No	No
Maturity category fixed effects	Yes	Yes	Yes
Borrower × creditor fixed effects	No	Yes	No
Issuer × bond type × placement fixed effects	No	No	Yes
Standard errors	Issuer	Issuer	Issuer

Panel B. Additional changes in priority provinces (pp)

Dependent variable	PC–bond gap (1)	PC yield (2)	Bond yield (3)
$\mathbb{1}\{\text{Priority province}\} \times \mathbb{1}\{\text{Year} = 2024\}$	1.042^{***} (0.288)	0.317 (0.240)	-0.725^{***} (0.143)
$\mathbb{1}\{\text{Priority province}\} \times \mathbb{1}\{\text{Year} = 2023\}$	0.251 (0.210)	0.584^{***} (0.182)	0.333^{***} (0.119)
Province wild p: 2024 gap	0.0063	—	—
Observations	2,296	2,296	2,296
Issuer fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Standard errors	Issuer	Issuer	Issuer

Notes: Each column is one regression of annual contractual rates (pp). Panel A reports 2023 and 2024 changes relative to 2022; earlier year terms remain estimated but are not shown. Issuer, channel, and maturity controls follow Appendix D.2; the three samples differ, so their changes are not additive. Panel B uses the same issuer-years in all columns, with issuer and year effects and a separate priority-province 2023 interaction. Its gap is additive. Parentheses contain issuer-clustered errors; stars indicate 10%, 5%, and 1% significance. Estimation samples: Panel A, 2018–2024; Panel B, 2019–2024.

Table 5
Prior reliance, refinancing intensity, and home LGFV allocation

Dependent variable	Home LGFV bond share (pp) (1)	Home LGFV bond share (pp) (2)	Home LGFV bond share (pp) (3)
Initial home overweight × refinancing intensity	−46.360*** (7.881)	−47.923*** (7.774)	—
Initial home overweight (through 2023Q2) × refinancing intensity	—	—	−47.541*** (7.838)
Log product assets	0.332 (0.644)	0.344 (0.643)	0.345 (0.642)
𝟙{Restricted redemption}	5.012** (2.254)	4.956** (2.259)	4.957** (2.258)
Product risk rating	−0.335 (1.643)	−0.357 (1.688)	−0.351 (1.689)
Log bank assets	4.687 (12.670)	7.792 (13.659)	7.798 (13.618)
Bank nonperforming-loan ratio	6.494 (3.998)	5.223 (3.549)	5.343 (3.585)
Bank cost–income ratio	0.255 (0.336)	0.173 (0.363)	0.174 (0.360)
Bank net interest margin	−4.501 (3.353)	−4.600 (3.203)	−4.508 (3.227)
Home-city GDP (RMB trillion)	−20.580** (9.759)	−24.420* (12.047)	−23.486* (11.857)
Home-city GDP growth (%)	−1.488** (0.537)	−1.541*** (0.540)	−1.493** (0.542)
Home-city LGFV debt/GDP	−22.410 (23.961)	−20.956 (24.057)	−20.922 (24.018)
Home-city income (RMB 10,000)	39.096*** (9.309)	38.962*** (9.548)	38.647*** (9.388)
Wild p: initial home overweight × refinancing	0.010	0.043	0.037
Observations	83,051	83,051	83,044
Initial home overweight × quarter indicators	No	Yes	Yes
Institution fixed effects	Yes	Yes	Yes
Home province × quarter fixed effects	Yes	Yes	Yes
Standard errors	Home province	Home province	Home province

Notes: Local-bank product-quarters; the outcome is the home-LGFV share of included debt (pp). Initial home overweight is the initial benchmark-adjusted home allocation across all included debt, and intensity is cumulative refinancing divided by the bank's home province's 2020 LGFV debt; both are fractions. Columns 2–3 include reliance-by-quarter slopes, not tabulated. Column 3 measures reliance through 2023Q2. Ordinary controls are displayed; institution and home-province-by-quarter effects absorb the baseline terms. Province-clustered errors are in parentheses; wild tests use 27 provinces. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table 6
Home-province LGFV holdings after the program

Specification	Baseline	Own-city split	Net buying	New purchases	City-risk interaction	Pre-period trend
Dependent variable	LGFV holdings / bond book (pp)	LGFV holdings / bond book (pp)	Net holdings change / current book (pp)	Log(1 + new purchases)	Log(1 + new purchases)	LGFV holdings / bond book (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\}$	-0.453^{**}	—	-0.127^{**}	-0.158^{**}	0.203^{*}	—
	(0.193)		(0.053)	(0.079)	(0.117)	
$\mathbb{1}\{\text{Own city}\} \times \mathbb{1}\{\text{Post}\}$	—	-1.936^{***}	—	—	—	—
		(0.626)				
$\mathbb{1}\{\text{Rest of home province}\} \times \mathbb{1}\{\text{Post}\}$	—	-0.242	—	—	—	—
		(0.179)				
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\} \times$ city risk	—	—	—	—	-0.149^{***}	—
					(0.051)	
$\mathbb{1}\{\text{Home buyer}\} \times$ trend	—	—	—	—	—	0.003
						(0.051)
Mean of dependent variable	0.784	0.784	-0.052	0.937	0.945	0.816
Observations	73,089	73,089	73,089	73,127	72,196	32,540
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Bank × issuer city fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bank × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Issuer city × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Bank city	Bank city	Bank city	Bank city	Bank city	Bank city

Notes: This table tests whether local buyers retreat relative to other buyers facing the same issuer-city bond supply. Estimates use equation (10). The unit of observation is a bank, LGFV issuer city, and quarter over 2022Q1–2025Q3; the panel contains 73,635 cells for 170 banks and 248 issuer cities and covers the cities whose LGFV bonds a bank holds at some point, with zero holdings in quarters without positions in the city. Home equals one when a local bank invests in a city of its own province, and Post equals one from 2023Q4. Holdings are the bank's LGFV holdings in the city divided by its current bond book, in percentage points. Net holdings change includes changes in continuing positions as well as entries and exits, is divided by the current bond book, and is winsorized at the 1st and 99th percentiles. New purchases enter as $\log(1 + \text{amount in RMB million})$. Column 2 separates the bank's own city from the rest of its home province. Column 5 interacts Home buyer × Post with the city's pre-program risk, the issue-size-weighted mean CSI spread of its LGFV bonds with maturities of one to five years over 2022Q3–2023Q2, in percentage points, for the 209 cities with at least five bond-quarter observations; the Home buyer × post coefficient in that column is evaluated at a zero spread. Column 6 replaces Post with a linear quarterly trend and uses 2022Q1–2023Q3. Standard errors clustered by bank city are in parentheses; clustering by issuer city or by bank-city pair gives standard errors of 0.151 and 0.146 in column 1. The mean holdings share in home cells is 3.91 before 2023Q4 and 3.43 afterward. Bank and destination-city characteristics are absorbed by the bank-by-quarter and city-by-quarter fixed effects. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q3 (columns 1, 2, 3, 4, 5); 2022Q1–2023Q3 (column 6).

Table 7
Private credit holdings by bank type and bank size

Asset category	National banks Pre	National banks Post	Local banks Pre	Local banks Post
Panel A. Composition of included debt holdings, percent				
Home-province LGFV bonds	0.4	0.3	24.7	17.4
Home-province private credit	2.1	1.9	6.1	13.9
Private credit from other provinces	22.0	26.6	3.2	5.7
Product-quarters	34,595	59,691	36,194	57,516
Panel B. Share of the sector's holdings in local banks' products, percent				
Home-province private credit			61.6	83.1
All private credit			17.3	31.3
Home-province LGFV bonds			97.4	97.1
Panel C. Local banks present in both periods, by pre-program size tercile				
Home-province private credit, smallest tercile			0.72	0.31
Home-province private credit, middle tercile			3.45	4.74
Home-province private credit, largest tercile			7.45	18.26
Home-province LGFV bonds, smallest tercile			25.1	20.8
Home-province LGFV bonds, middle tercile			37.7	25.5
Home-province LGFV bonds, largest tercile			20.7	15.0

Notes: PC denotes private credit. This table locates the observed PC expansion by bank type, borrower geography, and local-bank size group. Product-quarter portfolios with positive included debt holdings, 2022Q1–2025Q3. Pre ends in 2023Q3; Post begins in 2023Q4. All panels use holding-value weights. Panels A and B retain all bank groups and classify Guangfa and its subsidiary as national. Panel A reports class shares of each group's included debt holdings; Panel B reports the fraction of sector holdings held by local banks' products. Panel C retains 102 underlying local banks observed in both periods, with 25, 38, 39 banks in the smallest, middle and largest groups. Groups are fixed by FY2021 interest-bearing assets, using the common pre-announcement local-bank universe described in Section 2. Bank financials are assigned to the bank's own products and those of its wealth-management subsidiary; alternative names do not create additional banks. Missing FY2021 size is not filled. Private credit includes trust loans, leasing and other identified private credit.

Table 7a
Bank size groups and private-credit allocation

Dependent variable	Home PC share (pp) (1)	Total PC share (pp) (2)
Middle size tercile × Post	1.536	1.355
	(1.404)	(1.752)
Largest size tercile × Post	8.212***	10.464***
	(2.906)	(3.376)
Log product size	−1.952**	−2.986***
	(0.862)	(1.091)
Non-daily-open product	5.778*	10.572***
	(3.019)	(3.749)
Product risk level	3.842	4.604
	(2.914)	(3.846)
Bank nonperforming-loan ratio	−1.448	−3.888
	(5.666)	(6.473)
Bank cost-to-income ratio	−0.891	−1.222*
	(0.581)	(0.725)
Bank net interest margin	9.202	10.547
	(5.931)	(6.540)
City GDP	3.357	20.673
	(16.132)	(19.721)
City GDP growth	−1.041*	−1.234*
	(0.611)	(0.669)
City debt-to-GDP ratio	24.427	28.158
	(22.700)	(22.963)
City urban income	17.075*	22.989*
	(10.240)	(11.976)
Log bank assets	−10.152	−25.650
	(15.857)	(19.351)
Observations	85,181	85,181
Bank fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Standard errors	Parent bank	Parent bank

Notes: Each column is a product-quarter regression for local banks. Private-credit shares are percentages of disclosed bonds and private credit, not total WMP assets. The omitted group is the smallest FY2021 bank-asset tercile; groups follow Table 7. Post begins in 2023Q4. Bank effects absorb fixed group indicators and quarter effects absorb Post. Regressions use disclosed bond and private-credit value as weights, so the estimates describe the allocation of investment value. The displayed product, bank, and city controls include contemporaneous log bank assets. Parentheses contain bank-clustered standard errors; stars denote 10%, 5%, and 1% significance. Internet Appendix D.9 gives complementary size comparisons. Sample: 2022Q1–2025Q3.

Table 8
Bank size, private credit substitution, and relative returns

Dependent variable	Panel A. Refinancing intensity			Panel B. Post-program		
	Home PC share (pp)	Non-home PC share (pp)	Monthly return (bp)	Home PC share (pp)	Non-home PC share (pp)	Monthly return (bp)
	(1)	(2)	(3)	(4)	(5)	(6)
Log bank size × refinancing intensity	24.205***	14.531***	13.925***	—	—	—
	(4.877)	(2.733)	(5.265)			
Initial home overweight × refinancing intensity	5.021	13.152***	−43.242**	—	—	—
	(10.341)	(4.398)	(20.927)			
Log bank size × 𝟙{Post}	—	—	—	5.853***	2.329***	1.234*
				(0.866)	(0.412)	(0.725)
Initial home overweight × 𝟙{Post}	—	—	—	7.157**	4.363***	−0.936
				(3.095)	(1.240)	(3.056)
Log product assets	−2.172***	−1.122***	−0.927***	−2.151***	−1.115***	−0.938***
	(0.364)	(0.155)	(0.124)	(0.363)	(0.154)	(0.125)
𝟙{Restricted redemption}	5.538***	4.252***	9.008***	5.456***	4.218***	8.995***
	(1.711)	(0.778)	(0.672)	(1.721)	(0.776)	(0.679)
Product risk rating	3.442***	0.907**	−2.175***	3.563***	0.961**	−2.252***
	(1.004)	(0.460)	(0.841)	(1.005)	(0.462)	(0.845)
Log bank assets	−30.179***	−23.281***	14.410*	−25.758***	−22.166***	14.556*
	(9.590)	(6.046)	(8.658)	(9.830)	(6.004)	(8.713)
Bank nonperforming-loan ratio	−2.621	−4.356***	−8.821**	−0.336	−3.246**	−7.674**
	(2.474)	(1.400)	(3.885)	(2.356)	(1.418)	(3.850)
Bank cost–income ratio	−0.417*	−0.407**	0.105	−0.370*	−0.376*	0.219
	(0.226)	(0.189)	(0.282)	(0.216)	(0.192)	(0.287)
Bank net interest margin	16.147***	1.141	6.847*	13.616***	0.567	6.640*
	(3.389)	(1.602)	(3.752)	(3.187)	(1.607)	(3.707)
Home-city GDP (RMB trillion)	51.899***	15.888***	9.886	37.848***	11.336***	12.392
	(9.602)	(4.129)	(12.485)	(9.750)	(4.132)	(12.768)
Home-city GDP growth (%)	−3.070***	−0.469**	0.011	−1.711***	0.181	0.043
	(0.717)	(0.219)	(0.835)	(0.563)	(0.247)	(0.855)
Home-city LGFV debt/GDP	3.490	13.626**	−9.979	−12.207	4.744	−11.188
	(13.251)	(5.383)	(32.452)	(12.462)	(4.921)	(31.374)
Home-city income (RMB 10,000)	−43.525***	−17.132***	−17.165	−21.873**	−11.227***	−11.239
	(10.743)	(4.531)	(14.391)	(10.434)	(4.209)	(15.128)
Observations	83,027	83,027	206,770	83,027	83,027	206,770
Institution fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	Yes	Yes	No	Yes	Yes	No
Home province × month fixed effects	No	No	Yes	No	No	Yes
Standard errors	Bank city × quarter	Bank city × quarter	Bank city × month	Bank city × quarter	Bank city × quarter	Bank city × month

Notes: Each numbered column is one regression. Size is unstandardized log FY2021 bank assets in RMB trillion; prior reliance is initial home overweight. PC denotes private credit. Columns 1–3 (group A) interact both with cumulative refinancing divided by the bank's home province's 2020 LGFV debt; columns 4–6 (group B) interact them with Post. Outcomes are allocation shares (pp) or monthly returns (bp). All eleven controls are displayed, with institution and home-province-by-period effects. Allocation columns weight by included debt value; return columns are unweighted. Standard errors cluster by bank-city-by-period; underlying-bank clustered checks retain the size interactions ($p \leq 0.024$). Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3 (columns 1, 2, 4, 5); 2022Q2–2025Q3 (columns 3, 6).

Table 9
Private credit in new products, by bank size

First eligible portfolio	Smallest tercile	Middle tercile	Largest tercile
New products	253	1,877	5,248
Parent banks	13	25	33
Products holding home private credit	2	108	1,118
Products holding any private credit	2	134	1,586
Home private credit share (%)	0.78	6.31	27.46
Non-home private credit share (%)	0.00	2.73	14.74
Total private credit share (%)	0.78	9.04	42.20
Home LGFV bond share (%)	29.36	27.94	11.11
Home private credit share: first disclosure within one quarter (%)	0.59	5.75	27.29
Products in that restriction	234	1,737	4,611

Notes: PC denotes private credit. This table compares the first observed portfolios of products launched after the program by local banks in each size tercile. Launches in 2023Q4–2025Q3 are identified from income-start dates rather than first disclosure, and each product contributes its first eligible portfolio. Shares are holding-value weighted percentages of included debt holdings; counts are unweighted. Size terciles use fixed FY2021 parent-bank interest-bearing assets and the common local-bank cutoffs. Products of a bank's wealth-management subsidiary are assigned to the parent bank. The final two rows restrict to products whose first portfolio is disclosed in the launch quarter or the next quarter. No regressions enter this table.

Table 10
Bank size and WMP business growth after the program

Dependent variable	Log WMP balance				
Specification	Baseline	Initial reliance	Controls	Province–time	Size terciles
	(1)	(2)	(3)	(4)	(5)
Log bank size × 𝟙{Post}	0.152***	0.171***	0.155***	0.144***	—
	(0.039)	(0.044)	(0.030)	(0.052)	
Initial home overweight × 𝟙{Post}	—	0.190	0.288***	—	—
		(0.126)	(0.095)		
𝟙{Middle size tercile} × 𝟙{Post}	—	—	—	—	0.184*
					(0.093)
𝟙{Largest size tercile} × 𝟙{Post}	—	—	—	—	0.335***
					(0.088)
Log bank assets	—	—	0.098	—	—
			(0.226)		
Bank nonperforming-loan ratio	—	—	0.223***	—	—
			(0.066)		
Bank cost–income ratio	—	—	−0.013	—	—
			(0.009)		
Bank net interest margin	—	—	−0.064	—	—
			(0.080)		
Home-city GDP (RMB trillion)	—	—	0.985***	—	—
			(0.248)		
Home-city GDP growth (%)	—	—	−0.010	—	—
			(0.011)		
Home-city LGFV debt/GDP	—	—	−0.751*	—	—
			(0.403)		
Home-city income (RMB 10,000)	—	—	0.000	—	—
			(0.164)		
Wild-bootstrap p	< 0.001	< 0.001	< 0.001	0.008	< 0.001
Observations	571	479	395	494	571
Bank fixed effects	Yes	Yes	Yes	Yes	Yes
Half-year fixed effects	Yes	Yes	Yes	No	Yes
Home province × half-year fixed effects	No	No	No	Yes	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

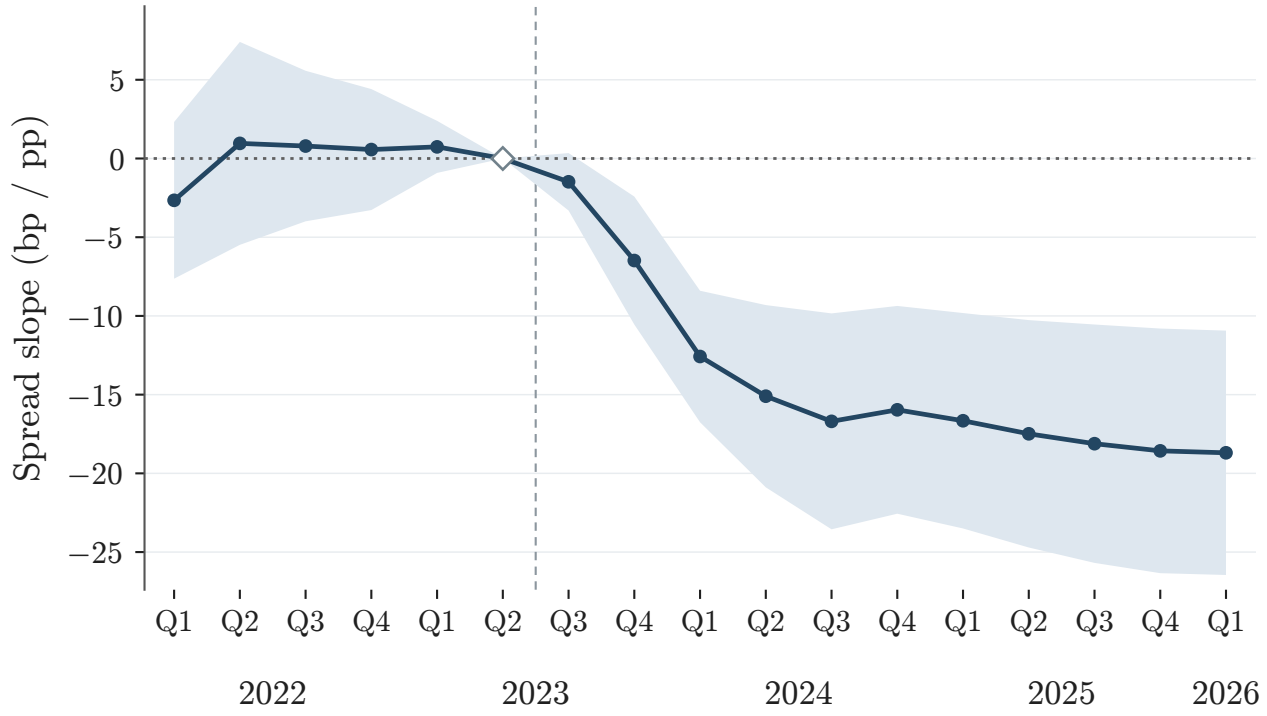
Notes: Each column is one bank-half-year regression. All columns use log positive observed WMP balances, combining the bank and subsidiary, with the source corrections described in Appendix E.1. Size is unstandardized log FY2021 bank assets in RMB trillion. Post begins at year-end 2023. The baseline includes bank and half-year effects, with no additional controls; later columns add reliance, the displayed bank and city controls, province-time effects, or fixed size terciles (smallest omitted). Missing reports are not zeros. Parentheses contain bank-clustered errors; stars indicate 10%, 5%, and 1% significance. Appendix E.1 discusses coverage. Estimation sample: 2022H1–2025H2.

Table 11
Initial bond reliance, private-credit expansion, and new WMP business

Dependent variable	Log new-product total	
	Initial characteristics (1)	+ Private-credit expansion (2)
Initial home LGFV bond share × 𝟙{Post}	−1.556***	−1.662***
	(0.521)	(0.528)
Log FY2021 bank size × 𝟙{Post}	0.224***	0.138
	(0.083)	(0.109)
Initial home private credit share × 𝟙{Post}	0.823	1.547**
	(0.554)	(0.711)
Home private credit share increase × 𝟙{Post}	—	1.363**
		(0.630)
Lagged log bank assets	−0.213	−0.213
	(0.735)	(0.735)
Lagged nonperforming-loan ratio	−0.459	−0.504*
	(0.285)	(0.285)
Lagged cost–income ratio	−0.018	−0.010
	(0.032)	(0.032)
Lagged net interest margin	0.268	0.252
	(0.296)	(0.289)
Lagged city GDP (RMB trillion)	1.554	1.697
	(1.037)	(1.025)
Lagged city GDP growth	−0.078	−0.080
	(0.051)	(0.052)
Lagged city LGFV debt/GDP	2.790***	3.033***
	(1.031)	(1.035)
Lagged urban income (RMB 10,000)	0.919*	0.894*
	(0.496)	(0.488)
Observations	647	641
Bank fixed effects	Yes	Yes
Launch-quarter fixed effects	Yes	Yes
Standard errors	Bank	Bank

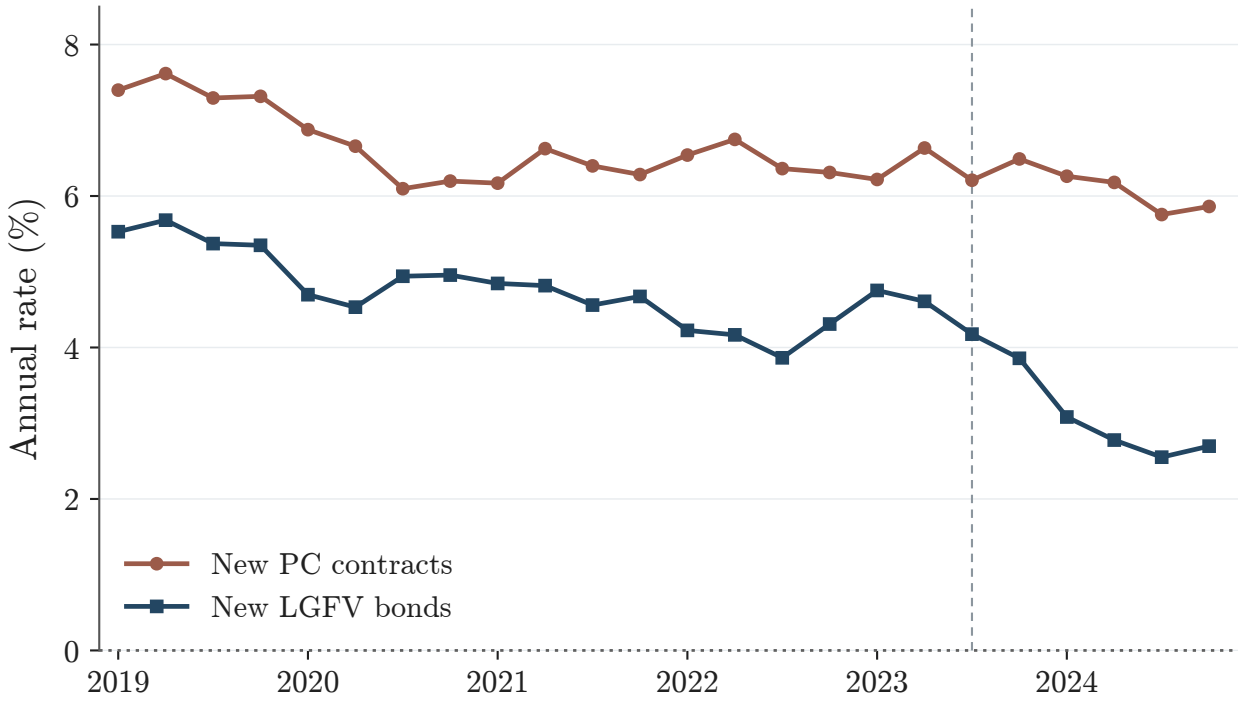
Notes: Each column is one regression of log total balances in products launched by a local bank in a quarter. Each product's balance is measured at the end of the following quarter; balances are summed before taking logs. Column 1 estimates equation (16); column 2 adds the home-private-credit share increase interacted with Post. Launches cover 2022Q2–2025Q2, excluding 2023Q3; Post begins in 2023Q4. Columns 1 and 2 cover 72 and 71 banks and 13,182 and 13,100 products, respectively. Initial shares use 2022Q1–2023Q2 holdings; expansion compares mean quarterly shares in that window with 2023Q4–2025Q3. Shares are raw fractions of disclosed bonds and private credit; size is raw log FY2021 interest-bearing assets. All slope coefficients are displayed. The eight controls lag one quarter; no product-composition controls enter. Each bank–launch quarter has equal weight, and unobserved cohorts are not zeros. Bank effects absorb fixed characteristics and quarter effects absorb Post. Parentheses contain conventional bank-clustered standard errors; stars denote 10%, 5%, and 1% significance. Appendix E.5 details construction and sample coverage.

Figure 1
Provincial LGFV spreads by final refinancing intensity



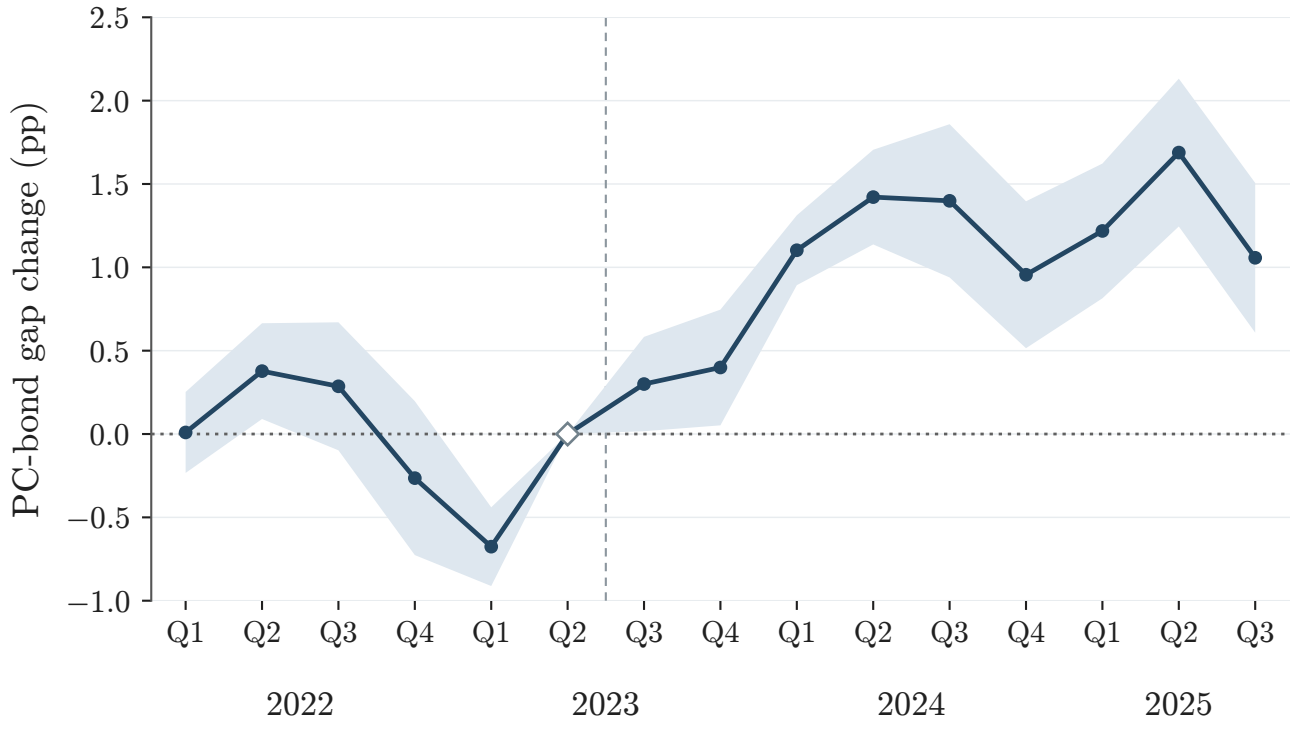
Notes: This event study plots the quarter-specific coefficients β_k in equation (D.1). Provincial LGFV spreads, in basis points, are regressed on final refinancing intensity interacted with quarter indicators, with province and quarter fixed effects; 2023Q2 is the omitted quarter. Final intensity is cumulative special-refinancing issuance at the end of the intensity series as a percentage of 2020 provincial GDP. Coefficients are basis points per percentage point of final intensity, without standardization. Spreads and intensity are winsorized at the 1st and 99th percentiles, and the sample covers 31 provinces over 2022Q1–2026Q1. The shaded bands are pointwise 95% confidence intervals based on standard errors clustered by province, and the dashed line marks 2023Q3.

Figure 2
Contractual rates on new LGFV financing



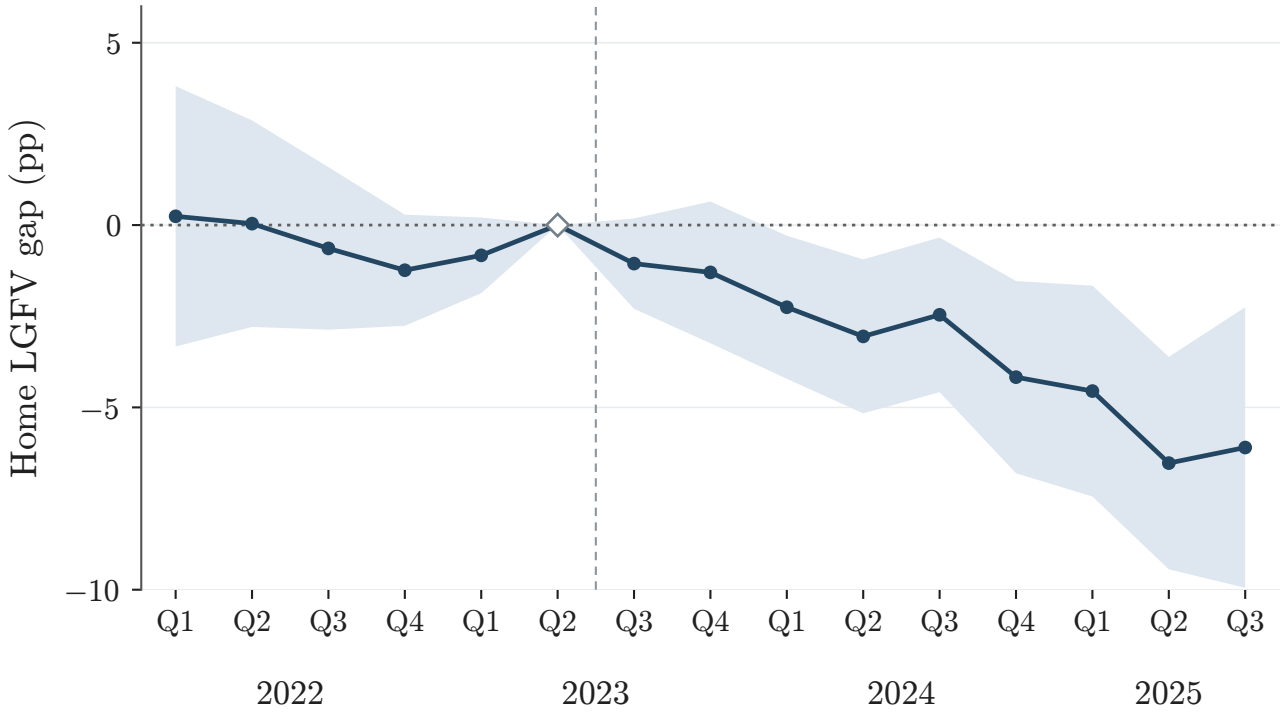
Notes: PC denotes private credit. This figure compares financing-yield opportunities in private credit and LGFV bonds. Quarterly annual contractual yields in percent on new PC contracts and new physical LGFV bond issues, 2019Q1–2024Q4. Each series uses financing reported by LGFV issuers in the deduplicated financing records and weights rate-visible observations by positive known original financing amounts; unknown amounts are not imputed. The series need not contain the same borrowers in a quarter and are not adjusted for changes in composition or tenor. The dashed vertical line marks the 2023Q3 announcement. Table 4 compares rates within reporting issuers and controls for tenor. Incomplete 2025 observations are excluded.

Figure 3
The private credit–bond rate gap within the same borrower



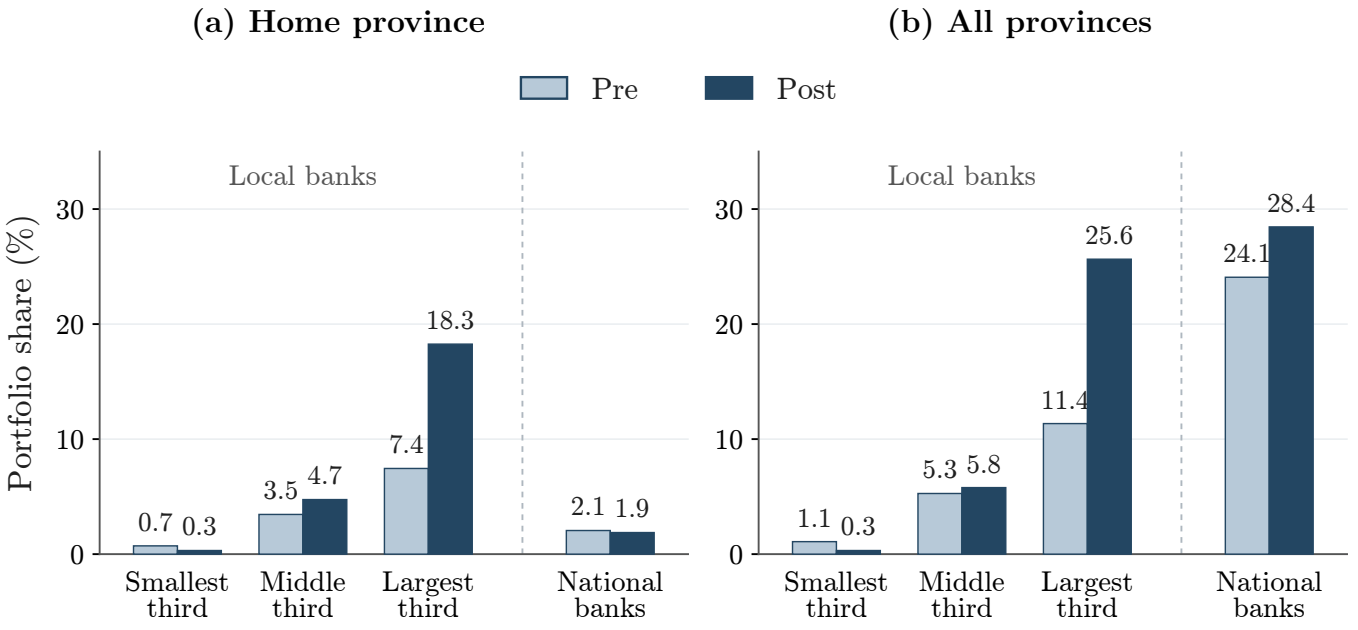
Notes: This within-borrower event study plots $\delta_q = \beta_q - \beta_{2023Q2}$ as defined in equation (8): changes in the annual contractual PC-minus-bond rate gap, in percentage points. Quarterly gap levels are estimated by equation (7), with borrower-by-calendar-month effects and continuous maturity interacted with every quarter; there are no borrower-by-debt-type effects. The plot starts in 2022Q1. The usable estimation sample is 2021Q1–2025Q3 (17,303 financing observations; 589 borrowers), with observed maturity required and fixed-effect singletons removed. Funded-borrower and reporting-issuer DM codes must coincide; affiliates are excluded. Before these restrictions, the pool contains borrower-quarters with both channels and borrowers with at least two matched quarters. Observations are equally weighted. Shading gives conventional pointwise 95% confidence intervals clustered by borrower province, using t_{25} critical values (26 provinces) and the covariance of the gap contrasts. The hollow diamond marks the reference; the vertical dashed line separates 2023Q2 from 2023Q3. The line stays solid through 2025. Coverage falls to 58, 35, and 21 borrowers in 2025Q1–Q3; 19 borrowers have both channels in the same month in 2025Q3. No usable 2025Q4 comparison remains. The PC extract's latest disclosure is April 20, 2026. Earlier changes, 2021Q1–2023Q1, reject joint zero ($p < 0.001$). Tables D4–D5 report specification, inference, and coverage checks.

Figure 4
Home-province gap in local banks' LGFV holdings by quarter



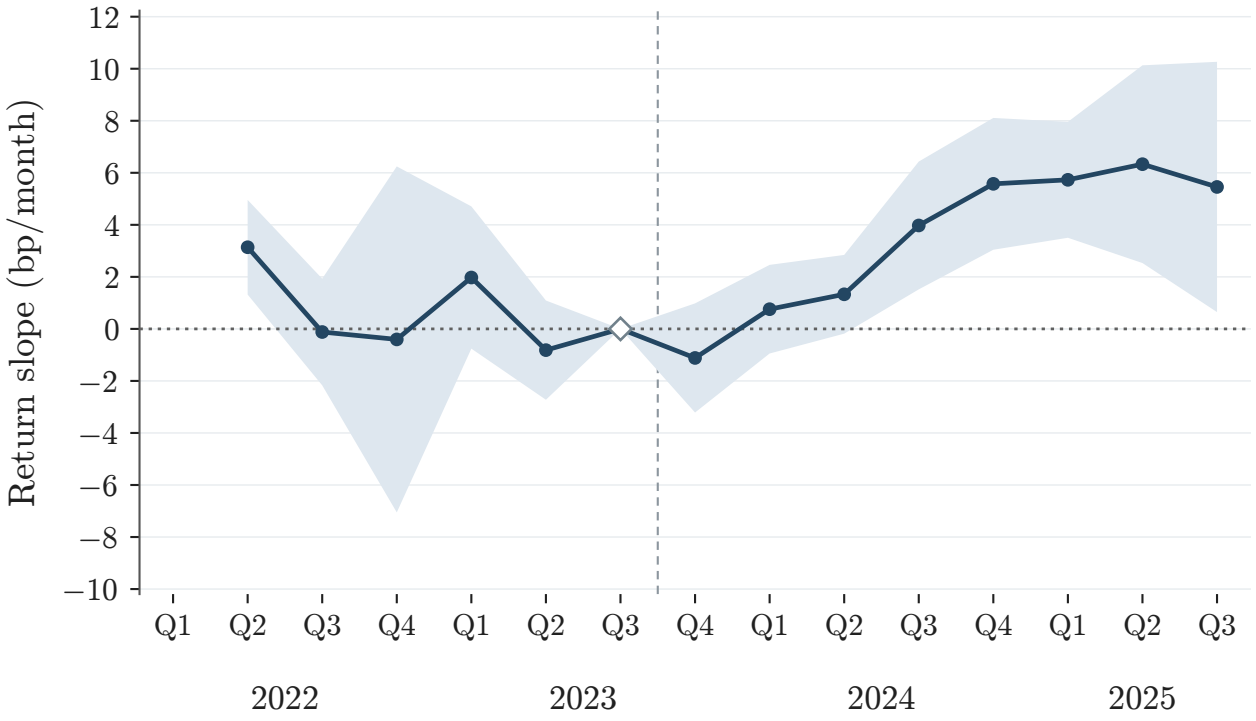
Notes: This event study plots the home-bond allocation coefficients η_k in equation (D.6). A buying institution's LGFV holdings in a province, as a percentage of its total bond holdings and winsorized at the 1st and 99th percentiles, are regressed on Home interacted with quarter indicators. Buyer-by-province, buyer-by-quarter, and destination-province-by-quarter fixed effects enter; 2023Q2 is omitted. The sample covers 2022Q1–2025Q3. The observation is a buyer, province, and quarter, so the coefficients measure changes in the home gap for an entire province rather than a single city as in Table 6. Home equals one for a local bank's own province. The shaded bands are pointwise 95% confidence intervals based on standard errors clustered by bank city, and the dashed line marks 2023Q3.

Figure 5
Private credit allocation before and after implementation



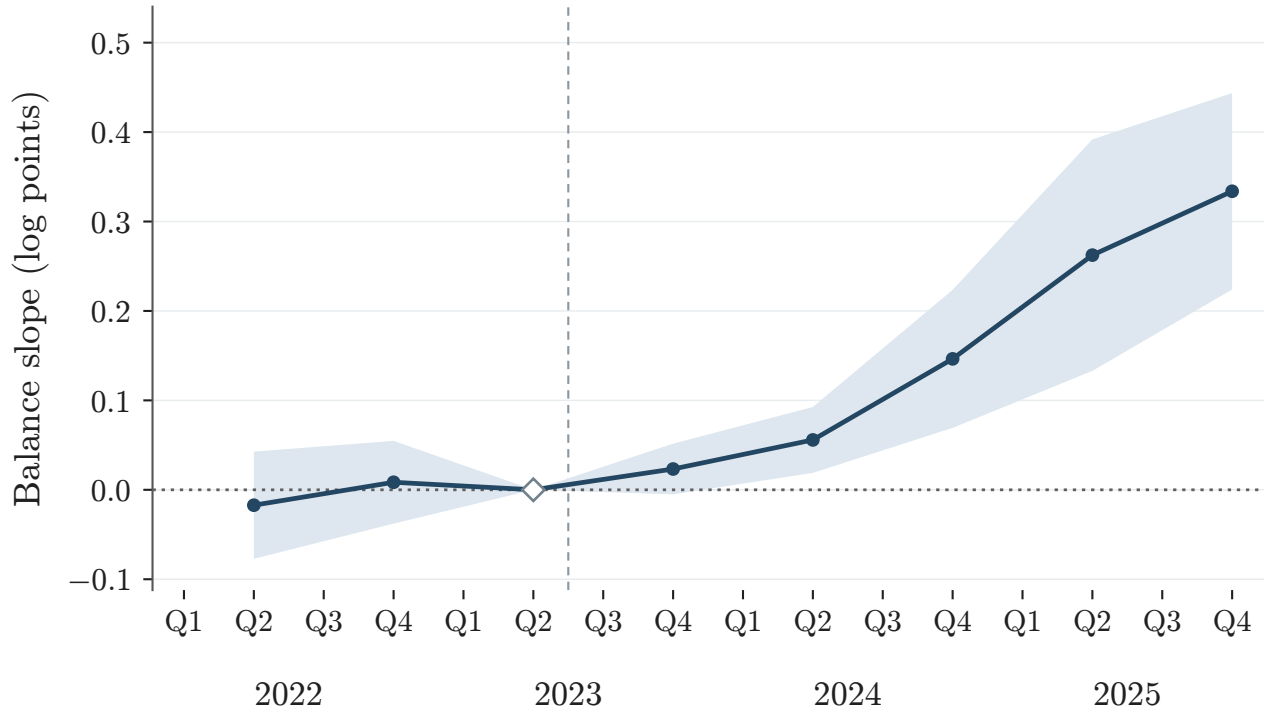
Notes: Bars are descriptive, holding-value-weighted portfolio shares. Both panels divide private credit by the same disclosed bond and private-credit holdings defined in Table 7; Panel (b) includes home and non-home private credit. Pre covers 2022Q1–2023Q3 and Post covers 2023Q4–2025Q3. Local-bank terciles use fixed FY2021 underlying-bank interest-bearing assets and retain the same 102 banks in both periods (25, 38, and 39 banks). The national-bank comparison uses the 18 banks in Table 7, Panel A. Home is the underlying bank's headquarters province.

Figure 6
Bank size and product returns by quarter



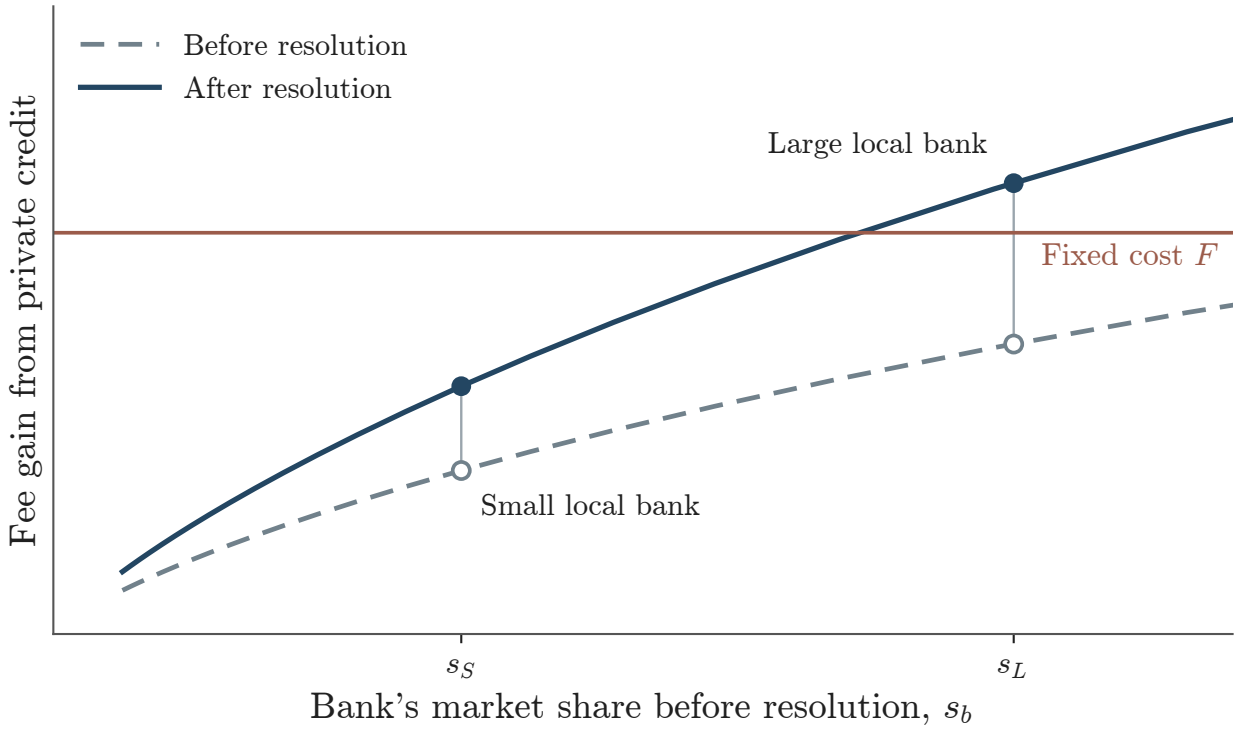
Notes: This event study replaces both refinancing interactions in equation (15) with log bank size interacted with quarter indicators. The outcome is monthly log adjusted-NAV return (bp) for local-bank products. Each coefficient measures the change in the size–return slope relative to 2023Q3, per log point of FY2021 underlying-bank interest-bearing assets in RMB trillion, without standardization. The regression includes the eleven time-varying controls, institution effects, and bank-home-province-by-month effects, but no prior-reliance interactions. The sample is 2022Q2–2025Q3 (206,770 NAV-series-months; 101 underlying banks). Shading shows pointwise 95% confidence intervals based on underlying-bank clustered errors. The hollow diamond marks the normalized reference quarter, not an estimated zero. The vertical line marks implementation in 2023Q4.

Figure 7
Bank size and total WMP balances



Notes: This event study replaces the size–Post interaction in equation (17) with log bank size interacted with half-year reporting-date indicators. The quarterly axis locates actual midyear and year-end reports at Q2 and Q4; no Q1 or Q3 balances are imputed. The log-balance regression includes bank and half-year fixed effects and no additional controls, for 89 local banks over 2022H1–2025H2 (571 bank-half-years). Each coefficient measures the change in the size–balance slope relative to the June 2023 report (2023Q2), per log point of FY2021 interest-bearing assets in RMB trillion, without standardization. Balances combine a bank's own products and those of its wealth-management subsidiary. Shading shows pointwise 95% confidence intervals based on bank-clustered errors. The hollow diamond marks the reference; the vertical line separates 2023Q2 and Q3. The joint test that the two 2022 coefficients are zero has a p-value of 0.41. Source: Wind institution-level WMP statistics, with the bank-disclosure correction described in Appendix E.1.

Figure 8
The scale threshold for private credit



Notes: This figure illustrates the entry condition in equation (30), using the parameters and equilibrium portfolios in Appendix B.6. The curves vary the reputation of an additional bank facing those portfolios and plot its optimized fee gain before resolution (dashed) and afterward (solid), against its bond-only market share before resolution. Entry is worthwhile when the gain covers F. The circles evaluate these curves at the small and large local banks' pre-resolution market shares: only the latter scale crosses the threshold. They illustrate the scale comparison; Table B1 reports the exact gains for the three equilibrium banks.

Internet Appendix

How Hidden Local Government Debt Shapes Shadow Banking Competition: Evidence from Wealth Management in China

The appendix separates supporting empirical exhibits (Internet Appendix A), the model solution and proofs (Internet Appendix B and Internet Appendix C), detailed specifications and supplementary results (Internet Appendix D), and checks of return measurement, balance reporting, holdings disclosure, and initial bond exposure (Internet Appendix E).

Internet Appendix A Supporting Evidence

A.1. Data, investment opportunities, and portfolio denominators

Table A1 defines the variables; Table A2 reports coverage by date and bank category; Table A3 summarizes products, holdings, and issuers. These tables establish what is observed and distinguish a share within one asset class from an allocation across bonds and private credit. The sample contains 171 underlying banks, with regression samples determined by the required data.

Table A15 compares LGFV issuance yields with those of other issuers. Across 20,172 unique renminbi fixed-rate bonds issued at par in 2022Q1–2023Q3, LGFV yields exceed other nonfinancial and financial issuers' yields by 122 and 149 basis points. Controlling for issue month, maturity, issue size, and call provisions retains gaps of 91 and 108 basis points; adding ratings gives 88 and 110. The comparison measures contractual yields at issuance.

Table A11 separates holding values, reported portfolio weights, and position counts. LGFV bonds represent 16.4% of local banks' summed reported weights versus 5.4% at national banks over 2022Q1–2025Q3. These broad disclosed-portfolio shares differ from the pre-program, bond-only shares in the introduction. Figure A3 plots both denominators; Appendix D.8 reconciles them. Figure A1 describes the wider investment menu, and Figure A2 supplies historical WMP balances. Figure A5 estimates both sides of the WMP–LGFV link: in 2024, LGFV bonds represented 10.2% of industry WMP assets and WMPs held 28.1% of the LGFV bond market. The extrapolation ends in 2024, the last verified market total.

A.2. Product design, performance, and borrower selection

Table A6 documents redemption choices: restricted products account for 80.2% of local-bank assets versus 47.3% at national banks. Table A7 estimates equation (3); restricted redemption widens the local–national home-allocation difference by 20.0 percentage points, including 7.7

points for LGFV bonds and 7.8 for private credit. Product design thus accommodates less liquid local investments.

Table A8 complements the return-premium evidence with reported downside outcomes. A 20-point increase in home-LGFV portfolio overweight is associated with about eight additional annual basis points and a 0.8-point lower probability of a negative month, against a 9.9% mean. These outcomes capture the performance visible in reported NAVs. Table A10 tests whether performance predicts subsequent flows: the top–bottom return-rank difference is 12.2% of beginning assets and 8.1 points in positive-flow probability. Appendices D.5 and D.6 give the designs and outcome definitions.

Tables A12 and A13 compare issuance pricing and subsequent issuer distress across otherwise similar bonds. Purchases proxied by issue-quarter ownership outside the issuer's city but within its province accompany better purchase terms; same-city ownership accompanies a larger yield concession and fewer later credit events. Table A4 varies distress horizons and aggregates to issuer-quarter: the same-city-minus-other-city contrast remains negative, with issuer-clustered p-values of 0.084, 0.054, and 0.115 in the alternative designs. Table A9 examines limits to support under refinancing pressure. A ten-point larger maturity wall predicts about 0.07 points less home net buying and a 1.16-point higher guarantee probability on new home purchases. These tests distinguish selective investment from unconditional borrower support. Designs are in Appendices D.4 and D.7.

A.3. Yield compression and replacement investments

Figure A4 shows provincial spread convergence; Table A5 relates spreads to cumulative refinancing/GDP with province and quarter effects. The refinancing coefficient is -17.67 basis points per percentage point of intensity; initial-spread convergence controls and province trends give -2.43 and -8.65. All three estimates indicate spread compression.

Table A14 asks whether private credit expansion offsets reduced home-bond holdings. With initial reliance, the eleven controls, institution effects, and home-province-by-quarter effects, the size–Post coefficient for the combined home allocation is 4.76 percentage points per log point of bank assets, or 3.3 points for a doubling. Larger banks therefore maintain more investment in the home borrower sector, rather than merely exchanging equal amounts of bonds and private credit.

Table A1
Variable definitions

Variable	Definition
$\mathbb{1}\{\text{Local bank}\}_b$	Local-bank indicator: equals one for banks other than state-owned and joint-stock commercial banks.
Within-category home bias$^n_{iq}$	Home holdings / all holdings within class n, minus the corresponding nationwide share. Requires positive class holdings (equation (2)).
Portfolio home overweight$^n_{iq}$	Home class-n holdings / all analyzed bonds and private credit, minus the corresponding nationwide share. Includes category zeros; without n, sums across classes (equation (D.12)).
Initial home overweight$_a$	Product-assets-weighted pre-program mean of portfolio home overweight for issuing institution a, 2022Q1–2023Q3.
Log bank size$_b$	Natural log of FY2021 underlying-bank interest-bearing assets (RMB trillion), without standardization.
Initial home LGFV share$_b$	Home LGFV bonds / included debt, holdings-value weighted over 2022Q1–2023Q2. Includes only standard bonds, without a market-weight adjustment.
$\mathbb{1}\{\text{Restricted redemption}\}_i$	Restricted-redemption indicator: equals one for products without daily redemption.
Monthly return	Log return computed from month-end adjusted NAVs in consecutive months, in basis points.
Trailing volatility	Annualized standard deviation of monthly returns over the twelve months ending in the outcome month.
Net flow$_{iq}$	Quarterly net flow: growth in inferred assets net of returns, in percent (equation (D.9)).
Yield concession$_{jq}$	Yield concession at issuance: 100 times the difference between the CSI valuation spread and the coupon-based issue spread, both stored in percentage points, yielding basis points.
Same-city ownership$_{jq}$	Share of bond j held by local banks in the issuer's city at the end of its issue quarter q, in percentage points of issue size.
Other-city provincial ownership$_{jq}$	Share held by local banks elsewhere in the issuer's province at the same date, in percentage points of issue size.
Distress	Equals 100 if the issuer records its first credit event within eight quarters after issuance, and zero otherwise.
$\mathbb{1}\{\text{Home buyer}\}_{ac}$	Equals one when WMP issuer a is a local bank or its subsidiary and invests in issuer city c of the underlying bank's headquarters province.
Maturity wall$_{cq}$	Scheduled principal on all LGFV bonds maturing in city c and quarter q, as a percentage of 2020 city budget revenue.
Refinancing$^{GDP}_{pq}$	Cumulative special-refinancing issuance / 2020 provincial GDP (percentage points). Final refinancing fixes this value at the last observed date.
Refinancing$^{debt}_{pq}$	Cumulative special-refinancing issuance / 2020 provincial LGFV debt (fraction), used for allocations and returns. Neither intensity measure is standardized.
$\mathbb{1}\{\text{Post}\}$	Equals one from 2023Q4 in quarterly data and from October 2023 in monthly data.
$h(m)$	The quarter preceding month m; exposures and controls for month m are measured at the end of that preceding quarter.
Log WMP balances$_{bs}$	Natural logarithm of the outstanding WMP balance of bank b at the end of half-year s, combining its own products and those of its wealth-management subsidiary.

Notes: Holdings cover 2022Q1–2025Q3. Included debt comprises bonds and identified private credit with known borrower locations, excluding central-government, central-bank, and policy-bank obligations. Other centrally administered issuers are never home in the primary return tests. Issuer credit events include defaults, rating warnings, and guarantor events, not necessarily default on the held bond. bp and pp denote basis points and percentage points.

Table A2
Sample coverage

Panel A. Overall coverage

Measure	Count
Source holding records	2,288,894
Holding records with a usable product identifier	2,279,252
WMP products	60,396
Product-quarter observations	253,484
Reported institution names	193
Underlying banks	171
Bank headquarters provinces	30
Quarters	15

Panel B. Annual coverage

Year	Banks	Source holding records	WMP products
2022	141	421,072	19,514
2023	161	579,703	26,021
2024	149	708,907	30,512
2025 (Q1–Q3)	140	579,212	29,527

Panel C. Bank categories

Bank group	Banks	WMP products	Product-quarters
National banks	18	27,799	131,535
Large state-owned	6	9,571	49,593
Nationwide joint-stock	12	18,228	81,942
Local banks	153	32,597	121,949
Regional commercial	100	23,817	91,937
Rural institutions	53	8,780	30,012
All banks	171	60,396	253,484

Notes: Coverage is 2022Q1–2025Q3. A holding record is a product-asset-quarter observation. Source-record counts include 9,642 records with blank or "NA" product registration codes; usable-record, product, and product-quarter counts exclude these codes. Reported institution names include wealth-management subsidiaries and alternative bank names; the parent-bank crosswalk groups them into 171 banks. Panel B counts banks and products once per year; annual counts therefore need not sum to full-period counts. Panel C uses the source bank-type codes and assigns subsidiary products to the parent bank's category. Regional commercial banks include provincial and city-focused banks, which the source does not code as mutually exclusive geographic subgroups. Rural institutions include rural commercial and cooperative institutions. Guangfa Bank's wealth-management subsidiary is classified with nationwide joint-stock banks. Estimation samples impose the restrictions stated in each regression table.

Table A3
Descriptive statistics

Variable	Observations	Mean	SD	Median	p90
Panel A. Holding-observation characteristics					
Position weight (%)	2,286,025	7.12	12.03	4.04	11.46
Holding value (RMB million)	2,278,335	59.22	345.74	10.39	113.89
Panel B. Standard-bond characteristics					
Issue yield (%)	1,162,002	4.09	1.19	4.00	5.58
Issue maturity (years)	1,162,491	5.09	3.06	5.00	10.00
Panel C. Additional bond-issue characteristics					
$\mathbb{1}\{\text{Guaranteed}\}$	57,429	0.10	0.30	0.00	1.00
$\mathbb{1}\{\text{Callable}\}$	57,429	0.09	0.29	0.00	0.00
Number of WMP holders	57,429	0.23	0.52	0.00	1.00
Issue-quarter WMP ownership (pp)	57,429	1.52	4.59	0.00	5.02
Panel D. Product-quarter characteristics in the NAV sample					
Log product assets	153,563	19.41	1.64	19.38	21.51
Product risk rating	153,563	2.15	0.44	2.00	3.00
Restricted redemption (0/1)	153,563	0.88	0.33	1.00	1.00
Panel E. Pre-implementation issuers' FY2021 assets (RMB billion)					
LGFV issuers	1,939	42.79	48.09	28.41	88.39
Other nonfinancial issuers	2,062	134.48	548.95	35.32	243.34
Financial issuers	570	525.79	2,765.50	52.90	560.29

Notes: Panel A describes holding records and Panel B standard-bond holding records with the characteristic available. Panel C describes all 57,429 unique issues in the 2022Q1–2025Q4 bond database, before estimation restrictions and including issues without WMP ownership; pricing regressions end in 2025Q3. Panel D counts each product-quarter in the return sample once. Log product assets is $\log(1 + \text{inferred assets in RMB})$; product risk is the numerical rating and p90 the 90th percentile. The cross-sectional SD of products' average monthly returns is 10.5 bp. Among local-bank product-quarters before October 2023, mean home and non-home LGFV credit shares are 0.263 (home p90: 0.722) and 0.180. Mean issue rates are 5.62% for home LGFV holdings and 4.93% for other-province holdings; the mean within-product gap is 0.33 pp. Panel E separately describes firms issuing RMB debt in the available Wind issuance file over 2022Q1–2023Q3, not only WMP-held issues. Each issuer receives equal weight and appears once, matched by normalized exact name to reported year-end 2021 total assets. No future balance-sheet values are substituted. The LGFV master identifies the first group; source issuer types identify financial institutions and other enterprises. Central and local governments, policy banks, and unclassified issuers are excluded. Assets are observed for 1,939 of 1,964 LGFV issuers (98.7%), 2,062 of 2,167 other nonfinancial issuers (95.2%), and 570 of 723 financial issuers (78.8%).

Table A4
Issuer distress after issuance: windows, cohorts, and units of observation

Dependent variable: subsequent issuer distress (percent).

Event window	Eight quarters	Four quarters	Through 2025Q4	Issuer-quarter; eight quarters
	(1)	(2)	(3)	(4)
Other-city provincial ownership (pp)	−0.007	0.029	0.027	0.013
	(0.043)	(0.027)	(0.034)	(0.064)
Same-city ownership (pp)	−0.112**	−0.038	−0.098**	−0.145*
	(0.056)	(0.025)	(0.048)	(0.088)
Log issue amount	0.770***	0.253*	0.683***	1.205***
	(0.259)	(0.147)	(0.186)	(0.435)
Issue maturity (years)	−0.351***	−0.195***	−0.329***	−0.450***
	(0.091)	(0.050)	(0.067)	(0.157)
𝟙{Guaranteed}	−0.676	−0.128	−0.346	−0.821
	(0.432)	(0.256)	(0.339)	(0.509)
𝟙{Callable}	−0.268	0.204	0.041	−0.517
	(0.474)	(0.302)	(0.349)	(0.822)
Number of WMP holders	0.607*	0.287	0.331	0.776
	(0.327)	(0.182)	(0.236)	(0.512)
Total WMP ownership (pp)	−0.072**	−0.053***	−0.058**	−0.125**
	(0.034)	(0.018)	(0.026)	(0.060)
𝟙{County-level issuer}	1.041	0.466	1.011	0.336
	(0.823)	(0.437)	(0.617)	(1.187)
𝟙{Prefecture-level issuer}	2.113**	1.420***	2.125***	1.690
	(0.972)	(0.536)	(0.718)	(1.399)
Coupon (%)	−0.544	−0.707*	−0.232	−0.773
	(0.456)	(0.402)	(0.428)	(0.890)
Public valuation spread (pp)	1.391***	1.352***	1.125***	2.090**
	(0.456)	(0.422)	(0.427)	(0.854)
Implied contrast from the same regression				
Same-city minus other-city ownership	−0.105*	−0.067**	−0.125**	−0.158*
	(0.064)	(0.031)	(0.054)	(0.090)
Issuers	1,692	1,880	1,880	1,660
Issuers with an event	47	42	51	46
Issuer-clustered p: ownership contrast	0.132	0.084	0.054	0.115
Observations	10,857	16,185	16,185	5,563
Issuer city × issue quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Robust	Robust	Robust	Robust

Notes: Each column is a separate regression of subsequent issuer distress on same-city and other-city provincial ownership and the displayed controls, with issuer-city-by-issue-quarter effects. Ownership is measured in percentage points of issue size; distress is 100 if the stated first credit event occurs and zero otherwise. Column 1 uses a complete eight-quarter window, column 2 a complete four-quarter window, column 3 events observed through 2025Q4, and column 4 aggregates to issuer-quarter. Parentheses and stars use heteroskedasticity-robust errors; the separate row reports issuer-clustered tests of the ownership contrast. Contrasts use the full covariance matrix. The baseline and alternative fixed effects appear in Table A13. Stars denote 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q4 (columns 1, 4); 2022Q1–2024Q4 (columns 2, 3).

Table A5
Refinancing intensity and provincial LGFV spreads

Trend controls	None	Pre-program spread trend	Province trends
Dependent variable	LGFV spread (bp) (1)	LGFV spread (bp) (2)	LGFV spread (bp) (3)
Refinancing / GDP (pp)	−17.665*** (3.223)	−2.434** (1.044)	−8.652** (3.800)
Pre-program spread × linear trend	—	−13.550*** (1.017)	—
Mean of dependent variable	195.694	195.694	195.694
Observations	496	496	496
Province fixed effects	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes
Province-specific trends	No	No	Yes
Standard errors	Province	Province	Province

Notes: This table tests whether provincial spread compression is greater where cumulative refinancing is more intensive. Estimates use equation (D.2). The unit of observation is a province-quarter. The dependent variable is the province's mean secondary-market LGFV spread in basis points. Refinancing intensity is cumulative special-refinancing issuance as a percentage of 2020 provincial GDP, without standardization. Spreads and intensity are winsorized at the 1st and 99th percentiles. Column 2 adds the 2022Q1–2023Q3 mean spread in basis points interacted with a linear quarterly trend, and column 3 includes a separate linear trend for each province. Standard errors clustered by province are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q4.

Table A6
Redemption terms by bank group

Institution group	Redemption	Product-quarter observations	Equal-weighted share (%)	Size-weighted share (%)
National banks	Daily	17,294	20.5	52.7
National banks	Non-daily	67,054	79.5	47.3
Local banks	Daily	2,376	2.8	19.8
Local banks	Non-daily	83,273	97.2	80.2

Notes: This table identifies the redemption structure accommodating local banks' credit portfolios. Entries report the number and share of product-quarters by bank group and redemption terms. Daily products allow redemption on every business day; non-daily products are periodic-open or closed-end. Size-weighted shares weight product-quarters by inferred product assets. Shares sum to 100 within each bank group.

Table A7
Portfolio home overweight and redemption terms

Asset class	All credit	LGFV	PC	Other standard
Dependent variable	Portfolio home overweight (fraction) (1)	Portfolio home overweight (fraction) (2)	Portfolio home overweight (fraction) (3)	Portfolio home overweight (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	−0.063**	0.009	−0.021*	−0.050***
	(0.026)	(0.013)	(0.012)	(0.019)
Log bank assets	−0.016***	−0.004	0.018***	−0.030***
	(0.005)	(0.004)	(0.005)	(0.006)
$\mathbb{1}\{\text{Local bank}\} \times$ restricted redemption	0.200***	0.077***	0.078***	0.045***
	(0.024)	(0.011)	(0.014)	(0.012)
Log product assets	−0.001	0.001	0.000	−0.002**
	(0.002)	(0.001)	(0.001)	(0.001)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.048***	−0.014***	0.010***	−0.045***
	(0.005)	(0.003)	(0.003)	(0.005)
Product risk rating	0.015***	0.000	0.012**	0.003
	(0.005)	(0.003)	(0.005)	(0.003)
Bank nonperforming-loan ratio (%)	−0.074***	−0.005	−0.044***	−0.022**
	(0.018)	(0.014)	(0.009)	(0.009)
Bank cost–income ratio (%)	−0.001***	−0.001**	−0.000	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Bank net interest margin (%)	0.040*	0.026*	0.007	0.004
	(0.022)	(0.016)	(0.010)	(0.011)
Home-city GDP (RMB trillion)	0.003	0.023***	−0.023***	0.003
	(0.008)	(0.006)	(0.003)	(0.004)
Home-city GDP growth (%)	0.013**	0.019***	−0.003	−0.003
	(0.005)	(0.004)	(0.002)	(0.003)
Home-city LGFV debt/GDP	0.189***	0.217***	0.019	−0.049***
	(0.023)	(0.018)	(0.014)	(0.017)
Home-city income (RMB 10,000)	−0.048***	−0.044***	0.007**	−0.010**
	(0.007)	(0.005)	(0.003)	(0.004)
Mean of dependent variable	0.191	0.092	0.031	0.069
Observations	169,997	169,458	168,327	169,997
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Bank city × quarter	Bank city × quarter	Bank city × quarter	Bank city × quarter

Notes: PC denotes private credit. This table tests whether restricted redemption is associated with a wider local–national home-credit gap. Estimates use equation (3). The unit of observation is a product-quarter. The dependent variable Portfolio home overweight$^{n}_{iq}$ is the product's home-province holdings of the asset class divided by the value of all its included debt holdings, minus the home province's holdings of the class as a share of nationwide included debt holdings, as a fraction; the first column pools all included debt holdings. Non-daily redemption equals one for products without daily redemption, so the local-bank coefficient measures the difference between local and national banks among daily-open products and the interaction measures the additional difference among other products. All columns use the sample with nonmissing controls. Standard errors clustered at the bank-city-by-quarter level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Estimation sample: 2022Q1–2025Q3.

Table A8
Local LGFV exposure and product returns

Panel A. Monthly returns and volatility

Dependent variable	Monthly NAV return (bp) (1)	Trailing annualized volatility (bp) (2)
Home LGFV portfolio overweight (fraction)	3.140***	−11.150***
	(0.783)	(2.203)
𝟙{Local bank}	−0.076	5.279***
	(0.627)	(1.568)
Mean of dependent variable	23.873	75.758
Product, bank, and city controls	Yes	Yes
Observations	554,555	554,555
Month fixed effects	Yes	Yes
Standard errors	Bank city × month	Bank city × month

Panel B. Downside outcomes

Dependent variable	Monthly excess return (bp) (1)	Downside volatility (bp) (2)	Drawdown (log gap ×100) (3)	Negative return (0/1) (4)	Bottom-decile return (0/1) (5)	Drop over 50 bp (0/1) (6)
Home LGFV portfolio overweight (fraction)	3.817***	−6.942***	−0.061***	−0.039***	−0.039***	−0.008***
	(0.784)	(1.213)	(0.009)	(0.009)	(0.008)	(0.002)
𝟙{Local bank}	0.259	1.929**	0.009	0.012	0.006	−0.001
	(0.618)	(0.893)	(0.007)	(0.008)	(0.007)	(0.002)
Mean of dependent variable	0.271	24.489	0.088	0.099	0.092	0.015
Product, bank, and city controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	558,455	558,455	558,455	558,455	558,455	558,455
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Bank city × month	Bank city × month	Bank city × month	Bank city × month	Bank city × month	Bank city × month

Notes: Each column estimates equation (D.10), retaining all eleven product, bank, and city controls. The exposure is home LGFV holdings as a fraction of analyzed bonds and private credit, less the corresponding nationwide portfolio weight. Exposure and controls lag one quarter. Only exposure and local-bank coefficients are displayed. Panel A uses the common return–volatility sample, excluding observations outside either variable's 1st–99th percentiles. Panel B uses a separate common sample: excess return subtracts the mean for the same redemption type; downside volatility uses negative monthly returns; drawdown is 100 times the trailing-peak minus current log NAV. The last three outcomes are indicators, so coefficients are probability changes. Volatility measures are annualized over up to twelve months, requiring two observations. Appendix D.6 gives the complete definitions. The sample is 2022Q2–2025Q4. Parentheses contain bank-city-by-month clustered standard errors; stars denote 10%, 5%, and 1% significance.

Table A9
Refinancing pressure and home-province allocation: all-issuance wall

Panel A. Allocation margins

Dependent variable	Log(1 + new purchases)	New purchases (%)	Net purchases (%)	𝟙{Buy}	Exits (%)	Holdings (%)
	(1)	(2)	(3)	(4)	(5)	(6)
𝟙{Home buyer} × maturity wall	0.00011	−0.00088	−0.0070***	0.00053	0.0045***	−0.0087*
	(0.00218)	(0.00120)	(0.0018)	(0.00046)	(0.0013)	(0.0046)
Mean of dependent variable	0.937	0.120	−0.023	0.261	0.155	0.671
Observations	73,127	73,089	66,013	73,127	66,013	73,089
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Bank × issuer city fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bank × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Issuer city × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Bank city	Bank city	Bank city	Bank city	Bank city	Bank city

Notes: Bank-city-quarter regressions. Home denotes a local bank investing in its own province. The wall is scheduled principal on all city LGFV bonds maturing in the quarter as a percentage of 2020 city budget revenue. Bank-city, bank-quarter, and issuer-city-quarter effects enter. The Buy indicator equals one for any new purchase. Net buying removes valuation changes; exits include nonrenewal. Parentheses contain bank-city clustered errors. Appendix D.4 defines each outcome. Stars indicate 10%, 5%, and 1% significance. Estimation samples: Panel A, 2022Q1–2025Q3 (columns 1, 2, 4, 6); 2022Q2–2025Q3 (columns 3, 5); Panel B, 2022Q1–2025Q3.

Panel B. Characteristics of purchased bonds

Dependent variable	Issuer rating (1–4)	𝟙{Guaranteed}	Maturity (years)	Issue spread (pp)	Log issue amount
	(1)	(2)	(3)	(4)	(5)
𝟙{Home buyer}	−0.1972***	0.0389**	−0.0676	0.1507***	−0.0901***
	(0.0316)	(0.0172)	(0.0599)	(0.0454)	(0.0254)
𝟙{Home buyer} × maturity wall	0.00119*	0.00116**	0.0027*	−0.00040	0.000042
	(0.00060)	(0.00046)	(0.0014)	(0.00080)	(0.000512)
Mean of dependent variable	2.8405	0.1540	3.6470	1.6174	2.0001
Observations	33,209	36,303	36,303	36,303	36,303
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Bank × quarter fixed effects	Yes	Yes	Yes	Yes	Yes
Issuer city × quarter fixed effects	Yes	Yes	Yes	Yes	Yes
Standard errors	Bank city	Bank city	Bank city	Bank city	Bank city

Notes: New bond purchases. Outcomes are characteristics of the bonds purchased; issuer rating, guarantees, maturity, spread, and size are not city characteristics. The regressions include Home, its wall interaction, bank-quarter and issuer-city-quarter effects, but not bank-city effects. The wall has the same percentage-point definition as in Panel A; neither panel standardizes it. Spreads are percentage points; guarantees are indicators. Parentheses contain bank-city clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table A10
Relative performance and subsequent investor flows

Performance ranks and alternative flow measures

Dependent variable	Net flow (% initial assets) (1)	Net flow (% initial assets) (2)	Symmetric net flow (pp) (3)	Positive-flow probability (pp) (4)
Prior-quarter performance: bottom fifth omitted				
$\mathbb{1}\{\text{Return quintile} = 2\}$	1.915***	1.913***	1.425***	1.715***
	(0.632)	(0.609)	(0.513)	(0.636)
$\mathbb{1}\{\text{Return quintile} = 3\}$	2.942***	2.955***	2.873***	4.471***
	(0.806)	(0.770)	(0.539)	(0.934)
$\mathbb{1}\{\text{Return quintile} = 4\}$	5.218***	5.449***	4.754***	5.800***
	(1.178)	(1.185)	(0.768)	(1.056)
$\mathbb{1}\{\text{Return quintile} = 5\}$	12.177***	12.815***	8.711***	8.146***
	(2.723)	(2.783)	(1.521)	(1.672)
Log beginning product assets	−7.512***	−7.773***	−4.611***	−3.795***
	(1.010)	(1.002)	(0.573)	(0.340)
Product risk rating	Absorbed	Absorbed	Absorbed	Absorbed
Restricted redemption	Absorbed	Absorbed	Absorbed	Absorbed
Log bank assets	4.122***	4.523***	3.307***	3.454**
	(1.136)	(1.303)	(0.825)	(1.462)
Bank nonperforming-loan ratio (%)	−1.112	−0.585	−0.852	0.919
	(2.048)	(3.376)	(1.504)	(2.418)
Bank cost–income ratio (%)	−0.153**	−0.135*	−0.130**	−0.082
	(0.069)	(0.080)	(0.055)	(0.060)
Bank net interest margin (%)	0.948	−0.293	−0.932	2.060
	(2.093)	(1.950)	(1.283)	(2.465)
Home-city GDP (RMB trillion)	−0.764	0.700	−0.699	2.301
	(0.766)	(2.095)	(0.613)	(1.434)
Home-city GDP growth (%)	−0.222	−0.206	−0.104	0.129
	(0.438)	(0.605)	(0.323)	(0.686)
Home-city LGFV debt/GDP	−1.385	−3.140	1.713	4.399
	(2.887)	(3.851)	(1.933)	(5.510)
Home-city income (RMB 10,000)	−0.036	−3.750**	−0.325	−3.055**
	(0.685)	(1.812)	(0.476)	(1.342)
$\mathbb{1}\{\text{Local bank}\}$	4.467*	3.805	5.177***	15.079***
	(2.652)	(3.268)	(1.969)	(3.914)
Underlying banks	129	129	129	129
Mean of dependent variable	1.764	1.760	−4.648	43.102
Top-fifth wild-bootstrap p	< 0.001	—	< 0.001	< 0.001
Observations	123,745	123,738	123,745	123,745
Category × quarter fixed effects	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression of next-quarter flows on prior-quarter return fifths within risk and redemption categories; the bottom fifth is omitted. Controls are displayed and dated before flows; category-by-outcome-quarter effects enter, and column 2 adds home-province-by-quarter effects. Parentheses contain parent-bank clustered errors. Ordinary flows are capped at 1/99; symmetric flows are bounded and uncapped. Positive flow is in percentage points. Appendix D.5 defines the measures. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q3–2025Q3.

Table A11
Composition of disclosed WMP holdings by bank group

Asset category	National banks	Local banks	All banks
Panel A. Mean quarterly reported value (RMB trillion)			
Standard LGFV bonds	0.15	0.27	0.43
Other standard bonds	0.91	0.47	1.39
Private credit	0.32	0.11	0.43
Other disclosed assets	5.34	1.41	6.75
Panel B. Portfolio-weight share (%)			
Standard LGFV bonds	5.4	16.4	11.6
Other standard bonds	16.3	23.4	20.3
Private credit	11.6	6.2	8.5
Other disclosed assets	66.7	54.0	59.6
Panel C. Holding-observation share (%)			
Standard LGFV bonds	8.6	25.6	16.6
Other standard bonds	31.4	37.3	34.2
Private credit	4.5	2.2	3.4
Other disclosed assets	55.4	34.9	45.8

Notes: This table places the debt-allocation regressions in the broader composition of disclosed product assets. Entries describe the composition of disclosed WMP holdings for products issued by national banks (state-owned and joint-stock commercial banks), local banks (all other banks), and all banks, pooled over 2022Q1–2025Q3. Each holding record is assigned to one asset category. Panel A reports the average quarterly disclosed value in RMB trillion, Panel B the share of disclosed portfolio weight, and Panel C the share of holding records, all within bank group. Portfolio weights are the position weights reported by products. Other disclosed assets include deposits, funds, trusts, and other investment vehicles.

Table A12
Local WMP ownership and issue pricing

Specification	LGFV: baseline	LGFV: controls	LGFV: province × quarter	LGFV: city × quarter	Non-LGFV
Dependent variable	Yield concession (bp)	Yield concession (bp)	Yield concession (bp)	Yield concession (bp)	Yield concession (bp)
	(1)	(2)	(3)	(4)	(5)
Other-city provincial ownership (pp)	−0.0740	−0.2456***	−0.2821***	−0.4039***	−0.0812
	(0.0883)	(0.0942)	(0.0994)	(0.1006)	(0.1334)
Same-city ownership (pp)	0.8596**	0.5311	0.3606	0.6088*	0.0293
	(0.3508)	(0.3422)	(0.3483)	(0.3261)	(0.0898)
Log issue amount	—	0.2738	−0.1392	−0.5084	1.2756***
		(0.5531)	(0.5284)	(0.5133)	(0.4907)
Issue maturity (years)	—	5.2971***	5.3751***	5.8458***	0.2808***
		(0.1970)	(0.2000)	(0.1897)	(0.0907)
𝟙{Guaranteed}	—	−9.2684***	−7.8799***	−7.9945***	1.9214*
		(0.8286)	(0.8151)	(0.8492)	(1.0386)
𝟙{Callable}	—	−0.0602	0.5658	2.6258**	−8.5134***
		(1.0159)	(1.0828)	(1.0473)	(0.6729)
Number of WMP holders	—	−1.9114***	−1.6755***	−1.2945**	−1.4005**
		(0.6499)	(0.6349)	(0.6413)	(0.6700)
Total issue-quarter WMP share (pp)	—	−0.0286	−0.0395	−0.0312	−0.0303
		(0.0645)	(0.0654)	(0.0656)	(0.0456)
𝟙{County-level issuer}	—	1.7540	7.7765***	7.0148***	—
		(1.6906)	(2.2917)	(2.0447)	
𝟙{Prefecture-level issuer}	—	1.6979	7.2669***	3.3418	—
		(1.7640)	(2.4089)	(2.0974)	
Coupon (%)	—	11.0794***	7.4932***	3.4766***	4.3664***
		(0.6636)	(0.5914)	(0.6285)	(1.1202)
Same city minus other city	0.9335***	0.7767**	0.6427*	1.0127***	0.1105
	(0.3611)	(0.3444)	(0.3579)	(0.3367)	(0.1525)
Equality test: $p(\beta_C = \beta_P)$	0.0097	0.0242	0.0725	0.0026	0.4685
Mean of dependent variable	19.4802	19.4802	19.4358	19.1572	11.7806
Observations	21,116	21,116	21,090	20,526	30,660
Issue-quarter fixed effects	Yes	Yes	No	No	No
Issuer province × quarter fixed effects	No	No	Yes	No	No
Issuer city × quarter fixed effects	No	No	No	Yes	Yes
Standard errors	Robust	Robust	Robust	Robust	Robust

Notes: Equation (6) compares same-city and other-city provincial ownership to assess information and borrower support. Observations are issues in 2022Q1–2025Q3; later issues lack holdings coverage and are excluded, not assigned zero ownership. Yield concession is 100 times the difference between the CSI valuation spread and the coupon-based issue spread, both in percentage points, giving bp. A larger concession means a lower issue yield, hence a higher price relative to the valuation benchmark. Ownership is issue-quarter-end WMP holdings, in pp of issue size, not primary orders. Same-city ownership is positive for 2.7% of LGFV issues; its conditional mean is 9.8 pp. Columns 1–4 use LGFVs and column 5 other issuers. The common LGFV sample has 21,116 issues; removing singletons leaves 20,526 in column 4. All nine controls are displayed; dashes mean not included or absorbed. The contrast is $\beta_C - \beta_P$. Parentheses and stars use heteroskedasticity-robust inference; issuer clustering gives contrast SE 0.355 and $p = 0.0044$ in column 4. Stars denote 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table A13
Local WMP ownership and subsequent issuer distress

Specification	LGFV: baseline	LGFV: controls	LGFV: province × quarter	LGFV: city × quarter	Non-LGFV
Dependent variable	Distress within 8 quarters (%)	Distress within 8 quarters (%)	Distress within 8 quarters (%)	Distress within 8 quarters (%)	Distress within 8 quarters (%)
	(1)	(2)	(3)	(4)	(5)
Other-city provincial ownership (pp)	−0.1018***	−0.1186***	−0.0561	−0.0072	−0.0026**
	(0.0279)	(0.0358)	(0.0392)	(0.0432)	(0.0011)
Same-city ownership (pp)	−0.1221***	−0.2163***	−0.0997**	−0.1117**	−0.0045***
	(0.0179)	(0.0374)	(0.0444)	(0.0560)	(0.0017)
Log issue amount	—	1.1398***	0.6484**	0.7698***	−0.0310*
		(0.2570)	(0.2565)	(0.2587)	(0.0164)
Issue maturity (years)	—	−0.5663***	−0.2167**	−0.3512***	−0.0162*
		(0.0900)	(0.0878)	(0.0906)	(0.0085)
𝟙{Guaranteed}	—	−0.3580	−0.6696*	−0.6761	−0.0705***
		(0.3952)	(0.3998)	(0.4316)	(0.0241)
𝟙{Callable}	—	−1.3133***	−0.7909	−0.2680	−0.0780
		(0.4775)	(0.5090)	(0.4737)	(0.0629)
Number of WMP holders	—	0.5738	0.7151*	0.6072*	0.0295
		(0.4056)	(0.3651)	(0.3266)	(0.0258)
Total issue-quarter WMP share (pp)	—	−0.0400	−0.0789**	−0.0721**	−0.0015
		(0.0330)	(0.0330)	(0.0338)	(0.0013)
𝟙{County-level issuer}	—	2.6641***	3.7025***	1.0408	—
		(0.5277)	(0.7866)	(0.8230)	
𝟙{Prefecture-level issuer}	—	4.6174***	4.1417***	2.1131**	—
		(0.5760)	(0.8690)	(0.9716)	
Coupon (%)	—	0.0466	−0.6141	−0.5440	0.0772*
		(0.5078)	(0.4694)	(0.4558)	(0.0448)
Public valuation spread (pp)	—	1.6083***	1.4589***	1.3906***	0.0286
		(0.4807)	(0.4580)	(0.4561)	(0.0322)
Same city minus other city	−0.0203	−0.0978***	−0.0436	−0.1045*	−0.0019*
	(0.0314)	(0.0370)	(0.0498)	(0.0635)	(0.0011)
p-value, issuer clusters	0.575	0.044	0.448	0.132	0.305
Mean of dependent variable	2.6679	2.6679	2.6712	2.7171	0.0338
Observations	11,170	11,170	11,156	10,857	14,776
Issue-quarter fixed effects	Yes	Yes	No	No	No
Issuer province × quarter fixed effects	No	No	Yes	No	No
Issuer city × quarter fixed effects	No	No	No	Yes	Yes
Standard errors	Robust	Robust	Robust	Robust	Robust

Notes: Equation (6) tests whether favorable same-city pricing accompanies lower subsequent issuer distress conditional on observed issue risk. Observations are bonds issued in 2022Q1–2023Q4 by previously undistressed issuers. The outcome equals 100 if the issuer records its first credit event within eight quarters, and zero otherwise. Events include defaults, rating warnings, and guarantor events, observed through 2025Q4; every bond has a complete window. Ownership and controls follow Table A12, adding the CSI valuation spread at issuance to control for ex ante risk. Columns 1–4 use LGFV bonds and column 5 other issuers. Same city minus other city reports $\beta_C - \beta_P$; the next row gives its issuer-clustered p-value. Parentheses contain heteroskedasticity-robust standard errors. Stars denote 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q4.

Table A14

Bank size and combined home local credit

Dependent variable	Home LGFV plus PC share (pp) (1)
Log bank size × 𝟙{Post}	4.761***
	(0.628)
Initial home overweight × 𝟙{Post}	−6.493**
	(2.879)
Log product assets	−2.289***
	(0.292)
𝟙{Restricted redemption}	12.213***
	(1.582)
Product risk rating	5.048***
	(0.909)
Log bank assets	−9.400
	(6.278)
Bank nonperforming-loan ratio	3.607
	(2.695)
Bank cost–income ratio	−0.143
	(0.178)
Bank net interest margin	5.396*
	(2.863)
Home-city GDP (RMB trillion)	10.200
	(9.296)
Home-city GDP growth (%)	−1.645***
	(0.565)
Home-city LGFV debt/GDP	−25.496**
	(12.169)
Home-city income (RMB 10,000)	46.474***
	(12.043)
Observations	83,027
Institution fixed effects	Yes
Home province × quarter fixed effects	Yes
Standard errors	Bank city × quarter

Notes: One regression of the combined home-LGFV and home-PC share of included debt (pp), for local-bank products. Size and prior reliance are interacted with Post, which begins in 2023Q4. Size is unstandardized log FY2021 parent-bank assets in RMB trillion. All eleven controls, institution effects, and home-province-by-quarter effects are included and displayed. Parentheses contain bank-city-by-quarter clustered errors; the size interaction has an underlying-bank clustered p-value of 0.00002. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3. Regressions weight observations by included debt value.

Table A15
LGFV bond issuance yields relative to other issuers

Panel A. Fixed-rate bonds issued at par, 2022Q1–2023Q3

	LGFV	Other nonfinancial	Financial
Mean issuance yield (%)	4.26	3.03	2.77
Median issuance yield (%)	3.92	2.79	2.75
Mean maturity (years)	2.28	1.79	2.12
Bond issues	8,044	10,832	1,296
Issuers	1,581	1,523	205

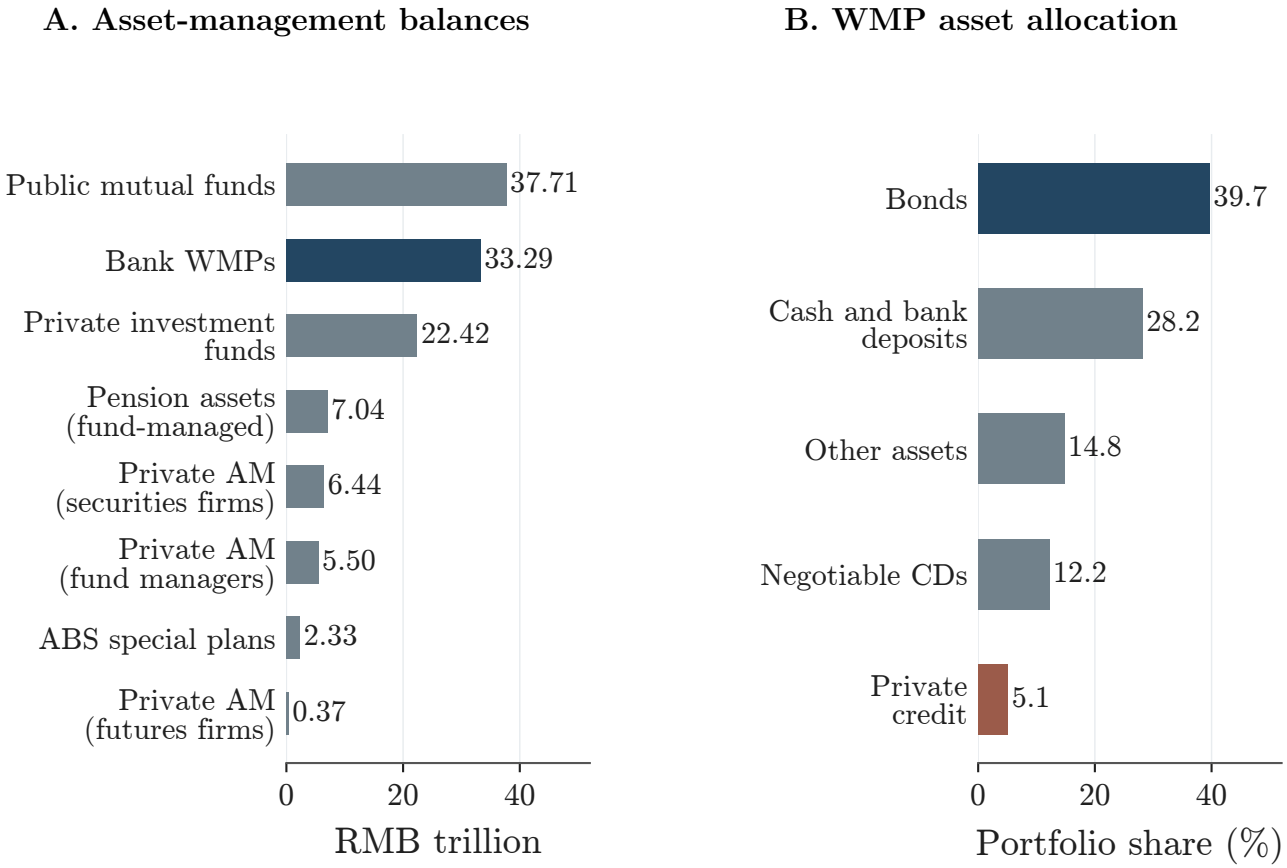
Panel B. Yield differences and bond characteristics

Dependent variable	Issuance yield (pp) (1)	Issuance yield (pp) (2)	Issuance yield (pp) (3)
$\mathbb{1}${LGFV issuer}	1.224***	0.910***	0.875***
	(0.069)	(0.089)	(0.093)
$\mathbb{1}${Financial issuer}	−0.265***	−0.173***	−0.226***
	(0.054)	(0.057)	(0.063)
Maturity (years)	—	0.282***	0.277***
		(0.023)	(0.028)
Maturity squared	—	−0.009***	−0.009***
		(0.001)	(0.001)
Log issue amount	—	−0.280***	−0.264***
		(0.038)	(0.036)
$\mathbb{1}${Callable}	—	0.378***	0.398***
		(0.099)	(0.099)
LGFV minus financial-issuer yield	1.489***	1.084***	1.101***
	(0.061)	(0.099)	(0.117)
Observations	20,172	20,172	20,172
Month fixed effects	No	Yes	Yes
Reported rating fixed effects	No	No	Yes
Standard errors	Issuer	Issuer	Issuer

Notes: Panel A describes unique CNY fixed-rate bonds issued at par in 2022Q1–2023Q3. Panel B regresses issuance yields (pp) on issuer-type indicators, with other nonfinancial issuers omitted. All columns use the same sample; later columns add the displayed issue characteristics, issue-month effects, and reported ratings (including an unreported category). Issuer assets need not be observed. CDs, convertibles, exchangeables, structured, perpetual, subordinated, floating-rate, and reset-rate bonds are excluded. There is no winsorization. Parentheses contain issuer-clustered errors; stars indicate 10%, 5%, and 1% significance. For these fixed-rate, at-par issues, the stated rate equals the contractual yield at issuance. Estimation sample: 2022Q1–2023Q3.

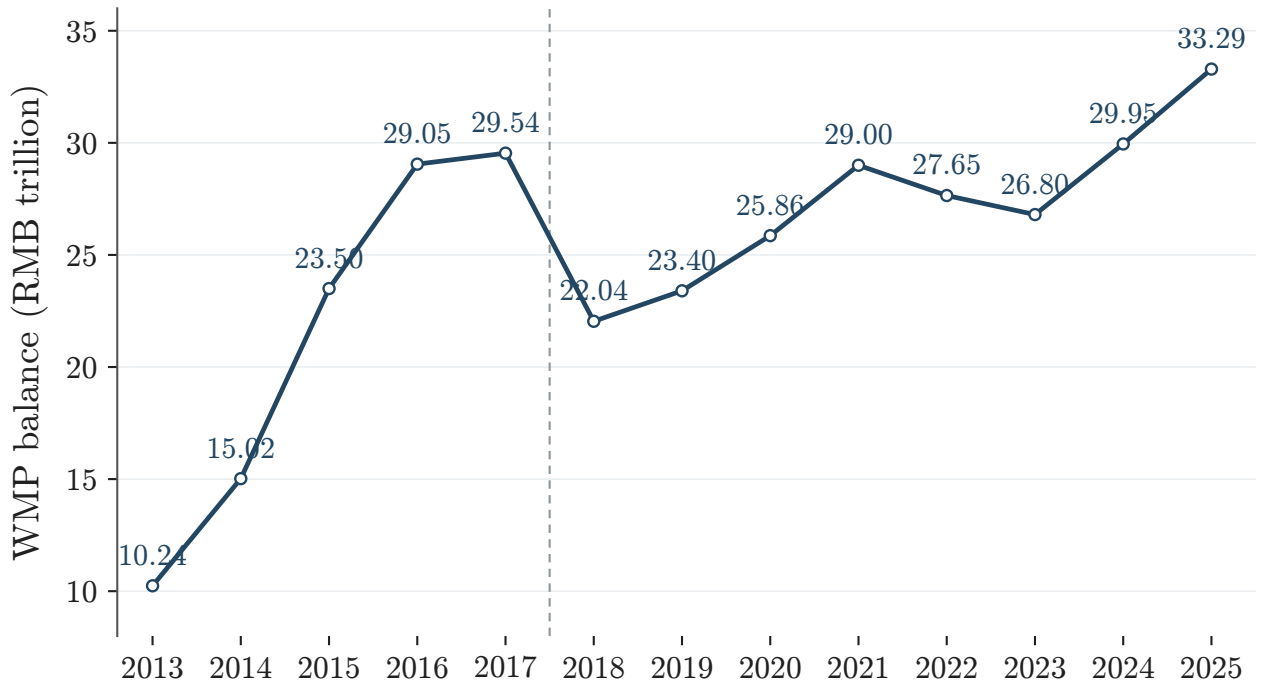
Figure A1

Chinese asset-management products and WMP asset allocation, 2025



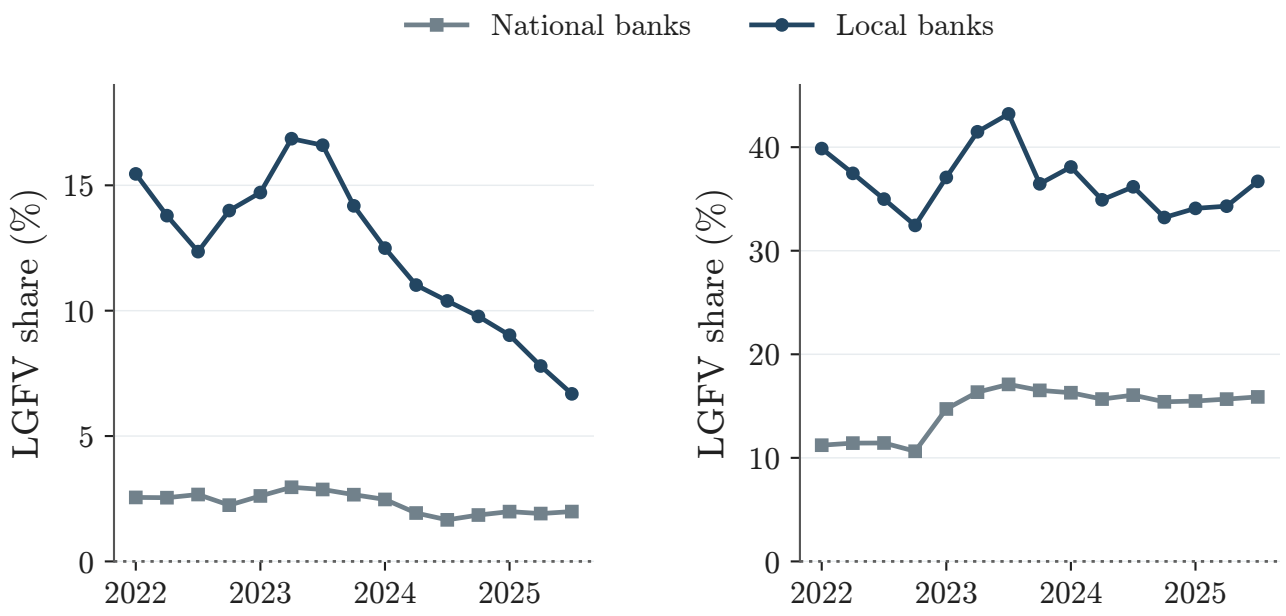
Notes: Panel A reports assets under management of Chinese asset-management products at the end of 2025, in RMB trillion, and Panel B reports the asset allocation of bank WMPs, in percent. Balances overlap through cross-investment. Sources: Asset Management Association of China and annual WMP market reports.

Figure A2
Bank WMP balances, 2013–2025



Notes: The figure plots outstanding bank WMP balances at year end, in RMB trillion. The dashed line marks a change in the statistical definition in 2018, when guaranteed and certain structured products were reclassified outside bank wealth management.

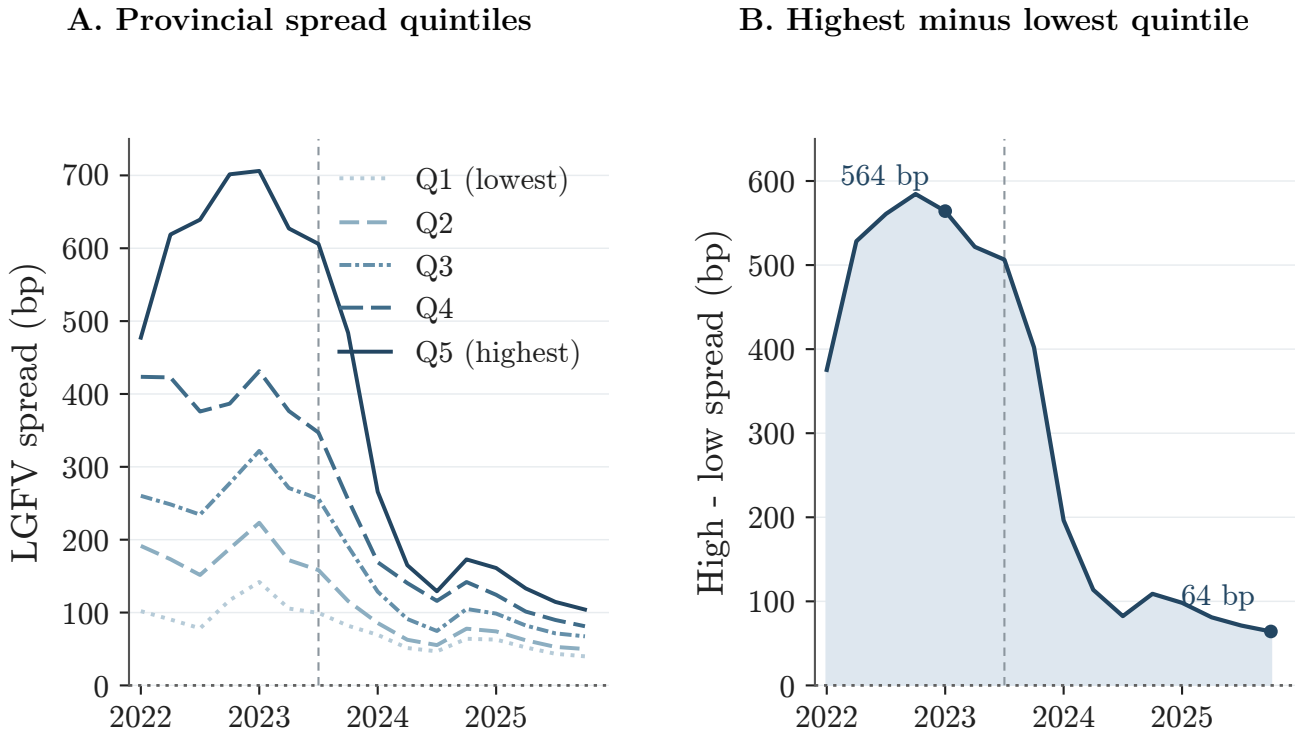
Figure A3
LGFV share of disclosed WMP holdings by bank group



Notes: The figure plots LGFV bonds as a share of the disclosed value of all holdings (Panel A) and of standard bonds excluding certificates of deposit (Panel B), by bank group, at each quarter end.

Figure A4

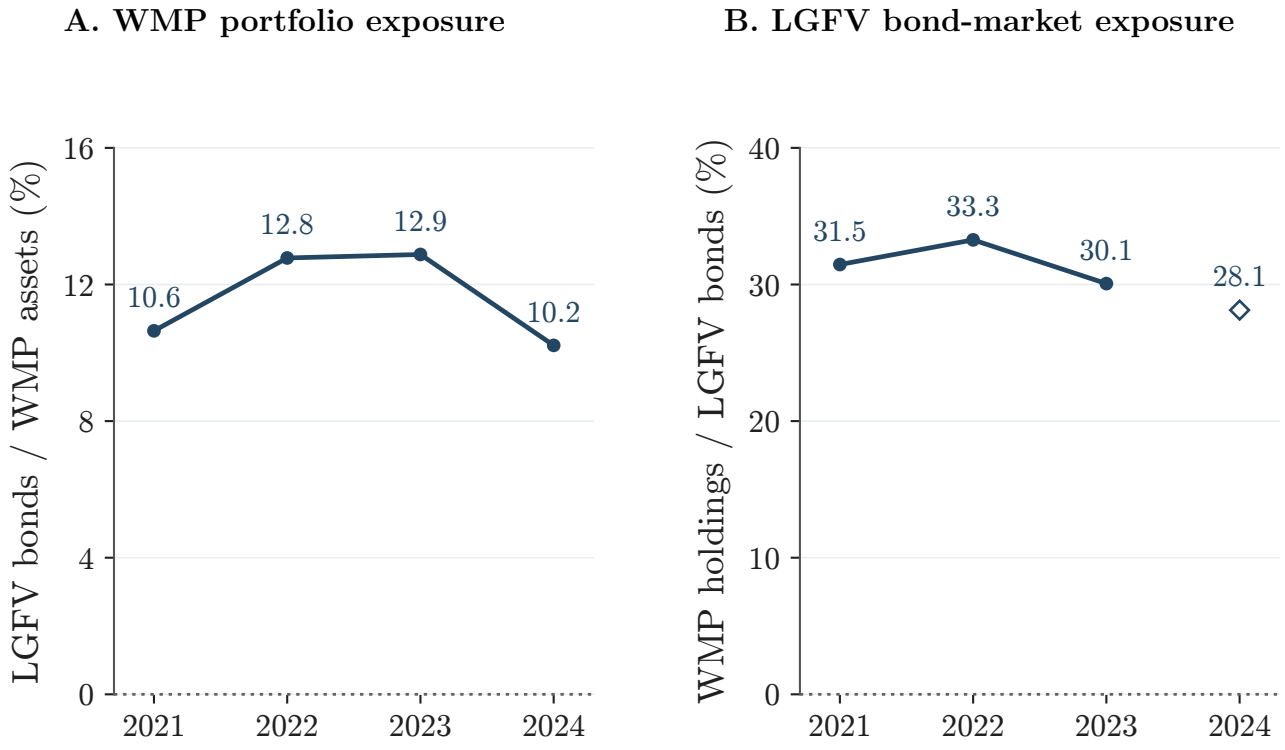
Provincial LGFV spreads by pre-program spread quintile



Notes: Provinces are grouped into quintiles of their mean LGFV spread over 2022Q1–2023Q2. Panel A plots the mean spread of each quintile, and Panel B plots the difference between the highest and lowest quintiles. The highest-minus-lowest gap falls from 564.2 bp in 2023Q1 to 64.2 bp in 2025Q4. The dashed line marks 2023Q3.

Figure A5

The scale of the WMP–LGFV bond link, 2021–2024



Notes: Both panels use the holdings-based extrapolation $\widehat{H}_t = B_t q_t$ in Internet Appendix D.8, not directly observed industry holdings. The left panel divides by official WMP net assets under management; the right divides by the external LGFV bond balance. The open diamond marks the 2024 balance-source change. Sources: Wind disclosures, annual WMP market reports, Lianhe Ratings (2024), and Zhang (2025).

Internet Appendix B
Local Reliance and Unequal Adjustment

This appendix derives the results of the model in Section 6; Internet Appendix C gives proofs. Subsection B.6 solves the full equilibrium numerically, without the first-order approximations used in the text.

B.1 Savers and market shares

Savers' risk aversion γ has a nondegenerate distribution G on $[\underline{\gamma}, \overline{\gamma}]$, where $0 < \underline{\gamma} < \overline{\gamma} < \infty$, with utility given by equation (18). Bounded support permits differentiation under the integral. Under the logit assumption on savers' tastes, the share of savers with risk aversion γ choosing bank b is equation (19). Holding rivals' portfolios fixed,

$$\frac{\partial s_b(\gamma)}{\partial x_b} = \beta s_b(\gamma)\big(1 - s_b(\gamma)\big)\big[\mu_B - \gamma\big(\sigma_B^2 x_b + \sigma_{B,PC} n_b\big)\big],$$
$$\frac{\partial s_b(\gamma)}{\partial n_b} = \beta s_b(\gamma)\big(1 - s_b(\gamma)\big)\big[\mu_{PC} - \gamma\big(\sigma_{B,PC} x_b + \sigma_{PC}^2 n_b\big)\big], \qquad \frac{\partial s_b(\gamma)}{\partial a_b} = \gamma\, s_b(\gamma)\big(1 - s_b(\gamma)\big). \tag{B.1}$$

The small-share approximation treats $s_b(\gamma)$ as small for every bank and saver type, so $s_b(\gamma)(1 - s_b(\gamma)) \approx s_b(\gamma)$. A bank's marginal savers then have approximately the same distribution as its savers; the distributions coincide in the small-share limit used for the analytic comparative statics. Subsection B.6 instead retains the full demand system, including a large national-bank share. For any two banks, and whatever their shares, equation (19) gives equation (28):

$$\log s_b(\gamma) - \log s_{b'}(\gamma) = \beta\big(R_b - R_{b'}\big) + \gamma\Big[\big(a_b - a_{b'}\big) - \frac{\beta}{2}\big(V_b - V_{b'}\big)\Big]. \tag{B.2}$$

B.2 Portfolio choice

Integrating equation (B.1) over G gives

$$\frac{\partial s_b}{\partial x_b} = \beta W_b\big[\mu_B - \bar{\gamma}_b\big(\sigma_B^2 x_b + \sigma_{B,PC} n_b\big)\big], \qquad \frac{\partial s_b}{\partial n_b} = \beta W_b\big[\mu_{PC} - \bar{\gamma}_b\big(\sigma_{B,PC} x_b + \sigma_{PC}^2 n_b\big)\big],$$

where $W_b = \int s_b(\gamma)(1 - s_b(\gamma))\, dG(\gamma) > 0$ and $\bar{\gamma}_b$ is defined in equation (22). The variance used in the sorting and scale conditions is $\mathrm{Var}_b(\gamma) = W_b^{-1} \int (\gamma - \bar{\gamma}_b)^2 s_b(\gamma)(1 - s_b(\gamma))\, dG(\gamma)$. Conditional on the investment regime, the fixed cost does not vary with the portfolio.

For completeness, let $z_b = (x_b, n_b)'$, $\mu = (\mu_B, \mu_{PC})'$, and let Σ be the covariance matrix of the two assets. At a constrained optimum with private credit access, the necessary conditions

are

$$\beta W_b(\mu - \bar{\gamma}_b \Sigma z_b) - \lambda_b \mathbf{1} + \nu_b = 0, \qquad \lambda_b(1 - \mathbf{1}' z_b) = 0, \qquad \nu_{bj} z_{bj} = 0,$$

with $z_b \geq 0$, $\mathbf{1}' z_b \leq 1$, $\lambda_b \geq 0$, and $\nu_b \geq 0$. Here $\mathbf{1} = (1, 1)'$, λ_b is the budget multiplier, and ν_{bj} enforces nonnegative investment in asset j. The bond-only problem restricts $n_b = 0$ and optimizes over $0 \leq x_b \leq 1$. The interior cases used in the text set the relevant multipliers to zero, giving equation (23). These conditions are necessary but not generally sufficient for an exact best response because market share need not be globally concave. The feasible sets are compact and share is continuous, so a maximizer exists within each regime; Subsection B.6 checks the computed choices numerically.

Holding marginal savers fixed gives the quadratic objective $Q_b(z) = \mu' z - \bar{\gamma}_b z' \Sigma z / 2$. The maintained covariance restrictions imply $\det \Sigma = \sigma_B^2 \sigma_{PC}^2 - \sigma_{B,PC}^2 > 0$, so Q_b is strictly concave. Its constrained maximizer is unique, and its interior choices are equations (24) and (26). The bond-only value is $\mu_B^2 / (2\bar{\gamma}_b \sigma_B^2)$. To see the gain from private credit, write $\chi = \sigma_{B,PC} / \sigma_B^2$ and $d = \sigma_{PC}^2 - \sigma_{B,PC}^2 / \sigma_B^2 > 0$. After optimizing the bond position, the additional value is $(\mu_{PC} - \chi \mu_B) n - \bar{\gamma}_b d n^2 / 2$. Its interior maximum is g_b in equation (27). If $\mu_{PC} \leq \chi \mu_B$ and the bond-only optimum is interior, the optimal private credit position and its gain are zero. If a portfolio constraint binds, the constrained solution replaces the interior formulas.

In the full demand system, a portfolio changes which savers are marginal, so an interior candidate best response solves equation (23) as a fixed point. The two regimes generally have different values $\bar{\gamma}_b^B$ and $\bar{\gamma}_b^{PC}$. The cross-regime formula $x_b^{PC} = x_b^B - \chi n_b^{PC}$ and quadratic gain g_b apply with a common, fixed $\bar{\gamma}_b$. Exact entry instead compares maximized shares, allowing the weights to differ. Subsection B.6 computes each portfolio with the marginal savers it attracts.

B.3 Sorting by reputation

Proposition B.1 (Sorting)**.** *In the small-share case, holding portfolios fixed,*

$$\frac{\partial \bar{\gamma}_b}{\partial a_b} = \mathrm{Var}_b(\gamma), \qquad \frac{\partial \log s_b}{\partial a_b} = \bar{\gamma}_b. \tag{B.3}$$

Along an interior bond-only optimum, holding rivals' offers fixed and allowing the bank's bond share to respond,

$$\frac{d\bar{\gamma}_b}{da_b} = \frac{\mathrm{Var}_b(\gamma)}{1 - \beta \sigma_B^2 \, (x_b^B)^2 \, \mathrm{Var}_b(\gamma) / \bar{\gamma}_b}, \qquad \frac{dx_b^B}{da_b} = -\frac{x_b^B}{\bar{\gamma}_b} \frac{d\bar{\gamma}_b}{da_b}, \tag{B.4}$$

with $d\bar{\gamma}_b / da_b > 0$ and $dx_b^B / da_b < 0$ when the denominator is positive. In the small-share case, if $a_b - a_{b'} > \beta \sigma_{PC}^2 / 2$, then $\bar{\gamma}_b > \bar{\gamma}_{b'}$ for any feasible portfolios; if both banks have interior bond-only choices, $x_b^B < x_{b'}^B$.

The bond-only portfolio behind Prediction 1 is

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b \, \sigma_B^2}. \tag{B.5}$$

Using the fixed-portfolio sorting derivative in equation (B.3), its dependence on reputation through marginal-investor risk aversion is

$$\frac{\partial \log x_b^B}{\partial a_b} = -\frac{1}{\bar{\gamma}_b} \frac{\partial \bar{\gamma}_b}{\partial a_b} = -\frac{\mathrm{Var}_b(\gamma)}{\bar{\gamma}_b} < 0. \tag{B.6}$$

Equation (B.4) allows the portfolio itself to adjust and gives the corresponding total derivative under its stated condition. The portfolio response reinforces the sorting: a bank that holds fewer bonds offers a safer portfolio, which attracts still more cautious savers. Outside the small-share case, the ordering still holds for all of a bank's savers but need not hold for its marginal savers, whose weights $s_b(\gamma)(1 - s_b(\gamma))$ fall where a bank serves most of a group. The numerical equilibrium in Subsection B.6, in which the national bank has a large share, confirms the ordering of $\bar{\gamma}_b$ for its parameters. The condition in the last statement of Proposition B.1 ensures that the reputation gap outweighs any difference in portfolio risk, because no feasible portfolio has variance above σ_{PC}^2. In the data, FY2021 bank assets proxy for the reputation and distribution network that determine a bank's share.

B.4 Scale and entry into private credit

Access to private credit changes the risk-adjusted return offered to saver type γ by $\Delta v_b(\gamma) = \Delta R_b - \frac{\gamma}{2}\Delta V_b$, comparing the bank's exact optimal portfolios with and without access. This change need not be positive for every saver type. Holding rivals' portfolios fixed, the fee gain is

$$G_b = f \int s_b(\gamma) \, \frac{(1 - s_b(\gamma))\left(e^{\beta \Delta v_b(\gamma)} - 1\right)}{1 - s_b(\gamma) + s_b(\gamma) e^{\beta \Delta v_b(\gamma)}} \, dG(\gamma), \tag{B.7}$$

where $s_b(\gamma)$ is its bond-only share before the portfolio change. Since access expands the feasible set, $G_b \geq 0$. A first-time entrant pays the cost if and only if $G_b \geq F$, taking entry at indifference. An already equipped bank pays no further cost and holds private credit whenever access strictly improves its optimum. Equivalently, $G_b = f(s_b^{PC} - s_b^B)$, the exact comparison in equation (30).

Proposition B.2 (Scale and entry)**.** *In the small-share case, the first-order approximation to the fee gain is $\widetilde{G}_b = f\beta s_b g_b$, with s_b the bond-only market share and g_b in equation (27). Along an interior bond-only optimum in a fixed competitive environment, allowing the bond*

share to respond to reputation,

$$\frac{d\log \widetilde{G}_b}{da_b} = \bar{\gamma}_b - \frac{1}{\bar{\gamma}_b}\frac{d\bar{\gamma}_b}{da_b}, \tag{B.8}$$

which is positive whenever $\mathrm{Var}_b(\gamma)\,(1+\beta R_b) < \bar{\gamma}_b^2$*, where* $R_b = \mu_B x_b^B$ *is expected excess return. If this condition holds along the interior optimum for all reputations between* a_S *and* a_L*, then* $\widetilde{G}_L > \widetilde{G}_S$*. In this first-order entry benchmark, the large bank enters whenever the small bank does, and for* $\widetilde{G}_S < F \leq \widetilde{G}_L$ *only the large bank enters.*

The condition bounds the dispersion of risk aversion among marginal savers relative to its mean. Holding portfolios fixed, it is $\mathrm{Var}_b(\gamma) < \bar{\gamma}_b^2$; the factor $1 + \beta R_b$ accounts for the portfolio response in equation (B.4). Under this condition, the proportional increase in the investor base exceeds the proportional decline in the value of the substitute to marginal savers. This establishes the scale channel in the approximation; the exact entry comparisons below use G_b, not $\widetilde{G}_b$. Because the cost is paid by the WMP business rather than each product, launching a new product does not remove it.

B.5 Debt resolution, substitution, and market shares

Debt resolution lowers the bond's expected excess return from $\mu_{B,0}$ to $\mu_{B,1} = (1-\tau)\mu_{B,0}$, with $0 < \tau < 1$. Subscripts 0 and 1 denote values before and after resolution. The return and risk of private credit are unchanged. To allow resolution to make bonds safer as well, the bond variance may fall from $\sigma_{B,0}^2$ to $\sigma_{B,1}^2 \leq \sigma_{B,0}^2$, retaining a positive-definite covariance matrix and interior conditional choices. The propositions hold the composition of each bank's marginal savers, $\bar{\gamma}_b$, fixed across the two states; Subsection B.6 lets it adjust.

Proposition B.3 (Bond retreat)**.** *A bank without private credit changes its bond share by*

$$\Delta x_b = \left[(1-\tau)\frac{\sigma_{B,0}^2}{\sigma_{B,1}^2} - 1\right] x_{b,0}^B, \tag{B.9}$$

which is proportional to its initial bond share and negative if and only if

$$\frac{\mu_{B,1}}{\sigma_{B,1}^2} < \frac{\mu_{B,0}}{\sigma_{B,0}^2}. \tag{B.10}$$

With unchanged bond risk, $\Delta x_b = -\tau x_{b,0}^B$*. A bank that holds private credit before and after resolution, with unchanged risks, changes its bond share by* $-\tau\sigma_{PC}^2\mu_{B,0}/\left[\bar{\gamma}_b(\sigma_B^2\sigma_{PC}^2 - \sigma_{B,PC}^2)\right]$*, which is also proportional to the bond return it loses.*

With unchanged bond risk, the supporting bond-retreat comparative static can also be written directly in primitives:

$$\Delta x_b = -\tau\, x_{b,0}^B = -\tau\, \frac{\mu_{B,0}}{\bar{\gamma}_b\, \sigma_B^2}. \tag{B.11}$$

Equivalently, $\partial^2 \Delta x_b / (\partial \tau\, \partial x_{b,0}^B) = -1$: a larger initial position produces a stronger response to resolution. This exposure interaction is examined in Section 4.4 and Table 5. This benchmark applies to banks that stay bond-only, holding marginal savers fixed. Condition (B.10) states that yield compression must outweigh any reduction in bond risk for banks to hold fewer bonds. The empirical reliance measure includes other home credit and subtracts a national benchmark; it proxies initial exposure to the affected investment.

For the remaining results, hold bond risk fixed. Write $g_{b,0}$ and $g_{b,1}$ for the gain from private credit before and after resolution, and $G_{b,0}$ and $G_{b,1}$ for the corresponding fee gains in equation (B.7), evaluated at equilibrium portfolios.

Proposition B.4 (Entry after debt resolution)**.** *Suppose the exact pre-resolution fee gains satisfy $G_{S,0} \le G_{L,0}$. The scale condition motivates this ordering, and Subsection B.6 verifies it directly. If $\mu_{PC} > (\sigma_{B,PC}/\sigma_B^2)\mu_{B,0}$, the fixed-composition gain from private credit rises with resolution,*

$$\frac{\partial g_b}{\partial \tau} = \frac{\sigma_{B,PC}}{\sigma_B^2}\, \mu_{B,0}\, n_b^{PC} > 0, \tag{B.12}$$

where n_b^{PC} is the conditional position in the fixed-composition benchmark. At a given market share, the first-order fee gain $\widetilde{G}_b = f\beta s_b g_b$ therefore rises. If the bond-only market share also changes, $d\widetilde{G}_b/d\tau = f\beta\big(s_b\, dg_b/d\tau + g_b\, ds_b/d\tau\big)$. The exact fee gains support entry by the national bank before resolution, entry by the large local bank only afterward, and no entry by the small local bank whenever

$$\max\{G_{L,0},\, G_{S,1}\} < F \le \min\{G_{L,1},\, G_{N,0}\}. \tag{B.13}$$

For an entrant, the combined share of bonds and private credit changes by

$$\Delta(x_b + n_b) = -\tau x_{b,0}^B + \frac{\sigma_B^2 - \sigma_{B,PC}}{\sigma_B^2}\, n_{b,1}, \tag{B.14}$$

which is positive when the new private credit position is large relative to the bond retreat.

The interval in condition (B.13) is nonempty when both upper bounds exceed its lower bound. This requires enough post-resolution fee income to make entry worthwhile for the large local bank, while the small bank's gain remains below the cost. Subsection B.6 verifies these comparisons with endogenous market shares rather than inferring them from the yield change alone.

The portfolio expression behind Prediction 2 combines the first-order entry benchmark with the fixed-composition private credit choice. After resolution,

$$\begin{aligned} n_{b,1}^{*} &= \mathbb{1}\{\widetilde{G}_{b,1} \geq F \text{ or } e_b = 0\}\, n_{b,1}^{PC}, \\ n_{b,1}^{PC} &= \frac{\mu_{PC} - (\sigma_{B,PC}/\sigma_B^2)(1-\tau)\mu_{B,0}}{\bar{\gamma}_b\Big(\sigma_{PC}^2 - \sigma_{B,PC}^2/\sigma_B^2\Big)}. \end{aligned} \tag{B.15}$$

Here $\widetilde{G}_{b,1} = f\beta s_{b,1}^{B} g_{b,1}$, with $s_{b,1}^{B}$ the post-resolution bond-only market share; e_b records whether the bank has already paid the initiation cost. The exact problem instead compares maximized shares using equation (30) and computes each portfolio with its own marginal savers. Substituting $g_{b,1}$ into the first-order entry benchmark gives, for a first-time entrant,

$$\frac{s_b}{\bar{\gamma}_b} \;\geq\; \frac{2F\Big(\sigma_{PC}^2 - \sigma_{B,PC}^2/\sigma_B^2\Big)}{f\beta\Big(\mu_{PC} - (\sigma_{B,PC}/\sigma_B^2)(1-\tau)\mu_{B,0}\Big)^2}. \tag{B.16}$$

In this approximation, s_b is the bank's bond-only share at the date of the comparison. Holding the portfolio used to evaluate investor composition fixed, equation (B.3) gives $\partial \log(s_b/\bar{\gamma}_b)/\partial a_b = \bar{\gamma}_b - \mathrm{Var}_b(\gamma)/\bar{\gamma}_b$, positive when $\mathrm{Var}_b(\gamma) < \bar{\gamma}_b^2$. Proposition B.2 gives the stronger condition when that portfolio adjusts. Holding market shares and marginal savers fixed as resolution intensifies, the right side of equation (B.16) falls: a lower bond return makes private credit more valuable and lowers the entry cutoff. The full numerical solution lets both market shares and marginal savers change.

For a bank that already used private credit, $e_b = 0$ and the entry condition is no longer binding. Subtracting its pre-resolution conditional position from equation (B.15), with risks and marginal-saver composition fixed and both portfolios interior, gives

$$\Delta n_b^{PC} = \frac{\tau\, \sigma_{B,PC}\, \mu_{B,0}}{\bar{\gamma}_b\Big(\sigma_B^2\sigma_{PC}^2 - \sigma_{B,PC}^2\Big)} > 0. \tag{B.17}$$

Positive covariance makes bonds and private credit substitutes in the portfolio: reducing bond yield raises the optimal private credit position without another setup payment. The model thus separates expansion by existing users from scale-dependent first entry. The observed increase in private credit can contain both; a newly launched WMP is not itself a first entry by its bank.

Proposition B.5 (Returns and market shares)**.** *Suppose $\bar{\gamma}_L > \bar{\gamma}_S$, as under the sorting condition, neither local bank holds private credit before resolution, and only the large bank enters afterward. Under the maintained fixed-composition and interior-portfolio conditions,*

the entrant's expected return changes relative to the small bank by

$$\Delta R_L - \Delta R_S = \tau(2-\tau)\,\mu_{B,0}\Big(x^B_{S,0} - x^B_{L,0}\Big) + \Big(\mu_{PC} - \frac{\sigma_{B,PC}}{\sigma_B^2}\mu_{B,1}\Big)n_{L,1} > 0, \tag{B.18}$$

and, in the small-share case, to first order in utility changes and holding the national bank's offer fixed, its market share grows faster:

$$\Delta \log s_L - \Delta \log s_S \simeq \beta\Big[\frac{\tau(2-\tau)}{2}\,\mu_{B,0}\Big(x^B_{S,0} - x^B_{L,0}\Big) + g_{L,1}\Big] > 0. \tag{B.19}$$

Both terms are positive. The small bank held more bonds (Proposition B.1), so it loses more bond income, and only the large bank gains from private credit.

The individual-bank decomposition behind Prediction 3 makes these two channels explicit. For either local bank b, which holds no private credit before resolution, holding marginal savers fixed,

$$\begin{aligned}\Delta R_b = \; & -\tau(2-\tau)\,\mu_{B,0}\,x^B_{b,0} \\ & + \mathbb{1}\{b \text{ enters}\}\Big(\mu_{PC} - \frac{\sigma_{B,PC}}{\sigma_B^2}\mu_{B,1}\Big)n^{PC}_{b,1}.\end{aligned} \tag{B.20}$$

The first term is lost bond income; the second is replacement income available only to the entrant. For the two local banks, in the small-share approximation, to first order and holding the national bank's offer fixed,

$$\begin{aligned}\Delta \log s_b - \Delta \log s_{b'} &\simeq \beta\Big(\Delta v_b - \Delta v_{b'}\Big), \\ \Delta v_b &= -\frac{\tau(2-\tau)}{2}\,\mu_{B,0}\,x^B_{b,0} + \mathbb{1}\{b \text{ enters}\}\,g_{b,1}.\end{aligned} \tag{B.21}$$

Here Δv_b is the change in risk-adjusted return evaluated at the bank's fixed marginal-investor risk aversion, and $g_{b,1}$ is the post-resolution gain from private credit. Finally, substituting equation (B.5) into the relative-return comparison gives

$$\begin{aligned}\Delta R_L - \Delta R_S = \; & \tau(2-\tau)\,\frac{\mu^2_{B,0}}{\sigma_B^2}\Big(\frac{1}{\bar\gamma_S} - \frac{1}{\bar\gamma_L}\Big) \\ & + \Big(\mu_{PC} - \frac{\sigma_{B,PC}}{\sigma_B^2}\mu_{B,1}\Big)n^{PC}_{L,1} > 0.\end{aligned} \tag{B.22}$$

The first term is positive under the marginal-investor sorting condition, and the second is positive when only the large local bank enters. At that bank, the conditional share $n^{PC}_{L,1}$ equals its actual private credit share $n_{L,1}$ in Proposition B.5. The national bank, which already holds private credit before resolution, is not part of this entrant–nonentrant comparison.

B.6 Numerical solution of the full equilibrium

The propositions use the small-share case, first-order approximations, and a fixed composition of marginal savers. We now compute a three-bank equilibrium of the full demand system for an illustrative parameterization. Savers' risk aversion is lognormal with log mean 1.0 and log standard deviation 0.6, truncated to $[0.4, 15]$; $\beta = 60$, $f = 1$, $\mu_B = 0.012$, $\mu_{PC} = 0.022$, $\sigma_B^2 = 0.012$, $\sigma_{PC}^2 = 0.040$, $\sigma_{B,PC} = 0.008$; reputations are $a_N = 0$, $a_L = -0.75$, and $a_S = -1.25$; and resolution lowers the bond return by $\tau = 0.5$. Each bank's portfolio is a numerically computed best response to its rivals' portfolios, with market shares obtained by integrating equation (19) over the distribution of risk aversion. The fixed cost $F = 0.0077$ lies in the interval of condition (B.13).

Table B1
Full equilibrium of the model

	Before resolution			After resolution		
	National	Large local	Small local	National	Large local	Small local
Bond share x_b	0.226	0.509	0.629	0.047	0.079	0.307
Private credit share n_b	0.125	0	0	0.160	0.266	0
Market share s_b	0.483	0.073	0.031	0.472	0.077	0.027
Marginal savers' risk aversion $\bar{\gamma}_b$	3.24	1.96	1.59	3.24	1.95	1.63
Expected excess return R_b (bp)	54.6	61.1	75.5	38.1	63.2	18.4
Fee gain from private credit	0.0130	0.0058	0.0032	0.0213	0.0090	0.0048

Notes: Numerically computed Nash equilibrium of the model in Section 6 with the illustrative parameters in the text. The fee gain compares optimized portfolios with and without private credit access, holding rivals' portfolios fixed; the fixed cost is 0.0077. Before resolution only the national bank's gain exceeds the cost; afterward the large local bank's gain does as well, the small bank's does not, and the national bank keeps access without paying again. In the counterfactual equilibrium restricting all three banks to bonds, their pre-resolution bond shares are 0.308, 0.510, and 0.629.

All three predictions hold in this numerical example (Table B1). Bond shares fall with reputation, and the more reputable banks' marginal savers are more risk averse. In the counterfactual with all three banks restricted to bonds, resolution lowers their bond shares by 0.154, 0.259, and 0.321, close to $\tau x_b^B = 0.154$, 0.255, and 0.315. The exact fee gain from private credit is larger at more reputable banks and rises after resolution; only the large local bank newly crosses the fixed cost. The large bank's expected return rises by 59 basis points relative to the small bank's, and the log of its market share relative to the small bank's rises by 0.17. Before resolution, the scale-condition ratio $\mathrm{Var}_b(\gamma)(1+\beta R_b)/\bar{\gamma}_b^2$ is 0.31 at the large local bank and 0.29 at the small one. These pointwise checks accord with the analytic mechanism; the exact fee comparisons establish the entry pattern for this example.

We check each bank before and after resolution, with and without private credit access. Holding rivals' equilibrium portfolios fixed, a search of the feasible set $\{x_b, n_b \geq 0,\ x_b + n_b \leq 1\}$ in steps of 0.002 finds no higher market share. Constrained optimizations from six starting portfolios agree within 10^{-7}; the solutions are feasible, have first-order residuals below 10^{-7},

and have negative-definite Hessians within each regime. Increasing integration accuracy from 400 to 800 Gauss–Legendre nodes leaves every reported table entry unchanged. These checks provide numerical evidence for the best responses, not a general uniqueness or global-optimality proof. The post-resolution large local bank's marginal risk aversion is 1.996 in the bond-only optimum and 1.953 with private credit, illustrating why the exact comparison allows the weights to differ.

Internet Appendix C
Proofs

C.1. Proof of Proposition B.1

In the small-share case, $s_b(\gamma) = \exp(u_b(\gamma))/D(\gamma)$, with $u_b(\gamma) = \gamma a_b + \beta(R_b - \gamma V_b/2)$ and $D(\gamma)$ the denominator in equation (19), whose dependence on bank b's own offer is negligible. The weights defining $\bar{\gamma}_b$ are then proportional to $\exp(\gamma t_b)\, h_b(\gamma)$, where $t_b = a_b - \beta V_b/2$ and $h_b(\gamma) = e^{\beta R_b}\, dG(\gamma)/D(\gamma)$ does not depend on a_b. This is an exponential tilting of a fixed distribution, so the derivative of its mean with respect to the tilt t_b is its variance, which gives $\partial\bar{\gamma}_b/\partial a_b = \mathrm{Var}_b(\gamma)$ holding the portfolio fixed. Similarly, $\partial \log s_b/\partial a_b = \int \gamma\, s_b(\gamma)\, dG / \int s_b(\gamma)\, dG = \bar{\gamma}_b$.

For a bank without private credit, $V_b = \sigma_B^2 (x_b^B)^2$ and $x_b^B = \mu_B/(\bar{\gamma}_b \sigma_B^2)$. Then $d\bar{\gamma}_b/da_b = \mathrm{Var}_b(\gamma)\, dt_b/da_b$, with $dt_b/da_b = 1 - \beta\sigma_B^2 x_b^B\, dx_b^B/da_b$ and $dx_b^B/da_b = -(x_b^B/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$. Substituting the last expression into the first and solving gives equation (B.4).

For the final statement, equation (B.2) shows that $s_b(\gamma)/s_{b'}(\gamma)$ increases with γ when $(a_b - a_{b'}) - \frac{\beta}{2}(V_b - V_{b'}) > 0$. Every feasible portfolio has $0 \le V \le (x\sigma_B + n\sigma_{PC})^2 \le \sigma_{PC}^2$, so $a_b - a_{b'} > \beta\sigma_{PC}^2/2$ guarantees the inequality whatever the portfolios. The distribution of bank b's savers then dominates that of bank b''s savers in the likelihood-ratio order. In the small-share case, a bank's marginal savers have the same distribution as its savers, so $\bar{\gamma}_b > \bar{\gamma}_{b'}$. If neither bank holds private credit, equation (24) gives $x_b^B < x_{b'}^B$. □

C.2. Proof of Proposition B.2

In the small-share case, the integrand in equation (B.7) is approximately $s_b(\gamma)\big(e^{\beta\Delta v_b(\gamma)} - 1\big)$. Linearizing in $\beta\Delta v_b$ gives $f\beta \int s_b(\gamma)\Delta v_b(\gamma)\, dG = f\beta s_b\big(\Delta R_b - \frac{\bar{\gamma}_b}{2}\Delta V_b\big)$, with the weights evaluated at the bond-only optimum. Maximizing this linearized demand gain is equivalent to maximizing Q_b at fixed $\bar{\gamma}_b$. Under the interior conditions, the improvement is g_b in equation (27), so the optimized first-order gain is $\widetilde{G}_b = f\beta s_b g_b$.

For the derivative, the bank's bond share maximizes s_b, so by the envelope theorem $d\log s_b/da_b = \partial \log s_b/\partial a_b = \bar{\gamma}_b$ in the small-share case. Since g_b is proportional to $1/\bar{\gamma}_b$, $d\log g_b/da_b = -(1/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$, giving equation (B.8). Equation (B.4) gives $d\bar{\gamma}_b/da_b = \mathrm{Var}_b(\gamma)/(1 - k_b)$, where $k_b = \beta R_b \mathrm{Var}_b(\gamma)/\bar{\gamma}_b^2$. The condition $\mathrm{Var}_b(\gamma)(1 + \beta R_b) < \bar{\gamma}_b^2$ implies $k_b < 1$ and is equivalent to $\bar{\gamma}_b^2(1 - k_b) > \mathrm{Var}_b(\gamma)$, so the derivative is positive. Integrating along an interior optimum under the stated condition gives $\widetilde{G}_L > \widetilde{G}_S$. Comparing these first-order gains with F establishes the entry ordering in the approximation. Exact entry requires the optimized-share comparison in equation (B.7). □

C.3. Proof of Proposition B.3

By equation (24) with $\bar{\gamma}_b$ fixed, $x^B_{b,1} = (1-\tau)\mu_{B,0}/(\bar{\gamma}_b\sigma^2_{B,1}) = (1-\tau)(\sigma^2_{B,0}/\sigma^2_{B,1})\,x^B_{b,0}$, which gives equation (B.9). The bracket is negative if and only if $(1-\tau)/\sigma^2_{B,1} < 1/\sigma^2_{B,0}$, which is condition (B.10). For a bank that holds private credit in both states, solving equation (23) gives $x_b = (\sigma^2_{PC}\mu_B - \sigma_{B,PC}\mu_{PC})/[\bar{\gamma}_b(\sigma^2_B\sigma^2_{PC} - \sigma^2_{B,PC})]$; with unchanged risks and μ_{PC}, the change is $-\tau\sigma^2_{PC}\mu_{B,0}/[\bar{\gamma}_b(\sigma^2_B\sigma^2_{PC} - \sigma^2_{B,PC})]$. □

C.4. Proof of Proposition B.4

With $\mu_B = (1-\tau)\mu_{B,0}$ in equation (27),

$$\frac{\partial g_b}{\partial \tau} = \frac{\big(\mu_{PC} - (\sigma_{B,PC}/\sigma^2_B)\mu_B\big)(\sigma_{B,PC}/\sigma^2_B)\mu_{B,0}}{\bar{\gamma}_b\big(\sigma^2_{PC} - \sigma^2_{B,PC}/\sigma^2_B\big)} = \frac{\sigma_{B,PC}}{\sigma^2_B}\,\mu_{B,0}\,n^{PC}_b,$$

using equation (26). It is positive because $\mu_{PC} - (\sigma_{B,PC}/\sigma^2_B)\mu_B \geq \mu_{PC} - (\sigma_{B,PC}/\sigma^2_B)\mu_{B,0} > 0$. Equivalently, by the envelope theorem, a change in μ_B changes the best risk-adjusted return by the bonds held, x^{PC}_b with private credit and x^B_b without it, so $\partial g_b/\partial\mu_B = x^{PC}_b - x^B_b = -(\sigma_{B,PC}/\sigma^2_B)n^{PC}_b$. Because the first-order fee gain $f\beta s_b g_b$ increases with g_b, it is larger after resolution at any given share and composition of marginal savers.

For the entry pattern, every F in the interval (B.13) satisfies $G_{N,0} \geq F$, $G_{L,0} < F \leq G_{L,1}$, and $G_{S,1} < F$. The maintained pre-resolution ordering $G_{S,0} \leq G_{L,0}$ also implies $G_{S,0} < F$. Thus only the national bank enters before resolution; afterward, the large local bank enters and the small bank remains out. All gains are evaluated against rivals' portfolios in the corresponding regime. The interval is nonempty precisely when $\max\{G_{L,0}, G_{S,1}\} < \min\{G_{L,1}, G_{N,0}\}$.

For an entrant, equation (26) gives $x_{b,1} = (1-\tau)x^B_{b,0} - (\sigma_{B,PC}/\sigma^2_B)n_{b,1}$, so $\Delta(x_b + n_b) = -\tau x^B_{b,0} + (1 - \sigma_{B,PC}/\sigma^2_B)n_{b,1}$, which is equation (B.14). □

C.5. Proof of Proposition B.5

Write $\chi = \sigma_{B,PC}/\sigma^2_B$ for the bonds displaced per unit of private credit. A bank without private credit earns $R_b = \mu_B x^B_b = \mu^2_B/(\bar{\gamma}_b\sigma^2_B)$, so its expected return changes by $\big((1-\tau)^2 - 1\big)\mu_{B,0}x^B_{b,0} = -\tau(2-\tau)\mu_{B,0}x^B_{b,0}$. After resolution, the entrant holds $x^B_{L,1} - \chi n_{L,1}$ in bonds and $n_{L,1}$ in private credit, so its expected return is $\mu_{B,1}x^B_{L,1} + (\mu_{PC} - \chi\mu_{B,1})n_{L,1}$, and its change is $-\tau(2-\tau)\mu_{B,0}x^B_{L,0} + (\mu_{PC} - \chi\mu_{B,1})n_{L,1}$. Subtracting the small bank's change gives equation (B.18). The second term is positive because the entrant holds private credit only when $\mu_{PC} > \chi\mu_{B,1}$, and the first is positive because $x^B_{S,0} > x^B_{L,0}$ by Proposition B.1.

In the small-share case, $\log s_b(\gamma) = u_b(\gamma) - \log D(\gamma)$. Holding the national bank's offer fixed, $D(\gamma)$ does not change to first order, because the local banks' own shares are small. A

change in the bank's portfolio then changes $\log s_b$ by β times the change in the risk-adjusted return to its savers, evaluated at their average risk aversion $\bar{\gamma}_b$. The bond-only risk-adjusted return at $\bar{\gamma}_b$ is $\mu_B x_b^B/2$, which changes by $-\tau(2-\tau)\mu_{B,0}x_{b,0}^B/2$, and the entrant gains $g_{L,1}$ in addition. Subtracting gives equation (B.19). □

Internet Appendix D
Specifications and Supplementary Evidence

This appendix supplies the measurement and specification details behind the argument. Subsections D.1–D.2 establish how private credit relates to LGFV borrowers and how the two financing channels are compared. Subsections D.3–D.7 explain the supporting investment, demand, and performance tests. Subsection D.8 reconciles market-wide exposure with shares of included debt holdings. Subsection D.9 and Tables D1, D2, and D3 separate initial specialization from subsequent replacement. Tables D4–D5 assess the financing-rate comparison; Figure D1 explains its 2023Q1 movement in rate levels. Return and business-balance checks are in Internet Appendix E.

D.1. Who borrows through private credit?

The substitution argument concerns a change in financing instrument within the local borrower sector. To identify who receives private credit financing, we extract borrower names from asset descriptions, locate them using company registration records, and match them to LGFV bond issuers over 2022Q1–2025Q3. These matches account for about 60% of local banks' home private credit value before the program and 59% afterward. Adding a keyword screen for financing-platform names raises the shares to 79% and 86%; among the largest FY2021 size tercile, the corresponding broad shares are 77% and 85%. Most of this substitution therefore remains within the LGFV sector, although sector-level matches do not trace each bond into a contract with the same legal borrower.

D.2. Spread compression and financing-rate comparisons

The descriptive issuance averages in Section 4 compare 7,128 pre-implementation LGFV issues in 2022Q1–2023Q3 with 9,763 post-implementation issues in 2023Q4–2025Q4, without conditioning on WMP ownership. Each issue receives equal weight. This sample differs from the issuer-type comparison in Table A15. We retain fixed-rate, at-par, nonstructured, nonconvertible, nonperpetual, nonsubordinated bonds, so the reported issue rate measures yield at issuance. Each issue's spread subtracts the Treasury-curve yield on its issue date at the nearest available maturity node. The before period includes the announcement quarter, consistent with implementation beginning in 2023Q4. These descriptive averages measure the change in yields available on new issues.

To trace the timing of compression, we interact final refinancing intensity with quarter indicators. For province p in quarter q, the event-study specification is

$$\text{Bond spread}_{pq} = \alpha_p + \alpha_q + \sum_{k \neq q_0} \beta_k \, \text{Final refinancing}_p^{GDP} \mathbb{1}\{q = k\} + \varepsilon_{pq}, \quad q_0 = 2023\text{Q2}. \tag{D.1}$$

Bond spread_{pq} is the province's secondary-market spread in basis points. Final intensity $\text{Final refinancing}_p^{GDP}$ is cumulative special-refinancing bond issuance at the last observed date, as a percentage of 2020 provincial GDP; it is fixed across quarters. Province and quarter effects remove persistent provincial differences and common movements. The plotted β_k measures additional spread change since 2023Q2 per percentage point of final intensity (Figure 1). The sample is 2022Q1–2026Q1; errors cluster by province. Spreads and intensity are winsorized at the 1st and 99th percentiles, and province effects absorb the intensity level.

Pre-announcement estimates are imprecise around zero. A province with one percentage point higher final intensity experiences 6.5 basis points more compression by 2023Q4 and 18.6 by 2025Q4. The difference deepens through 2024, indicating a persistent loss of additional yield where refinancing is greater.

The regression in Table A5 uses cumulative refinancing completed by each date rather than a fixed final exposure:

$$\text{Bond spread}_{pq} = \beta \, \text{Refinancing}_{pq}^{GDP} + \alpha_p + \alpha_q + \text{Trend controls}_{pq} + \varepsilon_{pq}. \tag{D.2}$$

where the outcome is the province-quarter LGFV bond spread in basis points and intensity is cumulative refinancing as a percentage of 2020 provincial GDP. The effects α_p and α_q compare changes within provinces after removing common quarterly movements. Without trend controls, the coefficient is -17.67 basis points per percentage point of refinancing/GDP. The term $\text{Trend controls}_{pq}$ adds either the pre-program spread interacted with a linear trend or province-specific linear trends, as reported in each column. These additions distinguish the refinancing association from convergence related to initial spreads and from existing provincial trends. Both reduce its magnitude but retain a negative association. Table A5 thus relates compression to refinancing, while Figure A4 shows the decline in the initial high–low spread gap.

To assess the income available from a substitute, the next comparison places private credit yields alongside bond issuance yields for the same reporting issuers. Both are annual contractual rates on new financing. For bonds, the source records the stated issue rate; these comparisons do not use secondary-market yields or infer changes in the contractual payments

on outstanding bonds. Table 4, Panel A, estimates

$$\begin{aligned}\text{Yield}_{jfq} = \alpha_{fq} + \alpha_{f,d(j)} + \alpha_{\text{mat}(j)} \\ + \sum_{t \neq 2022} \delta_t \, \mathbb{1}\{\text{Private credit}\}_j \mathbb{1}\{\text{Issue year}_j = t\} + e_{jfq}.\end{aligned} \tag{D.3}$$

Here j is a contract or bond issue, f its reporting issuer, and q its origination quarter; Yield_{jfq} is the annual contractual rate in percentage points. The functions $d(j)$ and $\text{mat}(j)$ identify its debt type and maturity bin. The effects α_{fq} compare financing within an issuer and quarter; $\alpha_{f,d(j)}$ absorb persistent differences between that issuer's financing channels. The year indicator uses the record's origination year, with t indexing calendar years. Thus δ_t measures the change in the private credit-minus-bond gap from 2022. The gap widens by 1.30 percentage points in 2024. This reporting-issuer sample can include financing by an issuer's affiliates, unlike the exact-borrower event study.

The separate-channel columns in Panel A instead regress each record's rate on origination-year indicators, with 2022 omitted. The private credit regression absorbs borrower-by-creditor and maturity-bin effects; the bond regression absorbs issuer-by-bond-type-by-placement and maturity-bin effects. Their year coefficients therefore compare rates within the relevant financing relationships or issue types. Both cluster errors by reporting issuer. Each channel regression uses its own sample; the gap regression estimates the relative change directly.

The quarterly comparison in Figure 3 instead holds the legal borrower fixed. It requires the funded borrower's DM code to equal the reporting issuer's code and matches bonds on that same code. Its initial pool contains borrower-quarters with both channels and borrowers with at least two matched quarters. We then require observed continuous maturity, restrict financing to 2021 onward, and remove borrower-month singletons. The resulting sample contains 17,303 financing records at 589 borrowers in 26 provinces. Contracts are dated by origination rather than disclosure; repeated contracts and physical bond issues are deduplicated, without further rate winsorization or outcome-based trimming. Because these restrictions can remove one channel, not every retained month or quarter still contains both. Mixed-channel months identify the direct rate gap, while other multi-observation months also inform the maturity slopes.

Equation (7) compares annual rates within borrower-by-calendar-month cells, controlling for continuous maturity with a separate slope in each quarter. It does not include borrower-by-debt-type effects. Quarterly coefficients estimate gap levels; the plotted changes subtract the 2023Q2 gap using the full covariance matrix. Table D4 asks whether the widening depends on maturity adjustment, financing frequency, or borrower composition. Column 2 allows separate maturity slopes by debt type and quarter and evaluates the gap at a common three-year maturity. Column 3 gives each borrower-quarter-debt-type cell total weight one, using inverse-frequency weights on the baseline sample. Column 4 retains the 101 borrowers with both channels in 2023Q2 and at least one later quarter after the baseline restrictions,

without requiring a balanced panel. All four comparisons show a wider gap in 2024Q4 and 2025Q3. The event study documents the evolution of the yield gap within legal borrowers, including the earlier movements visible before the announcement.

The pre-program dip in 2023Q1 raises a separate question: did private credit yields fall, or did bond issuance yields rise? Figure D1 plots both rates for the same 22 legal borrowers, each receiving equal weight in every quarter from 2022Q4 through 2023Q3. Each borrower has both channels in at least one common month of every quarter. Bond issuance yields rise from 4.25% in 2022Q4 to 4.78% in 2023Q1, then fall to 4.49% in Q2. Private credit rates instead move from 6.27% to 6.32% and 6.40%. The Q1 narrowing therefore reflects higher bond issuance yields rather than a collapse in private credit yields. A broader fixed set of 76 borrowers observed in both Q1 and Q2 gives the same direction: bond issuance yields fall 47 basis points while private credit rates rise 4. These descriptive comparisons fix borrowers and apply identical weights to the two channels, but contracts and maturities can change; the conditional regression separately adjusts for maturity.

Table D5 asks whether the late-period widening depends on how repeated financing records and regional dependence are treated. The baseline uses one-way province clustering, allowing arbitrary dependence across borrowers and months within a province. Figure intervals are $\hat{\delta}_q \pm t_{25,.975}\, SE(\hat{\delta}_q)$, with 26 province clusters. Additional inference checks use borrower-clustered errors, a delete-one-province jackknife, and a null-imposed province wild bootstrap with 9,999 Webb-weight draws. Bootstrap confidence sets invert bootstrap-t tests. For the jackknife, the variance is $(G-1)G^{-1}\sum_p(\hat{\delta}_{q,(-p)} - \bar{\delta}_{q,(-p)})^2$, where $G = 26$ and the bar is the mean leave-out estimate; intervals use t_{25}. All 26 leave-one-province estimates remain positive for each reported quarter. Panel A reports the main late-period contrasts and their province-bootstrap confidence sets.

The second regression column of Table D5 gives each borrower-month with both channels one observation. For legal borrower f in calendar month m, let Yield gap_{fm} be its mean PC yield minus mean bond issuance yield and Maturity gap_{fm} its mean PC maturity minus mean bond maturity. As in the main event study, $q(m)$ identifies the quarter containing month m. We estimate

$$\begin{aligned}\text{Yield gap}_{fm} = &\sum_q b_q \mathbb{1}\{q(m) = q\} \\ &+ \sum_q \eta_q \,\text{Maturity gap}_{fm} \mathbb{1}\{q(m) = q\} + u_{fm}.\end{aligned} \tag{D.4}$$

and report $b_q - b_{2023Q2}$. Each channel mean weights contracts or physical bond issues equally. Differencing removes the common borrower-month condition, so no borrower effect is added. This regression uses 2,525 borrower-months at 557 borrowers in 25 provinces. It tests whether the wider gap remains when each paired month contributes one observation rather than several financing records. The weighting and maturity adjustment differ from the contract-level

regression. Province-clustered errors and 9,999 Webb-weight province draws support widening in both reported quarters. Panel B reports the underlying financing records, borrowers, and provinces by quarter; no usable 2025Q4 comparison survives the monthly design. Together, these checks assess sampling uncertainty around the observed yield divergence.

The priority-province comparison in Table 4, Panel B, uses the same issuer-years for each channel and the difference between them:

$$\begin{aligned}\text{Rate}_{ft}^{d} = \alpha_f^d + \alpha_t^d + \beta_d \mathbb{1}\{\text{Priority province}\}_{p_f} \mathbb{1}\{t = 2024\} \\ + \psi_d \mathbb{1}\{\text{Priority province}\}_{p_f} \mathbb{1}\{t = 2023\} + u_{ft}^d.\end{aligned} \tag{D.5}$$

The index f again denotes a reporting issuer and t a calendar year. Superscript $d \in \{PC, B, PC - B\}$ labels the outcome: private credit yield, bond issuance yield, or their difference; it is not a date. The province p_f belongs to that issuer, not the WMP bank. The priority group comprises Tianjin, Inner Mongolia, Liaoning, Jilin, Heilongjiang, Guangxi, Chongqing, Guizhou, Yunnan, Gansu, Qinghai, and Ningxia. Issuer and year effects enter separately for each outcome, and the 2023 interaction allows a transition-year contrast. The coefficient β_d is the additional 2024 change in priority provinces relative to other provinces. Amount-weighted component rates are winsorized before subtraction, so their changes add to the gap on this common sample. Relative to other provinces, the 2024 gap widens by 1.04 percentage points: bond issuance yields fall an additional 0.73 points while the private credit rate change is positive but imprecise. The provincial comparison reinforces the distinction between compression in marketable bonds and the yield remaining on private credit.

D.3. Redemption terms and local holdings

The redemption comparison asks whether local banks place more home investments in products that allow them to hold less liquid assets (Table A7; equation (3)). The dependent variable is Portfolio home overweight$_{iq}^{n}$ in equation (D.12): the share of the product's included bonds and private credit invested in home class n, minus the corresponding province's share of nationwide included holdings. The all-credit column sums this tilt across classes, giving Portfolio home overweight$_{iq}$. Products with no position in a class retain a zero raw allocation before benchmark subtraction. Controls are product size and risk; bank assets, nonperforming loans, cost-to-income ratio, and net interest margin; and bank-home-city GDP, growth, LGFV debt to GDP, and income. Quarter effects remove common movements. The local-bank coefficient compares daily-open products; its interaction with restricted redemption measures how much wider the local–national allocation gap is in that segment. The additional gap is 20.0 percentage points overall, 7.7 for LGFV bonds, and 7.8 for private credit. Together with local banks' 80.2% asset share in restricted-redemption products (Table A6), the estimates link their product design to their home investment strategy.

D.4. Home-bond allocation and refinancing pressure

The destination comparison in equation (10) asks whether banks reduce home bond investment more than other buyers facing the same local bond supply (Table 6).

In this destination panel, a indexes the buying WMP-issuing institution (a bank or subsidiary), c the issuer city, and q the calendar quarter. Home identifies a local institution investing in a city of its underlying bank's home province; Post begins in 2023Q4. The effects μ_{ac}, μ_{aq}, and μ_{cq} hold persistent buyer–city ties, buyer-wide conditions, and issuer-city conditions fixed. The baseline outcome is LGFV bond holdings in the city as a percentage of the buyer's current bond holdings. Other columns measure net holdings changes relative to current bond holdings and log one plus new purchases in RMB million. The Home–Post coefficient therefore compares changes by home and outside buyers of the same city's bonds. The estimated 0.45-point decline is about 12% of the initial mean allocation to cities in the bank's home province. Replacing Home with separate own-city and rest-of-home-province indicators gives declines of 1.94 and 0.24 percentage points, respectively. Net holdings changes and new purchases also fall. Adding the interaction of Home buyer, Post, and the city's pre-program LGFV spread in percentage points to the new-purchase regression gives a larger purchase reduction in formerly higher-spread cities.

The related event study in Figure 4 aggregates issuer cities to provinces and replaces Home–Post with Home interacted with each quarter:

$$\begin{aligned}\text{LGFV bond share}_{apq} = &\sum_{k \neq q_0} \eta_k \, \mathbb{1}\{\text{Home buyer}\}_{ap} \mathbb{1}\{q = k\} \\ &+ \mu_{ap} + \mu_{aq} + \mu_{pq} + u_{apq},\end{aligned} \qquad q_0 = 2023\text{Q2}. \tag{D.6}$$

Province p is the investment destination. Its LGFV bonds are measured as a percentage of the buyer's total bonds, and Home identifies the underlying bank's own province. Buyer–province, buyer–quarter, and destination-province–quarter effects are the provincial counterparts of the city design. Each η_k is the change in the home–outside allocation gap since 2023Q2, not a holding level. The sample is 2022Q1–2025Q3; outcomes are winsorized at the 1st and 99th percentiles and errors cluster by bank city. Home levels and time-varying bank/province controls are absorbed. Pre-announcement confidence intervals include zero; the relative home allocation falls by about 2.3 percentage points in 2024Q1, 4.2 by 2024Q4, and 6.1 by 2025Q3. These province-wide changes differ from the per-city estimates in Table 6.

The maturity-wall test addresses a different source of adjustment: whether home banks provide additional financing when local borrowers have more principal due. In Table A9, the allocation specification is

$$\begin{aligned}\text{Outcome}_{acq} = \beta\, \mathbb{1}\{\text{Home buyer}\}_{ac} \times \text{Maturity wall}_{cq} \\ + \mu_{ac} + \mu_{aq} + \mu_{cq} + \varepsilon_{acq}.\end{aligned} \tag{D.7}$$

The buyer, city, and quarter indices are the same as in equation (10). The wall varies by issuer city c and quarter q: it is principal on all LGFV bonds maturing there as a percentage of 2020 city budget revenue. Home identifies local banks investing in their own province. Bank-by-city, bank-by-quarter, and issuer-city-by-quarter effects absorb persistent ties, common funding conditions, and destination shocks.

Panel B of Table A9 asks what kinds of bonds home banks add when maturities are high. For bond j newly appearing in buyer a's holdings in quarter q, the specification is

$$\begin{aligned}\text{Bond characteristic}_{jaq} = \beta_H\, \mathbb{1}\{\text{Home buyer}\}_{a,c(j)} \\ + \beta_W\, \mathbb{1}\{\text{Home buyer}\}_{a,c(j)} \times \text{Maturity wall}_{c(j),q} \\ + \mu_{aq} + \mu_{c(j),q} + \varepsilon_{jaq}.\end{aligned} \tag{D.8}$$

The function $c(j)$ gives the bond issuer's city. Each regression uses one purchased-bond characteristic as its outcome: issuer rating, guarantee status, maturity, issue spread, or log issue amount. The Home level is included because buyer–city effects are absent; the Wall level is absorbed by issuer-city-by-quarter effects. The coefficient β_W asks whether the home–outside difference in that characteristic changes with scheduled maturities. Both panels use bank-city clustered errors and the same unstandardized percentage-point wall measure.

The home–wall coefficients are -0.00696 for net buying and 0.00446 for exits, with both outcomes in percentage points. A ten-percentage-point larger wall therefore predicts about 0.070 points less net buying and 0.045 points more exits by home banks relative to other buyers of the same city's bonds. Exits include nonrenewal at maturity.

Among new purchases, the guarantee-indicator interaction is 0.00116 per percentage point of the wall. A ten-point larger wall thus widens the home–other buyer gap in guarantee probability by 1.16 percentage points. Issuer-rating and maturity interactions are also positive. Home banks therefore reduce net financing and seek more protection as repayments come due: their support for local borrowers is selective.

D.5. Product assets and net flows

The demand tests distinguish money entering a product from growth generated by its investments. We infer product assets as 100 times the median ratio of a disclosed position's

value to its percentage weight. Net flow then removes investment returns from asset growth:

$$\text{Net flow}_{iq} = 100\frac{AUM_{iq} - AUM_{i,q-1}(1+R_{iq})}{AUM_{i,q-1}}, \tag{D.9}$$

where R_{iq} is the simple quarterly return expressed as a fraction. A value of five means net inflows equal 5% of beginning assets. We apply the baseline 1st–99th percentile exclusion to monthly log returns before compounding, require three consecutive monthly returns per share class, and average class-level quarterly returns within each product. Both assets and returns must be observed at the required consecutive quarter ends. Flow tails are capped at the global 1st and 99th percentiles before selecting complete regression observations. These disclosure-based flows capture net investor transactions among continuing observed products; redemptions from products that disappear from the holdings data are outside this sample.

The ranking comparison in Table A10 uses equation (5) in the main text. Products are ranked within their prior risk-rating and redemption category $g(i, q-1)$, with category-by-outcome-quarter effects. Peer sets include national and local banks, require at least 20 products and three underlying banks, and are formed before selecting on subsequent flow availability. Tied returns receive midranks. Column 2 also absorbs home-province-by-outcome-quarter effects. These are cross-bank comparisons without bank effects.

The eleven control variables are log product assets, risk, redemption, log bank assets, nonperforming-loan ratio, cost-to-income ratio, net interest margin, and home-city GDP, GDP growth, LGFV debt/GDP, and urban income; we also include a local-bank indicator. Product assets are measured at the beginning of the flow quarter, and the remaining controls in the preceding quarter. Risk and redemption levels are absorbed by the category effects. Standard errors are clustered by underlying bank, combining a bank and its wealth-management subsidiary. The outcome sample covers 2022Q3–2025Q3; earlier observations supply lags.

Alternative outcomes test whether a few unusually large flow rates drive the ranking result. Write $\widehat{A}_{iq} = AUM_{i,q-1}(1+R_{iq})$. Symmetric flow is $200(AUM_{iq} - \widehat{A}_{iq})/(AUM_{iq} + \widehat{A}_{iq})$, bounded between -200 and 200 without tail capping. Positive flow is $100\,\mathbb{1}\{AUM_{iq} > \widehat{A}_{iq}\}$, so its coefficient is a percentage-point difference in the probability of net inflows. The main top–bottom flow difference is 12.2% of beginning assets, and the positive-flow probability differs by 8.1 points. The full-market ranking comparison measures the performance–flow relationship across disclosed products.

A linear benchmark replaces the quintile indicators with the previous quarter's return in percentage points, retaining the controls and category-by-quarter effects. Its coefficient is 9.381 (SE 2.064; 123,804 observations): ten basis points more quarterly return predicts next-quarter inflows higher by 0.94% of beginning assets. The ranking comparison spans a larger return difference, averaging 1.275% in the top fifth versus 0.246% in the bottom fifth. Both designs use the same return construction and lagged controls, with samples determined by their respective data requirements.

This flow comparison and the bank-balance comparison measure different margins. The 10.3-basis-point annualized size-related return gap in Table 8 is about 2.6 quarterly basis points. Applying the linear coefficient gives $9.381 \times (10.3/4/100) = 0.24$ percentage points more next-quarter inflow, not the roughly 11% relative WMP-balance growth in Table 10. The former concerns transactions in continuing observed products; the latter includes new products, maturities, and investment performance over several reporting dates. We use the flow evidence to establish the commercial relevance of returns and the balance evidence to document business divergence, rather than treating one as a quantitative decomposition of the other.

D.6. Full-period returns and downside outcomes

The return premium describes average reported performance. To assess what savers observed beyond that average, Table A8 also examines volatility and downside outcomes. Its full-period specification is

$$\begin{aligned}\text{Outcome}_{im} = \beta\, \text{Portfolio home overweight}_{i,h(m)}^{LGFV} + \gamma'\, \text{Controls}_{i,h(m)} \\ + \lambda\, \mathbb{1}\{\text{Local bank}\}_b + \alpha_m + \varepsilon_{im}.\end{aligned} \tag{D.10}$$

The dependent variable Outcome_{im} is monthly NAV return or one of the reported volatility and downside measures. The exposure $\text{Portfolio home overweight}^{LGFV}$ is the home-LGFV bond share of included holdings minus the corresponding provincial benchmark, measured as a fraction at the preceding quarter end. The eleven baseline controls, a local-bank indicator, and month effects enter; errors cluster by bank-city-by-month cells. Returns are consecutive-month changes in log adjusted NAV, and trailing annualized volatility uses the twelve-month window ending in the current month, with at least two available returns. Panel B measures excess returns, downside volatility, drawdowns, and the frequency of negative or unusually low monthly returns, as defined in the table notes.

The question is whether the strategy delivers higher reported performance only by exposing savers to more frequent visible losses. The full-period monthly return coefficient is 3.14 basis points per unit of tilt (SE 0.78), or about 8 annual basis points for a 20-percentage-point difference. For the same difference, the probability of a negative month is 0.8 percentage points lower, against a 9.9% mean. These are associations in reported NAVs, conditional on bank type and the other controls. They describe the full-period experience of the local-bond strategy; Table 3 separately estimates how its return premium changes after the program.

D.7. Pricing, distress, and sample timing

The ownership comparisons in Tables A12 and A13 use equation (6) in Section 3.5. Bond j is observed in its issue quarter q, and $c(j)$ identifies its issuer city. The outcome is either the

public valuation yield minus the issue yield in basis points or the subsequent issuer-distress indicator scaled to 100. Both ownership shares are in percentage points of issue size, measured at issue-quarter end. The main specification includes issuer-city-by-issue-quarter effects; the tables also report comparisons with issue-quarter or issuer-province-by-quarter effects in their place. The contrast $\beta_C - \beta_P$ shifts ownership from other-city provincial banks to same-city banks while holding their combined share fixed. Bond and ownership controls are listed in the tables; distress adds the public valuation spread to compare issuers with similar observed ex ante risk.

The pricing sample ends in 2025Q3, the last quarter of holdings coverage. We exclude 977 otherwise eligible LGFV issues in 2025Q4 rather than treating unobserved ownership as zero. The common LGFV sample has 21,116 issues, with 20,526 remaining after removing singleton city-quarter cells. With bond and ownership controls, the same-city-minus-other-city contrast is 0.78 basis points with issue-quarter effects and 1.01 with issuer-city-by-quarter effects. For the latter, the robust standard error is 0.337 ($p = 0.0026$) and the issuer-clustered standard error is 0.355 ($p = 0.0044$). Table A12 uses the holdings-covered sample throughout. Its non-LGFV column gives a small, imprecise contrast of 0.11 basis points. The pricing difference between the two local ownership groups is therefore concentrated in LGFV bonds, where local public-financing relationships are most relevant.

D.8. The scale of the WMP–LGFV bond link

The introduction reports the share of identified standard-bond holdings invested in LGFV bonds. We sum reported LGFV bond values and divide by the reported value of all positions identified as standard bonds within each bank group over 2022Q1–2023Q3, retaining government and policy-bank bonds and excluding private credit. Using valid product identifiers and the current underlying-bank classification gives 38.5% for local banks and 13.2% for national banks. This broader sample contains 46,772 local-bank and 45,346 national-bank product-quarters. By contrast, Table D1 uses matched product-quarters with positive included debt holdings: 36,194 and 34,595, respectively. Within that matched local-bank sample, LGFV bonds account for 40.4% of identified standard-bond value, 47.5% after restricting the denominator to eligible standard bonds, and 43.1% after adding eligible private credit. The table's holding-value weights also reproduce 43.1%. The bond-only and debt-allocation shares therefore describe different portfolios, not conflicting estimates of one share.

For market context, we reconstruct the year-end 2021–2024 top-ten holdings cross-sections from the original Wind exports. We retain one representative record per product registration code and deduplicate positions within that record, preserving identical investments made by distinct products. We restrict amounts to yuan-denominated positions, identify bonds by asset type, and match issuers to the January 4, 2026 LGFV roster. Unidentified bond issuers remain in the denominator but not the LGFV numerator. In 2024, the retained disclosures

contain RMB 0.3859 trillion of LGFV bonds out of RMB 2.3459 trillion of bonds, including negotiable certificates of deposit: a ratio of sums of 16.45%, not an equally weighted average of product ratios.

For each year t, let q_t be this disclosed LGFV-to-bond ratio, B_t the industry's bond assets including negotiable certificates of deposit, and A_t its net assets under management, from the annual WMP market reports. Estimated LGFV bond holdings are $\widehat{H}_t = B_t q_t$, and their share of WMP assets is $\widehat{H}_t/A_t$. The annual WMP market report published in 2025 gives year-end 2024 values of $B_t = 18.60$ and $A_t = 29.95$ trillion yuan; using unrounded holdings gives $\widehat{H}_t = 3.06$ trillion yuan, or 10.2% of net assets under management. Equivalently, the rounded inputs give $16.45\% \times (18.60/29.95) = 10.22\%$. The extrapolation applies the disclosed bond mix to the industry's bond assets. It excludes private credit and indirect exposure through funds and other investment vehicles.

This total-asset benchmark is not a weighted average of the 43.1% and 12.6% in Table D1. Those are holding-value-weighted shares for local and national banks in 2022Q1–2023Q3, conditional on positive included debt holdings. On a common date and sample, the accounting identity is LGFV/AUM = (LGFV/Debt) × (Debt/AUM), where Debt is the value of included debt holdings. Total WMP assets also include deposits, funds, central-government and policy-bank securities, and other assets outside these holdings. The industry benchmark uses a later date and the year-end bond-disclosure sample, so the initial holding shares cannot be directly aggregated into Figure A5.

The reverse ratio is $\widehat{H}_t/O_t$, where O_t is the published year-end LGFV bond balance. Lianhe Ratings (2024) reports Wind balances of RMB 9.81, 10.62, and 11.48 trillion for 2021–2023. Zhang (2025) reports RMB 10.88 trillion for 2024, implying a 28.1% benchmark holding share that year. Figure A5 leaves the 2024 reverse-ratio point unconnected to mark the change in source. These ratios describe two different economic exposures. Under the extrapolation, LGFV bonds account for about one-tenth of WMP assets in 2024, but WMPs hold more than a quarter of the LGFV bond market. A portfolio allocation within a large savings industry can therefore be an important financing source for the borrowers, even though local bonds do not dominate the industry's total assets.

D.9. Initial credit specialization and subsequent substitution

The allocation tests distinguish two comparisons. Before the program, do smaller banks invest more in LGFV bonds and larger banks more in private credit across the full banking market? Afterward, do larger local banks increase private credit more than smaller local rivals? Table D1 describes the initial portfolios, Table 1 estimates the all-bank size relationship, and Table D2 tests its subsequent change within local banks. Table D3 compares the largest tenth with the remaining local banks.

The initial comparison uses equation (1). Outcomes are percentage-point shares of included bonds and private credit, retaining products with zero holdings in a class. The fixed size measure Log bank size$_b$ is log FY2021 assets in RMB trillion and replaces contemporaneous assets, with the other ten covariates grouped as Controls. Quarter effects remove common movements. The regression weights by the value of included holdings and clusters by parent bank. It does not include a national-bank indicator, so the size coefficient captures differences both across bank types and within them. Across this market, larger banks hold less in LGFV bonds, especially home bonds, and more in private credit overall. The relationship with home private credit is much smaller because national banks hold most of their private credit outside their headquarters provinces.

We use two ratios that answer different questions. The first measures geographic concentration within an asset class. Product i is observed in quarter q, j indexes its disclosed positions, and n labels the asset class. Let w_{ijq} be the position's holding weight, p_b the underlying bank's home province, p_j the borrower's province, and $\mathcal{C}_{iq}^n$ the product's positions in class n. Then

$$\text{Home share}_{iq}^n = \frac{\sum_{j \in \mathcal{C}_{iq}^n} w_{ijq} \mathbb{1}\{p_j = p_b\}}{\sum_{j \in \mathcal{C}_{iq}^n} w_{ijq}}, \tag{D.11}$$

is the class's home share. Within-category home bias is its deviation from the market-portfolio weight:

$$\text{Within-category home bias}_{iq}^n = \text{Home share}_{iq}^n - \text{Province share}_{p_b,q}^n.$$

Here $\text{Province share}_{p_b,q}^n$ is the fraction of nationwide disclosed class-n holdings invested in borrowers in the bank's home province. Both the product ratio and the benchmark use class-n holdings alone as their denominator. Zero within-category home bias means investing at market weights within that class. The measure uses products with positive holdings in the class.

The second ratio measures the share of all included debt holdings invested in each home asset class:

$$\text{Home allocation}_{iq}^n = \frac{\sum_{j \in \mathcal{C}_{iq}^n} w_{ijq} \mathbb{1}\{p_j = p_b\}}{\sum_{j \in \mathcal{C}_{iq}} w_{ijq}}, \tag{D.12}$$

$$\text{Portfolio home overweight}_{iq}^n = \text{Home allocation}_{iq}^n - \text{Province allocation}_{p_b,q}^n.$$

Here $\mathcal{C}_{iq}$ pools the analyzed bonds and private credit, and $\text{Province allocation}_{pq}^n$ is the fraction of nationwide holdings across these classes invested in class n from borrowers in province p. Thus the product ratio and its benchmark both use the broader debt denominator, rather than class n alone. A raw Home allocation$^n = 0.20$ means RMB 20 in that home asset class per RMB 100 of analyzed bonds and private credit. Products with no holdings in the class have zero raw allocation and remain in the sample before the benchmark is subtracted. Neither measure uses total WMP assets. Within-category home bias^n measures

where a product invests in an asset class; Portfolio home overweight^n also reflects how much it invests in that class. Interpretation therefore follows each measure's denominator: the asset class for within-category home bias, and analyzed bonds and private credit for portfolio home overweight. The total Portfolio home overweight sums across classes. Combined home exposure, used later, includes home LGFV bonds and home PC but not home other bonds.

The geographic-specialization comparison in Table 2 uses equation (2) in the main text. The outcome is within-category home bias: the product's within-class home allocation in excess of its home province's weight in the nationwide market portfolio. Positive class holdings are required. The coefficient compares local with national banks, conditional on the controls listed in the table.

For subsequent adjustment within local banking, Table D2 uses equation (12) in the main text, with total or home PC as a percentage of included debt holdings as the outcome. Post begins in 2023Q4. The baseline uses the initial-size comparison's controls, holding-value weights, and parent-bank clusters, but only local-bank product-quarters. Bank-effect versions absorb fixed size and retain contemporaneous bank assets as a control.

Table D2 reports the pre-program size coefficient, its Post interaction, and their sum. The home private-credit slope rises from 4.72 to 12.18 percentage points per log point of bank assets. The 7.47-point increase implies about 5.2 percentage points more expansion at a bank twice as large. Columns 3–4 add bank fixed effects to compare changes within banks.

Table 7a uses the fixed size groups in equation (11), with the smallest tercile as the reference. Bank and quarter effects remove persistent bank differences and common changes. The regressions retain the eleven product, bank, and city controls and weight product-quarters by disclosed bond and private-credit value. The interaction coefficients compare the middle and largest groups' changes in portfolio shares with the smallest group's change.

Table D3 compares the 12 banks in the top tenth of FY2021 interest-bearing assets among the 121 local banks in the regression sample with the remaining banks. It replaces the two group–Post interactions in equation (11) with a top-tenth indicator interacted with Post, retaining bank and quarter effects and the same controls. The estimated difference in home private-credit shares is 14.29 percentage points with investment-value weights and 9.49 points with equal bank-quarter weights. Under the latter scheme, product weights remain proportional to disclosed investment value within each bank-quarter but sum to one, so every observed bank-quarter contributes equally.

Table D1
Initial credit portfolios by bank type and bank size

Share (%)	All provinces		Home province	
	LGFV bonds	PC	LGFV bonds	PC
Panel A. Bank type: all available banks				
National banks	12.6	24.1	0.4	2.1
Local banks	43.1	9.3	24.7	6.1
Panel B. FY2021 size terciles: local banks present in both periods				
Smallest tercile	38.6	1.1	25.1	0.7
Middle tercile	53.3	5.3	37.7	3.5
Largest tercile	40.6	11.4	20.7	7.4

Notes: PC denotes private credit. This table documents the initial allocation levels underlying the all-bank and local-bank size comparisons. Entries are percent of included debt holdings, not all standard bonds or total WMP assets, value weighted, over 2022Q1–2023Q3. All-province shares include home and non-home positions; home means the parent bank's headquarters province. The denominator excludes deposits, funds, other investment vehicles, and central-government, central-bank, and policy-bank obligations. Panel A contains 34,595 national-bank product-quarters from 18 banks and 36,194 local-bank product-quarters from 117 banks. Panel B uses the same 102 local banks and fixed FY2021 interest-bearing-asset terciles as Panel C of Table 7: 25, 38, and 39 banks, contributing 1,965, 9,617, and 22,911 pre-period product-quarters. No positive-holding condition is imposed. Table 1 reports controlled comparisons on complete-case samples. Before implementation, local banks hold 65.0% of the sample's LGFV bonds and national banks hold 82.7% of its PC. After implementation, the corresponding fractions are 62.6% and 68.7%. Ending the initial comparison before the July announcement gives the same descriptive sorting: local and national LGFV shares are 42.4% and 12.2%, while PC shares are 9.5% and 24.0%.

Table D2
Larger local banks and the expansion into private credit

Dependent variable	Total PC share (pp) (1)	Home PC share (pp) (2)	Total PC share (pp) (3)	Home PC share (pp) (4)
Log bank size	7.382*** (2.542)	4.716*** (1.757)	—	—
Log bank size × 𝟙{Post}	10.153*** (2.673)	7.469*** (2.203)	8.288*** (2.110)	6.033*** (1.702)
Log product assets	−1.780** (0.889)	−1.176* (0.640)	−2.896*** (1.080)	−1.887** (0.856)
𝟙{Restricted redemption}	9.036** (3.509)	4.724* (2.808)	10.623*** (3.722)	5.819* (2.955)
Product risk rating	7.813** (3.427)	6.627** (2.946)	4.784 (3.759)	3.968 (2.853)
Bank nonperforming-loan ratio	−5.396** (2.691)	−6.814*** (2.091)	−3.132 (5.828)	−0.880 (5.325)
Bank cost–income ratio	−0.009 (0.250)	0.006 (0.153)	−1.197* (0.643)	−0.867 (0.526)
Bank net interest margin	−0.963 (3.662)	0.361 (2.957)	8.924 (6.043)	8.055 (5.514)
Home-city GDP (RMB trillion)	−5.989*** (1.627)	−6.074*** (1.116)	7.876 (19.243)	−5.811 (15.541)
Home-city GDP growth (%)	−0.727 (0.655)	−0.821 (0.557)	−0.889 (0.584)	−0.788 (0.526)
Home-city LGFV debt/GDP	3.766 (3.764)	4.860 (3.512)	23.120 (20.629)	20.716 (20.528)
Home-city income (RMB 10,000)	2.880*** (0.936)	2.119*** (0.733)	14.448 (11.020)	10.758 (9.791)
Log bank assets	—	—	−18.928 (16.534)	−5.388 (13.793)
Implied contrast from the same regression				
Post-program size gradient	17.535*** (3.300)	12.185*** (2.509)	—	—
Wild p: log bank size × Post	< 0.001	< 0.001	< 0.001	0.017
Observations	85,182	85,182	85,181	85,181
Quarter fixed effects	Yes	Yes	Yes	Yes
Parent bank fixed effects	No	No	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one local-bank allocation regression (pp), with quarter effects, the displayed controls, and holdings weights. Size is unstandardized log FY2021 parent-bank assets in RMB trillion. Post begins in 2023Q4. Columns 1–2 estimate the initial size slope and its change; the post slope sums them using their covariance. Columns 3–4 add bank effects, absorb fixed size, and retain current bank assets. Parentheses contain parent-bank clustered errors; wild tests use 4,999 Rademacher draws. Stars indicate 10%, 5%, and 1% significance. Appendix D.9 provides details. Estimation sample: 2022Q1–2025Q3.

Table D3
Private-credit allocation at the largest local banks

Dependent variable	Home private-credit share (pp)	
Weighting	Investment value	Equal bank-quarter
	(1)	(2)
Largest tenth × Post	14.291***	9.486**
	(3.721)	(3.713)
Product, bank, and city controls	Yes	Yes
Observations	85,181	85,181
Bank fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Standard errors	Parent bank	Parent bank

Notes: Each column is a separate local-bank product-quarter regression. The outcome is home private credit as a percentage of disclosed bonds and private credit. Largest tenth identifies 12 of the 121 banks in the original regression sample, ranked by FY2021 interest-bearing assets; the other banks are the comparison group. Post begins in 2023Q4. Both columns include the control group in Table 7a, omitted here for compactness. Investment-value weights use disclosed bond and private-credit value. Equal-bank-quarter weights sum to one within each bank-quarter and weight its products by their disclosed investment value. Parentheses contain bank-clustered standard errors; stars denote 10%, 5%, and 1% significance. Sample: 2022Q1–2025Q3.

Table D4
The exact-borrower PC–bond rate gap: alternative comparisons

Dependent variable: contractual borrowing yield (percent).

Specification	Baseline	Debt-specific maturity	Equal cells	Reference cohort
	(1)	(2)	(3)	(4)
Private credit–bond gap at baseline	1.904***	1.749***	1.789***	1.848***
	(0.101)	(0.102)	(0.114)	(0.107)
$\mathbb{1}\{\text{Private credit}\} \times \mathbb{1}\{\text{Quarter} = 2023Q1\}$	−0.676***	−0.700***	−0.665***	−0.528***
	(0.115)	(0.104)	(0.132)	(0.131)
$\mathbb{1}\{\text{Private credit}\} \times \mathbb{1}\{\text{Quarter} = 2024Q4\}$	0.955***	1.060***	1.337***	0.681**
	(0.214)	(0.248)	(0.204)	(0.303)
$\mathbb{1}\{\text{Private credit}\} \times \mathbb{1}\{\text{Quarter} = 2025Q3\}$	1.057***	1.197***	1.067***	1.107**
	(0.218)	(0.217)	(0.239)	(0.502)
Observations	17,303	17,303	17,303	8,088
Quarter-specific maturity controls	Yes	Debt-specific	Yes	Yes
Borrower × calendar month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Borrower province	Borrower province	Borrower province	Borrower province

Notes: Each column is one regression of annual contractual rates in percentage points, with borrower-by-calendar-month effects and quarter-specific maturity slopes. The reference gap is PC minus bonds in 2023Q2; other rows are changes from that gap. Column 2 allows debt-specific slopes and evaluates the gap at three years; column 3 gives borrower-quarter-debt-type cells equal weight; column 4 retains the 101 reference-quarter borrowers with later matched financing. Affiliates are excluded. Only key contrasts are tabulated; all quarterly interactions and maturity slopes remain estimated. Parentheses contain province-clustered standard errors; stars use t_{G-1} tests at 10%, 5%, and 1%. The full baseline path is in Figure 3. Matching, singleton removal, and coverage are detailed in Appendix D.2. Estimation sample: 2021Q1–2025Q3.

Table D5
Uncertainty in the exact-borrower financing-rate gap

Panel A. Selected regression contrasts and inference

Dependent variable	Contractual yield (pp) (1)	Paired PC–bond yield gap (pp) (2)
$\mathbb{1}\{\text{Private credit}\} \times \mathbb{1}\{\text{Quarter} = 2024Q4\}$	0.955^{***} (0.214)	—
$\mathbb{1}\{\text{Quarter} = 2024Q4\}$	—	1.182^{***} (0.245)
$\mathbb{1}\{\text{Private credit}\} \times \mathbb{1}\{\text{Quarter} = 2025Q3\}$	1.057^{***} (0.218)	—
$\mathbb{1}\{\text{Quarter} = 2025Q3\}$	—	1.009^{***} (0.217)
Inference for the corresponding quarter contrast		
2024Q4: 95% province wild-bootstrap set	[0.472, 1.676]	[0.699, 1.899]
2025Q3: 95% province wild-bootstrap set	[0.440, 1.883]	[0.414, 1.761]
2024Q4: Province wild-bootstrap p	0.0039	0.0017
2025Q3: Province wild-bootstrap p	0.0056	0.0039
Observations	17,303	2,525
Quarter-specific maturity controls	Contract maturity	Maturity difference
Borrower × calendar month fixed effects	Yes	No
Standard errors	Borrower province	Borrower province

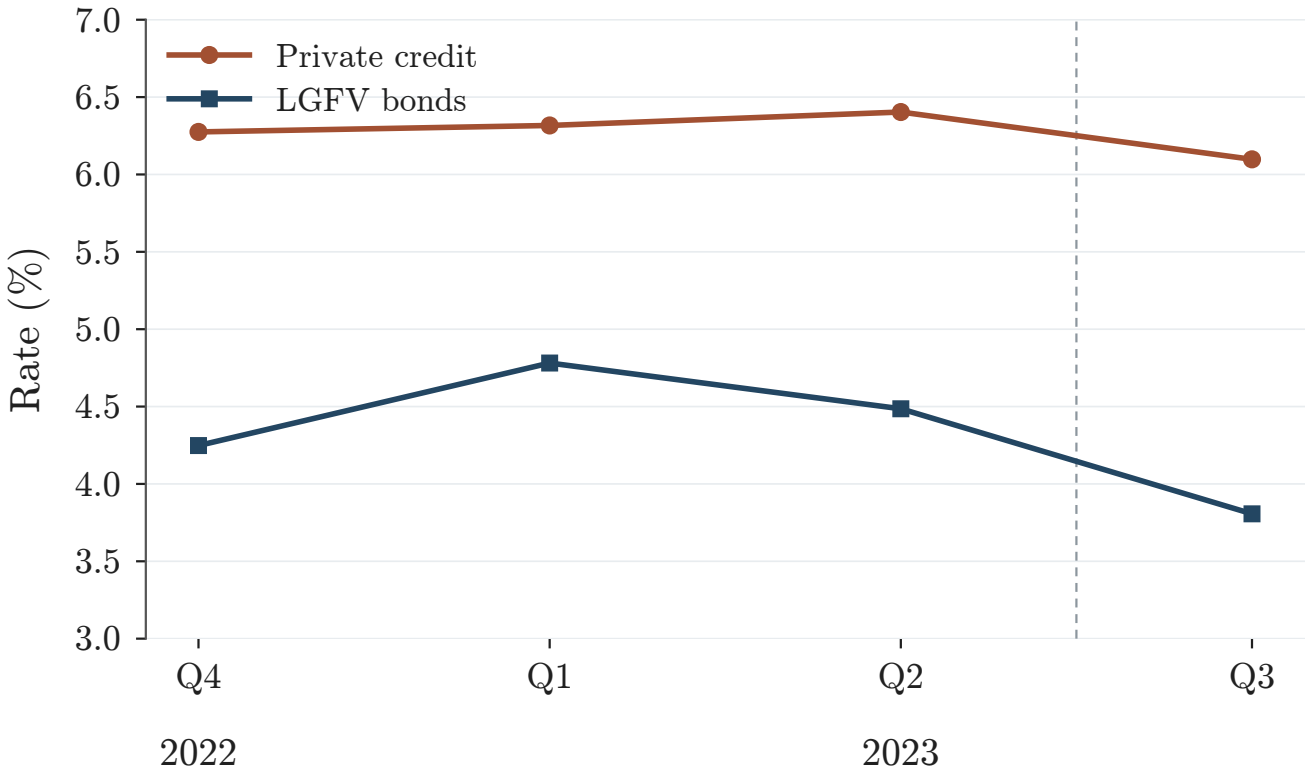
Panel B. Financing coverage (descriptive)

Coverage	2024Q4	2025Q2	2025Q3
PC contracts	360	138	81
Bond issues	140	66	40
Borrowers	75	35	21
Borrowers with both channels in a month	58	33	19
Provinces	13	11	9
Paired borrower-months	68	39	24

Notes: Column 1 uses Table D4' contract-level model; column 2 regresses one borrower-month PC-minus-bond gap on quarter indicators and quarter-specific maturity differences, with no borrower effect. Entries are changes from 2023Q2 (annual pp). Parentheses contain province-clustered errors. Confidence sets invert null-imposed bootstrap-t tests with 9,999 Webb-weight province draws. All quarterly terms remain estimated. Panel B is coverage, not a regression. Appendix D.2 gives the other inference checks. Stars indicate 10%, 5%, and 1% significance. The two regression samples contain 26 and 25 borrower-province clusters, respectively.

Figure D1

Contractual rates for a fixed set of borrowers



Notes: Annual contractual yields on PC and bonds for the same 22 legal borrowers in every quarter from 2022Q4 through 2023Q3, using 799 financing records in seven provinces. Each borrower has both channels in at least one common calendar month of every quarter. Within each borrower-month, contracts or physical bond issues receive equal weight within their channel. Matched monthly means are averaged within borrower-quarter using $h = n_{PC}n_B/(n_{PC} + n_B)$, where n_{PC} and n_B count records in the two channels; the same weights apply to both series. Each borrower then receives weight 1/22 in each quarter. These are observed uncapped contractual rates, without regression or seasonal adjustment. Borrowers and their quarterly weights are fixed, but contracts and maturities can change. The shorter diagnostic window is required by common-borrower coverage; it is not the full sample in Figure 3. Internet Appendix D.2 discusses the Q1–Q2 comparison.

Internet Appendix E
Return, Balance, and Mechanism Checks

This appendix checks return measurement, balance reporting (E.1), holdings disclosure (E.2), and return losses by initial home-LGFV exposure (E.3).

Table E1, Panel A, asks whether bond income explains part of the disappearing home-LGFV return premium. On identical observations, we add the portfolio's holdings-weighted original bond issue yield and its Post interaction to the return model. The home-LGFV Post coefficient changes from −6.33 to −2.57 monthly basis points per unit share: for a 20-point allocation difference, the annual loss falls from about 15 to 6 basis points. The attenuation is consistent with lower bond income contributing to the loss of the home-LGFV return premium. The post-program private credit premium remains positive.

The other columns increase rate coverage, restrict product age, control for volatility, or retain one NAV series. Each preserves the negative LGFV premium change and positive post-program private credit premium. Across additional tail treatments, these estimates range from −6.03 to −6.61 and from 3.96 to 4.64 monthly basis points per unit share, respectively.

Table E2 asks whether the result survives differences in when NAVs recognize gains and losses. Complete six- and twelve-month windows retain annualized home-LGFV premium losses of about 16 and 13 basis points for a 20-point exposure difference, alongside post-program home private credit premiums of 13 and 12 basis points (columns 2 and 4). Raw-return and incumbent-product estimates provide additional comparisons.

Geographic classification gives a further check. Counting centrally administered issuers in the bank's province as home, rather than excluding them from home exposure, changes other-bond shares in 27,244 product-quarters and private credit shares in 328; home LGFV shares are unchanged. The home LGFV level and Post coefficients become 8.31 and −5.32, versus 8.85 and −6.13 in the primary model. The non-home LGFV level is 2.29, with home–non-home equality rejected at $p < 0.001$.

The complete-window specification is the return model in equation (4), with performance measured over $H = 6$ or 12 months rather than one month:

$$\begin{aligned}\text{Window return}_{iw}^{(H)} = {} & \beta_H' \,\text{Asset shares}_{i,h(w)} \\ & + \delta_H' \,\text{Asset shares}_{i,h(w)} \times \mathbb{1}\{\text{Post}\}_w \\ & + \gamma_H' \,\text{Controls}_{i,h(w)} + \lambda_H \,\mathbb{1}\{\text{Local bank}\}_b + \alpha_w + \varepsilon_{iw}.\end{aligned} \tag{E.1}$$

Here w is the first month of the return window and $h(w)$ the preceding quarter, using the same lag convention as the monthly model. For each NAV series, we sum all H monthly log returns in basis points and divide by H; we then average complete series within product to obtain Window return$_{iw}^{(H)}$. Exposure shares and the eleven controls are fixed at the quarter end before the window, and α_w absorbs common window-date conditions. Coefficients are estimated separately for each horizon. Thus δ_H measures the change in an exposure's average-monthly-return premium, not the total return over H months. Table E2 gives the transition exclusions and April–March comparison dates; errors cluster by underlying bank.

Table E1
Home credit exposure and returns: robustness

Dependent variable: monthly product return (bp).

Panel A. Bond-income controls

Specification	Rate sample No carry (1)	Rate sample Add carry (2)	50% coverage No carry (3)	50% coverage Add carry (4)
Home LGFV bond share	9.137***	4.186***	9.411***	4.320**
	(1.909)	(1.578)	(2.015)	(1.723)
Home LGFV bond share × 𝟙{Post}	−6.331***	−2.572*	−6.885***	−3.051*
	(1.511)	(1.438)	(1.559)	(1.547)
Home private credit share	7.813***	5.967***	8.179***	6.524**
	(1.436)	(1.390)	(2.983)	(2.828)
Home private credit share × 𝟙{Post}	−2.185	−0.778	2.039	2.672
	(1.984)	(1.949)	(3.674)	(3.529)
Implied effects from the same regression				
Home private credit premium after the program	5.628***	5.189***	10.217***	9.196***
	(1.535)	(1.388)	(3.409)	(3.059)
Product, bank, and city controls	Yes	Yes	Yes	Yes
Observations	512,509	512,509	438,024	438,024
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. NAV and sample checks

Specification	Mature products (1)	Add volatility (2)	One NAV series (3)	No trimming (4)
Home LGFV bond share	9.587***	9.314***	9.245***	8.512***
	(1.735)	(2.160)	(2.005)	(1.938)
Home LGFV bond share × 𝟙{Post}	−6.460***	−5.743***	−6.692***	−6.609***
	(1.684)	(1.705)	(1.523)	(1.718)
Home private credit share	6.776***	6.263***	7.274***	7.598***
	(1.271)	(1.213)	(1.385)	(1.467)
Home private credit share × 𝟙{Post}	−2.089	−1.369	−3.328*	−3.637**
	(1.854)	(1.692)	(1.840)	(1.559)
Implied effects from the same regression				
Home private credit premium after the program	4.687***	4.894***	3.945***	3.961***
	(1.484)	(1.247)	(1.348)	(1.089)
Product, bank, and city controls	Yes	Yes	Yes	Yes
Observations	373,465	377,835	438,077	567,855
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column estimates the monthly return model in Table 3, with all five exposure shares, their Post interactions, the local-bank indicator, eleven controls, and month effects. Only focal estimates are displayed; parentheses contain parent-bank clustered errors. Panel A fixes the sample within each no-carry/add-carry pair; carry is the portfolio-weighted mean bond issue rate and its Post interaction. Panel B retains products at least four quarters old with lifecycle controls, adds trailing volatility and its Post interaction, retains one NAV series, or restores untrimmed observations of baseline products. Stars indicate 10%, 5%, and 1% significance. Appendix E describes further tail checks.

Table E2
Return premiums over complete measurement windows

Dependent variable: average monthly return (bp).

Specification	6 months Raw (1)	6 months Winsorized (2)	12 months Raw (3)	12 months Winsorized (4)	6 months Incumbents (5)	12 months Incumbents (6)
Home LGFV bond share	9.538**	7.267***	5.267**	5.927***	6.607**	5.596**
	(4.081)	(2.070)	(2.069)	(1.949)	(2.647)	(2.435)
Home LGFV bond share × 𝟙{Post}	−9.465**	−6.840***	−5.470**	−5.544**	−6.604**	−5.169*
	(3.942)	(2.242)	(2.140)	(2.201)	(2.935)	(2.693)
Home private credit share	5.972***	5.050***	5.857***	6.246***	2.782	5.904***
	(2.080)	(1.806)	(1.905)	(1.774)	(2.014)	(1.662)
Home private credit share × 𝟙{Post}	−0.023	0.534	−1.084	−1.152	5.068	−2.141
	(2.553)	(2.413)	(2.533)	(2.435)	(4.572)	(2.128)
Implied effects from the same regression						
Home LGFV bond premium after the program	0.073	0.427	−0.203	0.383	0.003	0.427
	(2.113)	(1.346)	(2.514)	(2.296)	(1.360)	(1.404)
Home private credit premium after the program	5.949***	5.585***	4.773**	5.095**	7.851**	3.763**
	(1.561)	(1.305)	(2.215)	(1.981)	(3.322)	(1.606)
Product, bank, and city controls	Yes	Yes	Yes	Yes	Yes	Yes
Other shares and Post interactions	Yes	Yes	Yes	Yes	Yes	Yes
Observations	53,469	53,469	11,210	11,210	24,097	3,900
Return-window date fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression with the full controls and asset shares of Table 3; only the home-LGFV and home-PC estimates are displayed. Parentheses contain parent-bank clustered errors. All monthly returns must be observed; exposure and controls precede the window, and complete share-class windows are averaged within product. Winsorization is at 1/99. Incumbents were observed before the program. Windows spanning July–September 2023 are excluded; annual April–March windows compare 2022–2023 with 2024–2025. Post-program premiums sum the level and Post interaction from the same column, using their covariance. Stars indicate 10%, 5%, and 1% significance.

E.1. Coverage and interpretation of bank-level WMP balances

The balance checks ask whether smaller banks' weaker WMP growth reflects changing reporting coverage or the different organizational forms of large and small banks. The specification is equation (17) in the main text.

The balance panel uses source-verified corrections to the Wind entries. Yuhang Rural Commercial Bank's June 2025 balance is RMB 16.909 billion.[50] Ruifeng's December 2022, 2023, and 2024 balances are RMB 14.247, 12.797, and 9.582 billion; Fudian's December 2022 and 2023 balances are RMB 21.154 and 21.450 billion. These replace entries approximately twice the reported totals.[51] All balance regressions and the balance event study use this corrected panel. Missing reports remain missing, and verified contractions are retained.

Pre-subsidiary status indicates a subsidiary-registered product observed by 2023Q2, rather than a licensing date. Group-time regressions replace common half-year effects with subsidiary-status-by-half-year effects. The separate-slope version also allows a different size–Post coefficient in each group. Both retain bank effects and bank-clustered inference.

Coverage rises from 56 size-matched local banks in 2022H1 to 86 in 2024H2, then falls to 53 in 2025H2. Table E3 addresses changing composition by excluding the final period, retaining the same 29 banks at every date, and excluding the nine banks with pre-existing subsidiary products. The size-related growth difference remains positive in each sample. Allowing subsidiary banks either a separate Post shift or their own half-year effects gives a coefficient of 0.105 log balance points per log point of assets (SE 0.040–0.041), implying about 7.5% greater relative growth for a doubling of size. The flexible-time wild p-value is 0.003.

Separate size–Post slopes are 0.099 without pre-existing subsidiary products (SE 0.042; wild $p = 0.006$) and 0.237 with them (SE 0.084; wild $p = 0.186$). Requiring an observed pre-announcement portfolio leaves 494 bank-half-years at 78 banks. With group-specific time effects, the size coefficient is 0.096 (SE 0.034; wild $p = 0.012$); the separate slope without subsidiary products is 0.085 (SE 0.036; wild $p = 0.030$). These comparisons hold subsidiary status fixed.

Splitting banks at RMB50 billion of June 2023 WMP balances preserves the size gradient in both groups (Table E3, Panel B). With bank and common half-year effects, the size–Post coefficients are 0.095 below the cutoff and 0.256 at or above it (wild $p = 0.026$ and 0.049), implying 6.8% and 19.4% greater relative growth for a doubling of parent-bank size.

[50] The bank's first-half 2025 report, Table 3, gives 169.09 hundred-million yuan and 191 products; its year-end report confirms the June amount. The Wind entry was 0.0169.

[51] Ruifeng's 2022 annual report and 2024 annual report, including its 2023 comparative, verify these amounts. Fudian's 2022 annual report and 2023 WMP report verify its totals. The latter reports the same balance under two different classifications; these totals must not be added together.

Table E3
WMP balance growth: reporting coverage and sample checks

Dependent variable: log WMP balance.

Panel A. Reporting coverage

Specification	Baseline (1)	Exclude 2025H2 (2)	Same banks all dates (3)	Exclude subsidiaries (4)	Subsidiary × Post (5)
Log bank size × 𝟙{Post}	0.152***	0.119***	0.135***	0.099**	0.105**
	(0.039)	(0.038)	(0.041)	(0.042)	(0.040)
𝟙{Pre-subsidiary} × 𝟙{Post}	—	—	—	—	0.306***
					(0.085)
Observations	571	521	232	506	571
Bank fixed effects	Yes	Yes	Yes	Yes	Yes
Half-year fixed effects	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. Organizational composition and the WMP-balance cutoff

Specification	Subsidiary × time (1)	Separate size slopes (2)	Known status group-time (3)	Cutoff matched (4)	Below RMB50bn (5)	At/above RMB50bn (6)
Log bank size × 𝟙{Post}	0.105**	—	0.096***	0.151***	0.095*	0.256***
	(0.041)		(0.034)	(0.040)	(0.049)	(0.075)
Log bank size × 𝟙{Post}: subsidiary parents	—	0.237***	—	—	—	—
		(0.084)				
Log bank size × 𝟙{Post}: other parents	—	0.099**	—	—	—	—
		(0.042)				
Wild p: log bank size × Post	0.003	—	0.012	< 0.001	0.026	0.049
Observations	571	571	494	496	349	147
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Half-year fixed effects	No	No	No	Yes	Yes	Yes
Subsidiary × time fixed effects	Yes	Yes	Yes	No	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: The outcome is log observed WMP balances. Each column is one bank-half-year regression with the displayed regressors and stated fixed effects; no additional bank or city controls enter. Errors cluster by bank. Size is unstandardized log FY2021 bank interest-bearing assets in RMB trillion. Group-time columns allow pre-program subsidiary status separate half-year paths; column B2 estimates both group-specific size slopes jointly. The same-bank sample contains 29 banks at every date. Columns B4–B6 fix groups using June 2023 WMP balances: 70 matched banks, 50 below RMB50bn, and 20 at or above. All 50 below-cutoff banks lack observed pre-program subsidiary products. The cutoff is not in parent-bank assets and does not identify bank-specific regulatory caps. Missing balances are neither zeros nor assigned to a group. Wild tests use 4,999 bank-level Rademacher draws. Appendix E.1 explains definitions and the verified balance corrections. Stars use conventional 10%, 5%, and 1% tests.

E.2. Top-ten disclosure coverage

Top-ten holdings can cover different fractions of product assets. A home-LGFV share calculated from those holdings could therefore vary with disclosure as well as investment choice. Table E4 addresses this concern in two ways: controlling for observed disclosure differences and removing the disclosed-debt denominator. It revisits three headline comparisons, without restricting the sample to high-coverage products or imposing a common position cutoff.

The three comparisons retain their original specifications: initial home-LGFV allocation on log bank size, ten controls, and quarter effects across all banks (equation (1)); home private credit allocation on log bank size, its Post interaction, and the same controls within local banks (equation (12)); and monthly returns on five lagged allocation shares, their Post interactions, eleven controls, a local-bank indicator, and month effects (equation (4)). The coefficients of interest are log size, log size–Post, and home-LGFV share–Post, respectively. Allocations retain holding-value weights; returns are unweighted. All errors cluster by bank, and each baseline–adjusted pair uses identical observations.

For each product, Coverage is the sum of all reported position weights, expressed as a fraction: disclosed positions totaling 40% give Coverage = 0.40. Log product assets is the natural log of product assets, and Positions counts disclosed positions. Adjusted regressions add

$$\begin{aligned}\text{Disclosure controls}_{it} = &\sum_{k=1}^{3}(\eta_k \,\text{Coverage}_{it}^{k} + \zeta_k \,\text{Log product assets}_{it}^{k}) + \eta_N \,\text{Positions}_{it} \\ &+ \sum_{k=1}^{3}(\tilde{\eta}_k \,\text{Coverage}_{it}^{k} + \tilde{\zeta}_k \,\text{Log product assets}_{it}^{k}) \times \mathbb{1}\{\text{Post}\}_t \\ &+ \tilde{\eta}_N \,\text{Positions}_{it} \times \mathbb{1}\{\text{Post}\}_t.\end{aligned} \tag{E.2}$$

Here t is the outcome quarter or month; in return regressions, Coverage, Log product assets, and Positions come from the preceding quarter end. Terms already in the baseline enter only once. Cubic polynomials allow nonlinear size and coverage relationships; Post interactions allow those relationships to change after implementation.

Panel A retains the original disclosed-debt shares. The initial size coefficient changes from −5.458 to −5.706: a bank twice as large holds 3.8 versus 4.0 percentage points less home LGFV bonds. The private credit size–Post coefficient changes from 7.469 to 6.766, implying 5.2 versus 4.7 points more expansion at a local bank twice as large. The home-LGFV return interaction changes from −6.133 to −5.863 monthly basis points per unit share. For a 20-point allocation difference, the annual premium loss is therefore 14.7 versus 14.1 basis points. All three adjusted estimates remain significant at 1%.

Panel B instead sums the reported home holdings as weights in product assets, removing the disclosed-debt denominator. Its return regression includes all six disclosed debt categories and their Post interactions, with other assets as the reference. The three disclosure-adjusted coefficients are −2.683, 3.175, and −8.603, all significant at 1%. Both adjustments preserve the headline patterns without reconstructing undisclosed holdings.

Table E4
Disclosure coverage and exposure denominators

Panel A. Shares of analyzed bonds and private credit

Dependent variable	Home LGFV share (pp)		Home PC share (pp)		Monthly return (bp)	
Specification	Baseline	Adjusted	Baseline	Adjusted	Baseline	Adjusted
	(1)	(2)	(3)	(4)	(5)	(6)
Log bank size	−5.458***	−5.706***	—	—	—	—
	(1.123)	(1.065)				
Log bank size × 𝟙{Post}	—	—	7.469***	6.766***	—	—
			(2.203)	(1.419)		
Home LGFV bond share × 𝟙{Post}	—	—	—	—	−6.133***	−5.863***
					(1.501)	(1.296)
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Disclosure controls	No	Yes	No	Yes	No	Yes
Observations	68,331	68,331	85,182	85,182	554,555	554,555
Quarter fixed effects	Yes	Yes	Yes	Yes	No	No
Month fixed effects	No	No	No	No	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. Reported weights in product assets

Dependent variable	Home LGFV share (pp)		Home PC share (pp)		Monthly return (bp)	
Specification	Baseline	Adjusted	Baseline	Adjusted	Baseline	Adjusted
	(1)	(2)	(3)	(4)	(5)	(6)
Log bank size	−2.863***	−2.683***	—	—	—	—
	(0.703)	(0.656)				
Log bank size × 𝟙{Post}	—	—	3.451**	3.175***	—	—
			(1.428)	(0.983)		
Home LGFV bond share × 𝟙{Post}	—	—	—	—	−8.097***	−8.603***
					(2.366)	(2.594)
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Disclosure controls	No	Yes	No	Yes	No	Yes
Observations	68,331	68,331	85,182	85,182	554,555	554,555
Quarter fixed effects	Yes	Yes	Yes	Yes	No	No
Month fixed effects	No	No	No	No	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column in each panel is one regression; PC denotes private credit. Only the named focal coefficients are displayed. Baseline controls comprise ten product, bank, and city variables in columns 1–4 and eleven in columns 5–6; the full baseline regressors, including other exposure shares and interactions, are retained. Columns 1–2 use all banks before implementation; 3–4 use local banks before and after; 5–6 use monthly returns. Designs follow equations (1), (12), and (4); adjusted columns add equation (E.2). Allocations are in pp; returns in monthly bp. Log bank size is log FY2021 underlying-bank interest-bearing assets; log bank assets is the time-varying control. Coverage is the sum of all disclosed position weights, not availability of issue yields. Panel B uses reported product-asset weights directly (fractions in return regressions), not independently observed full portfolios. Its return regression includes all six disclosed debt categories and their Post interactions. All other controls, allocation weights, and underlying-bank clustering are unchanged. Products with no disclosed holdings in a category remain in the regressions. Stars denote 10%, 5%, and 1% significance.

E.3. Initial home-LGFV reliance and return erosion

Table E5 tests whether banks initially more exposed to home LGFV bonds experience larger subsequent return declines, conditional on size. We average NAV series within product-month and products within bank-month, giving each bank-month equal weight. Return observations come from the corrected CNY NAV sample, retain its global 1/99 exclusion, and need no contemporaneous holdings match. Initial exposure is measured from pre-program holdings. By comparison, the return columns of Table 8 use equally weighted NAV-series-months matched to lagged product characteristics and control for broad initial home overweight. These designs measure initial bond-exposure losses and size-related product-return differences, respectively. Columns 3–4 here hold the 93-bank sample fixed while adding eight prior-quarter controls.

The home-exposure coefficient is −10.69 monthly basis points per unit share after adding pre-program non-home LGFV share–Post, and −11.07 with quarter-specific size slopes (both wild $p < 0.001$). The exposure result therefore persists when accounting for general LGFV reliance or allowing the size–return relationship to change by quarter: banks with greater initial home-LGFV exposure subsequently experience larger product-return declines.

Table E5
Initial home-LGFV exposure and WMP-return erosion

Dependent variable: bank-average monthly WMP return (bp).

	Broad sample (1)	Province comparison (2)	Matched sample (3)	Add controls (4)
Initial home LGFV bond share × 𝟙{Post}	−10.963***	−12.639***	−10.301***	−10.528***
	(2.657)	(4.079)	(3.355)	(3.410)
Wild p: initial home LGFV share × Post	< 0.001	0.002	0.002	0.003
Initial log bank size × Post control	Yes	Yes	Yes	Yes
Lagged bank controls	No	No	No	Yes
Lagged home-city controls	No	No	No	Yes
Observations	2,821	2,821	2,006	2,006
Bank fixed effects	Yes	Yes	Yes	Yes
Subsidiary status × month effects	Yes	Yes	Yes	Yes
Home province × month effects	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column uses monthly log returns (bp), July 2022–September 2025, excluding July–September 2023. Exposure is the 2022Q1–2023Q2 home-LGFV standard-bond share of included debt, not total WMP assets. For a ten-point exposure difference, multiply its coefficient by 0.10; multiply again by 12 to annualize. The table displays the exposure coefficients; initial log bank size interacted with Post remains a control in every column. Initial size is log FY2021 bank interest-bearing assets. Column 4 also includes lagged bank assets, nonperforming loans, cost–income ratio, net interest margin, and home-city GDP, growth, LGFV debt/GDP, and income. Post begins October 2023; controls lag one quarter. Products are equally weighted within bank-months, as are bank-months in estimation. Parentheses show bank-clustered standard errors in monthly units; wild tests use 4,999 draws. Stars denote 10%, 5%, and 1% significance.

Table E5 establishes the initial-exposure comparison; Table E6 asks whether its conclusion depends on product weighting or return coverage. All columns of Table E6 use equation (14), add home-province-by-month effects, and include initial log size–Post and the eight lagged controls. Columns 1–2 change only the weighting on identical bank-months: each product receives equal weight in column 1, whereas each bank-month receives equal weight in column 2. Thus banks with more products do not receive extra weight in the second comparison. Column 3 adds returns from products without matched holdings on those same bank-months; column 4 also admits the remaining eligible bank-months. Initial bond exposure stays fixed throughout.
For a ten-percentage-point difference in initial home-LGFV exposure, the annualized return losses are 8.2 and 15.2 basis points under the two weights (bank-level wild $p = 0.038$ and 0.005). Broader NAV coverage gives losses of 18.2 basis points on the same bank-months and 18.1 on the expanded sample. More exposed banks therefore lose more return whether the comparison represents the average observed product or gives each bank equal weight, and whether or not subsequent products have matched holdings. The controls and initial exposure are unchanged across these checks.

Table E6
Initial bond exposure: product weighting and return coverage

Dependent variable: monthly WMP log return (bp).

	Holdings-matched products		Broader NAV products	
	Equal products (1)	Equal banks (2)	Same bank-months (3)	All eligible bank-months (4)
Initial home LGFV bond share × 𝟙{Post}	−6.871*	−12.644***	−15.192**	−15.078***
	(3.647)	(4.286)	(5.987)	(5.514)
Bank-clustered p: exposure × Post	0.063	0.004	0.013	0.008
Wild p: exposure × Post	0.038	0.005	—	—
Initial log bank size × Post control	Yes	Yes	Yes	Yes
Lagged bank controls	Yes	Yes	Yes	Yes
Lagged home-city controls	Yes	Yes	Yes	Yes
Observations (bank-months)	1,562	1,562	1,562	1,655
Bank fixed effects	Yes	Yes	Yes	Yes
Subsidiary status × month effects	Yes	Yes	Yes	Yes
Home province × month effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression. Exposure, size, timing, and the eight prior-quarter controls follow Table E5. The exposure is a raw fraction of analyzed disclosed bonds and private credit, not total WMP assets. NAV series are first averaged within product-month, then products within bank-month. All regressors are bank-month variables. Column 1 weights each bank-month by its product count, exactly reproducing equal-product coefficients; columns 2–4 weight bank-months equally. This common aggregation also uses bank-months for the finite-sample adjustment to standard errors. Columns 1–3 contain the same 80 identifying banks; the matched sample in columns 1–2 contains 124,612 product-months. Column 3 adds NAV products without requiring a holdings match. Column 4 includes the remaining eligible bank-months, giving 86 banks. Shared NAV observations have identical returns and retain the original global 1/99 exclusion rule. Only the initial-exposure coefficient is displayed; all stated controls remain included. For a ten-point exposure difference, multiply by 0.10×12 to obtain annualized basis points. Parentheses contain bank-clustered standard errors. Wild tests use 4,999 bank-level draws; dashes indicate not computed. Stars use conventional 10%, 5%, and 1% tests.

E.4. Private-credit expansion and WMP performance

The allocation results establish that larger local banks expanded private credit more. This comparison asks whether banks making greater use of that substitute also preserved more return and WMP business, conditional on their initial portfolios and size. For underlying bank b at date t, we estimate

$$\begin{aligned}\text{Outcome}_{bt} = {} & \beta_P\, \Delta\text{Home PC share}_b \times \mathbb{1}\{\text{Post}\}_t \\ & + \beta_L\, \text{Initial home LGFV share}_b \times \mathbb{1}\{\text{Post}\}_t \\ & + \beta_S\, \text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_t \\ & + \beta_H\, \text{Initial home PC share}_b \times \mathbb{1}\{\text{Post}\}_t \\ & + \gamma'\, \text{Lagged controls}_{bt} + \alpha_b + \alpha_t + u_{bt}.\end{aligned} \tag{E.3}$$

Here PC denotes private credit. For returns, t is a month and the outcome is the bank's average monthly WMP log return in basis points, constructed as in Table E5. For balances, t is a half-year reporting date and the outcome is log outstanding WMP balances. Post begins in 2023Q4. Bank and date effects absorb fixed bank characteristics and the common Post change. Inference clusters by underlying bank; Table E7 uses conventional bank-clustered p-values and significance stars.

Within each bank-quarter, the home-private-credit share divides home-private-credit holding value by the value of analyzed bonds and private credit. Thus products are value-weighted within a bank. We average these quarterly shares over 2022Q1–2023Q2 and over 2023Q4–2025Q3, then take the latter minus the former to obtain ΔHome PC share$_b$. Equal weighting across quarters prevents growth in the disclosed portfolio from mechanically giving later quarters more weight. Banks need at least two observed holdings quarters in each period; missing holdings are not zeros. Initial home-LGFV and private-credit shares use holding-value aggregation over the pre-program window. Initial size remains raw log FY2021 interest-bearing assets. All eight lagged controls are shown in columns 2–4.

The return estimates separate initial exposure from subsequent adjustment. In column 2, ten points more initial home-LGFV exposure predicts 15.8 annualized basis points more return decline ($p < 0.001$), while a ten-point larger private-credit increase is associated with 11.0 basis points less decline ($p = 0.039$). Return weighting addresses a separate measurement choice. On identical products and bank-months with lagged product assets available, a ten-point private-credit increase corresponds to 14.8 annualized basis points in the equally weighted product-return average ($p = 0.034$), versus 4.0 when returns are weighted by prior-quarter product assets ($p = 0.611$). These estimates relate observed portfolio adjustment to subsequent performance, conditional on initial portfolios, size, and controls.

The balance columns use the corrected reports described in Appendix E.1. Column 3 includes initial portfolios and size but omits the private-credit-expansion term; column 4 adds it. Both use the same eight bank and city controls, calendar half-year effects, and equal bank–reporting-date weights. Annual controls are matched to quarters as described in Appendix E.5; each specification retains all banks with usable pre- and post-program observations. Column 3 contains 331 observations at 51 banks, and column 4 contains 325 observations at 49 banks with sufficient holdings to measure expansion.

A ten-point larger private-credit increase is associated with 7.2% greater proportional WMP-balance growth in column 4 ($p = 0.024$). The size coefficient falls from 0.103 to 0.067 when expansion is included, implying relative growth of 7.4% and 4.7% for a doubling of bank size. On column 4's observations, the initial-characteristics specification gives a size coefficient of 0.107, so sample attrition does not account for this reduction. Initial home-LGFV exposure has positive coefficients in the total-balance regressions, unlike its negative coefficients for new-product totals in Table 11. The latter measure newly launched business; total outstanding balances also contain the stock of older products.

Table E7
Private-credit expansion, returns, and WMP balances

Dependent variable	Monthly return (bp)		Log WMP balance	
	Baseline	Controls	Initial characteristics	+ PC expansion
	(1)	(2)	(3)	(4)
Home private credit share increase × 𝟙{Post}	8.127*	9.189**	—	0.692**
	(4.488)	(4.376)		(0.296)
Initial home LGFV bond share × 𝟙{Post}	−10.685***	−13.172***	0.242*	0.200*
	(3.007)	(3.758)	(0.122)	(0.119)
Log FY2021 bank size × 𝟙{Post}	1.128	0.298	0.103***	0.067**
	(0.764)	(0.759)	(0.025)	(0.028)
Initial home private credit share × 𝟙{Post}	8.097*	6.956	0.972***	1.174***
	(4.189)	(5.171)	(0.295)	(0.284)
Lagged log bank assets	—	−0.871	0.226	0.228
		(5.174)	(0.264)	(0.255)
Lagged nonperforming-loan ratio	—	−2.728	0.214***	0.224**
		(2.319)	(0.067)	(0.103)
Lagged cost–income ratio	—	−0.435	−0.016*	−0.013
		(0.274)	(0.009)	(0.009)
Lagged net interest margin	—	2.304	−0.022	−0.064
		(2.758)	(0.081)	(0.087)
Lagged city GDP (RMB trillion)	—	15.929**	1.070***	0.990***
		(6.210)	(0.270)	(0.259)
Lagged city GDP growth	—	−0.629*	−0.018	−0.014
		(0.352)	(0.011)	(0.011)
Lagged city LGFV debt/GDP	—	10.854	−0.745*	−0.776
		(11.202)	(0.430)	(0.473)
Lagged urban income (RMB 10,000)	—	6.684**	−0.002	−0.069
		(3.341)	(0.168)	(0.172)
Observations	2,369	1,851	331	325
Bank fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	No	No
Half-year fixed effects	No	No	Yes	Yes
Standard errors	Bank	Bank	Bank	Bank

Notes: Each column is one DiD regression. Columns 1–2 and 4 estimate equation (E.3); column 1 omits the eight controls and column 3 omits private-credit expansion. All slope coefficients are displayed. Shares are raw fractions of analyzed bonds and private credit; size is raw log FY2021 bank interest-bearing assets. For a ten-point share difference, multiply monthly-return coefficients by 0.10×12 for annualized bp; the balance difference is $100[\exp(0.10\beta) - 1]\%$. Returns cover July 2022–September 2025, excluding 2023Q3; balances use available June/December reports in 2022–2025. Post begins in 2023Q4. Controls lag one quarter. Balance columns contain 51 and 49 banks with usable observations before and after the program; column 4 also requires post-program holdings. Observations receive equal bank-date weight. Fixed bank characteristics and Post are absorbed by the fixed effects. Parentheses contain conventional bank-clustered standard errors; stars indicate 10%, 5%, and 1% significance. Appendix E.4 defines the variables and discusses the comparisons.

E.5. Measuring new WMP business

Table 11 measures new business as the total balances of observed products launched by the same underlying local bank in the same quarter. For each product launched in quarter q, we use total assets at the end of quarter $q+1$, constructed from the holdings-based asset measure in Appendix D.5. The measurement window retains products aged 90–183 days. We sum these positive balances within bank–launch quarter and then take the log. The outcome therefore combines how many products the bank launches with how much business those products accumulate. It is neither a bank's total outstanding WMP balance nor net subscriptions. Missing products or cohorts are not assigned zero balances.

Equation (16) compares cohort totals before and after the program using initial home-LGFV exposure, initial home-private-credit exposure, and bank size interacted with Post. Column 2 adds actual home-private-credit expansion interacted with Post, as defined in Appendix E.4. Initial shares aggregate holding values over 2022Q1–2023Q2, requiring at least two observed quarters; their denominator is analyzed disclosed bonds and private credit, not total WMP assets. Measuring expansion additionally requires at least two observed quarters during 2023Q4–2025Q3. Size is unstandardized log FY2021 interest-bearing assets. Launches cover 2022Q2–2025Q2, excluding 2023Q3, and Post begins with launches in 2023Q4.

Both columns include bank and launch-quarter effects and the eight prior-quarter bank and home-city controls displayed in the table. Each bank–launch quarter receives equal weight, and standard errors cluster by underlying bank. The outcome captures total new business across product designs, so the regressions do not condition on the number or composition of newly launched products. The control vector is the same as in the total-WMP-balance comparison in Table E7; each outcome uses its full eligible sample rather than an intersection of banks reporting both outcomes.

The bank and city controls originate in annual records. We match each observed bank-year record to all four quarters of that year before applying the one-quarter lag. This prevents a quarter without a portfolio disclosure from losing an otherwise available annual control record. The match does not introduce new values or additional across-year filling. The same matching rule is used for the total-WMP-balance columns of Table E7.

Column 1 contains 647 cohorts, 13,182 products, and 72 banks. Column 2 contains 641 cohorts, 13,100 products, and 71 banks; the excluded bank lacks sufficient post-program holdings to measure expansion. Both require usable cohorts before and after the program. Re-estimating column 1 on column 2's sample gives reliance and size coefficients of -1.525 and 0.216, close to -1.556 and 0.224 in the full sample. Thus requiring expansion data does not drive the change in the coefficients.

Initial reliance predicts lower new-product totals: ten points more home-LGFV exposure corresponds to a 14.4% relative decline in column 1 and 15.3% in column 2. A ten-point larger private-credit increase corresponds to 14.6% greater relative growth in column 2. A doubling of initial bank size corresponds to 16.8% greater growth before expansion is included; the additional size coefficient is smaller once expansion is included. These comparisons link initial bond exposure and subsequent substitution to aggregate new business, without requiring each individual product to become larger or smaller.